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ROBOT CONTROLLER

RC90 Series

Manual

Rev.5 ENM231C5630F

Original instructions

RC90-B

ROBOT CONTROLLER RC90 Series Rev.5

ROBOT CONTROLLER

RC90 Series Manual

Rev.5

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FOREWORD

Thank you for purchasing our robot products.
This manual contains the information necessary for the correct use of the robot controller.
Please carefully read this manual and other related manuals before installing the robot system.
Keep this manual handy for easy access at all times.

The robot system and its optional parts are shipped to our customers only after being subjected to the strictest quality controls, tests, and inspections to certify its compliance with our high performance standards. Please note that the basic performance of the product will not be exhibited if our robot system is used outside of the usage conditions and product specifications described in the manuals.

This manual describes possible dangers and consequences that we can foresee. Be sure to comply with safety precautions on this manual to use our robot system safety and correctly.

TRADEMARKS

Microsoft, Windows, and Windows logo are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries. Other brand and product names are trademarks or registered trademarks of the respective holders.

TRADEMARK NOTATION IN THIS MANUAL

Microsoft® Windows® 8 operating system
Microsoft® Windows® 10 operating system
Microsoft® Windows® 11 operating system
Throughout this manual, Windows 8, Windows 10 and Windows 11 refer to above respective operating systems. In some cases, Windows refers generically to Windows 8, Windows 10 and Windows 11.

NOTICE

No part of this manual may be copied or reproduced without authorization.
The contents of this manual are subject to change without notice.
Please notify us if you should find any errors in this manual or if you have any comments regarding its contents.

MANUFACTURER

SEIKO EPSON CORPORATION

CONTACT INFORMATION

Contact information is described in “SUPPLIERS” in the first pages of the following manual:

Robot System Safety Manual Read this manual first

DISPOSAL

When disposing this product, dispose in accordance with the laws and regulations of each country.

Regarding battery disposal

The battery removal/replacement procedure is described in the following manuals:
Maintenance Manual

For European Union customers only



The crossed out wheeled bin label that can be found on your product indicates that this product and incorporated batteries should not be disposed of via the normal household waste stream. To prevent possible harm to the environment or human health please separate this product and its batteries from other waste streams to ensure that it can be recycled in an environmentally sound manner. For more details on available collection facilities please contact your local government office or the retailer where you purchased this product. Use of the chemical symbols Pb, Cd or Hg indicates if these metals are used in the battery.

This information only applies to customers in the European Union, according to DIRECTIVE 2006/66/EC OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL OF 6 September 2006 on batteries and accumulators and waste batteries and accumulators and repealing Directive 91/157/EEC and legislation transposing and implementing it into the various national legal systems.
For other countries, please contact your local government to investigate the possibility of recycling your product.

For Users in Taiwan region



Please separate used batteries from other waste streams to ensure that it can be recycled in an environmentally sound manner. For more details on available collection facilities please contact your local government office or the retailer where you purchased this product.

For California customers only

The lithium batteries in this product contain
Perchlorate Material - special handling may apply,
See www.dtsc.ca.gov/hazardouswaste/perchlorate

Before Reading This Manual

CAUTION	■ Necessity of organizational measures for cyber security The following organizational measures should be taken to address cyber security risks: - Perform risk analysis based on security threats and vulnerabilities related to your organization's assets. - Develop a security policy to address risks and educate and train appropriate personnel. - Create guidelines for how to respond when security issues arise and make them known throughout your organization. ■ Security measures for network connections Epson robot systems are designed to be used within a closed local area network. Please refrain from connecting to networks with Internet access. If a connection to the Internet is required, please implement the necessary technical measures* to protect from malicious attacks and vulnerabilities over the Internet. *: These measures include, but are not limited to, access controls, firewalls, data diodes, and so on.

NOTE

Do not connect the followings to the TP/OP port of RC90-B. Connecting to the followings may result in malfunction of the device since the pin assignments are different.

OPTIONAL DEVICE dummy plug

Operation Pendant OP500

Operator Pendant OP500RC

Jog Pad JP500

Teaching Pendant TP-3**

Operator Panel OP1

NOTE

For RC90-B, be sure to install the EPSON RC+7.0 to the development PC first, then connect the development PC and RC90-B with the USB cable.

If RC90-B and the development PC are connected without installing the EPSON RC+7.0 to the development PC, [Add New Hardware Wizard] appears. If this wizard appears, click the <Cancel> button.

NOTE

Concerning the security support for the network connection:

The network connecting function (Ethernet) on our products assumes the use in the local network such as the factory LAN network. Do not connect to the external network such as Internet.

In addition, please take security measure such as for the virus from the network connection by installing the antivirus software.

NOTE

Security support for the USB memory:

Make sure the USB memory is not infected with virus when connecting to the Controller.

Control System Configuration

This manual explains with the following combinations of Controllers and software.

Robot Controller RC90-B:

		RC90-B controller firmware
		Ver.7.4.2.0 or later
EPSON RC+ 7.0	Before Ver.7.4.1	!!!
	Ver.7.4.2 or later	OK

OK: Compatible All functions of the EPSON RC+ 7.0 and the Controller are available.

!!!: Compatible Connection is OK. We recommend using EPSON RC+7.0 Ver. 7.4.2 or later.

NOTE

Manual PDF for this robot system is available from EPSON RC+ 7.0 Ver. 7.0.2

Each Manipulator is compatible from following version.

LS3-B****		RC90-B controller firmware
		Ver.7.4.4.2 or later
EPSON RC+ 7.0	Before Ver.7.4.3	!!!
	Ver.7.4.4 or later	OK

OK: Compatible All functions of the EPSON RC+ 7.0 and the Controller are available.

!!!: Compatible Connection is OK. We recommend using EPSON RC+7.0 Ver. 7.4.2 or later.

LS3-B401S-V1		RC90-B controller firmware
		Ver.7.5.1.2 or later
EPSON RC+ 7.0	Before Ver.7.5.1A	!!!
	Ver.7.5.1B or later	OK

OK: Compatible All functions of the EPSON RC+ 7.0 and the Controller are available.

!!!: Compatible Connection is OK. We recommend using EPSON RC+7.0 Ver. 7.5.1B or later.

LS6-B****		RC90-B controller firmware
		Ver.7.4.3.1 or later
EPSON RC+ 7.0	Before Ver.7.4.2	!!!
	Ver.7.4.3 or later	OK

OK: Compatible All functions of the EPSON RC+ 7.0 and the Controller are available.

!!!: Compatible Connection is OK. We recommend using EPSON RC+7.0 Ver. 7.4.2 or later.

LS6-B602S-V1		RC90-B controller firmware Ver.7.5.0.5 or later
EPSON RC+ 7.0	Before Ver.7.5.0	!!!
	Ver.7.5.0 R3 or later	OK

OK: Compatible All functions of the EPSON RC+ 7.0 and the Controller are available.

!!!: Compatible Connection is OK. We recommend using EPSON RC+7.0 Ver. 7.4.2 or later.

LS10-B****		RC90-B controller firmware Ver.7.4.2.1 or later
EPSON RC+ 7.0	Before Ver.7.4.1	!!!
	Ver.7.4.2 or later	OK

OK: Compatible All functions of the EPSON RC+ 7.0 and the Controller are available.

!!!: Compatible Connection is OK. We recommend using EPSON RC+7.0 Ver. 7.4.2 or later.

LS20-B****		RC90-B controller firmware Ver.7.4.5 or later
EPSON RC+ 7.0	Before Ver.7.4.4	!!!
	Ver.7.4.5 or later	OK

OK: Compatible All functions of the EPSON RC+ 7.0 and the Controller are available.

!!!: Compatible Connection is OK. We recommend using EPSON RC+7.0 Ver. 7.4.2 or later.

The Manuals of This Product

The following are typical manual types for this product and an outline of the descriptions.

Safety Manual (book, PDF)

This manual contains safety information for all people who handle this product. The manual also describes the process from unpacking to operation and the manual you should look at next.

Read this manual first.

- Safety precautions regarding robot system and residual risk
- Declaration of conformity
- Training
- Flow from unpacking to operation

RC90 series Manual (PDF)

This manual explains the installation of the entire robot system and the specifications and functions of the controller. The manual is primarily intended for people who design robot systems.

- The installation procedure of the robot system (specific details from unpacking to operation)
- Daily inspection of the controller
- Controller specifications and basic functions

LS-B series Manual (PDF)

This manual describes the specifications and functions of the Manipulator. The manual is primarily intended for people who design robot systems.

- Technical information, functions, specifications, etc. required for the Manipulator installation and design
- Daily inspection of the Manipulator

Status Code/Error Code List (PDF)

This manual contains a list of code numbers displayed on the controller and messages displayed in the software message area. The manual is primarily intended for people who design robot systems or do programming.

RC90 series Maintenance Manual (PDF)
LS-B series Maintenance Manual (PDF)

This manual describes the details of maintenance etc. The manual is intended for people who perform maintenance.

- Daily inspection
- Replacement and repair of maintenance parts
- The method of firmware update and controller setting backup etc.

EPSON RC+ 7.0 User's Guide (PDF Manual)

This manual describes general information about program development software.

EPSON RC+ 7.0 SPEL+ Language Reference (PDF Manual)

This manual describes the robot programming language "SPEL+".

Other Manual (PDF Manual)

Manuals for each option are available.

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Installation

This section describes outline from unpacking to operation of Robot System and design of Robot System.

1. Safety

Please read “Safety Manual” and check safety considerations before installing the robot system or before connecting cables.
Keep this manual handy for easy access at all times.

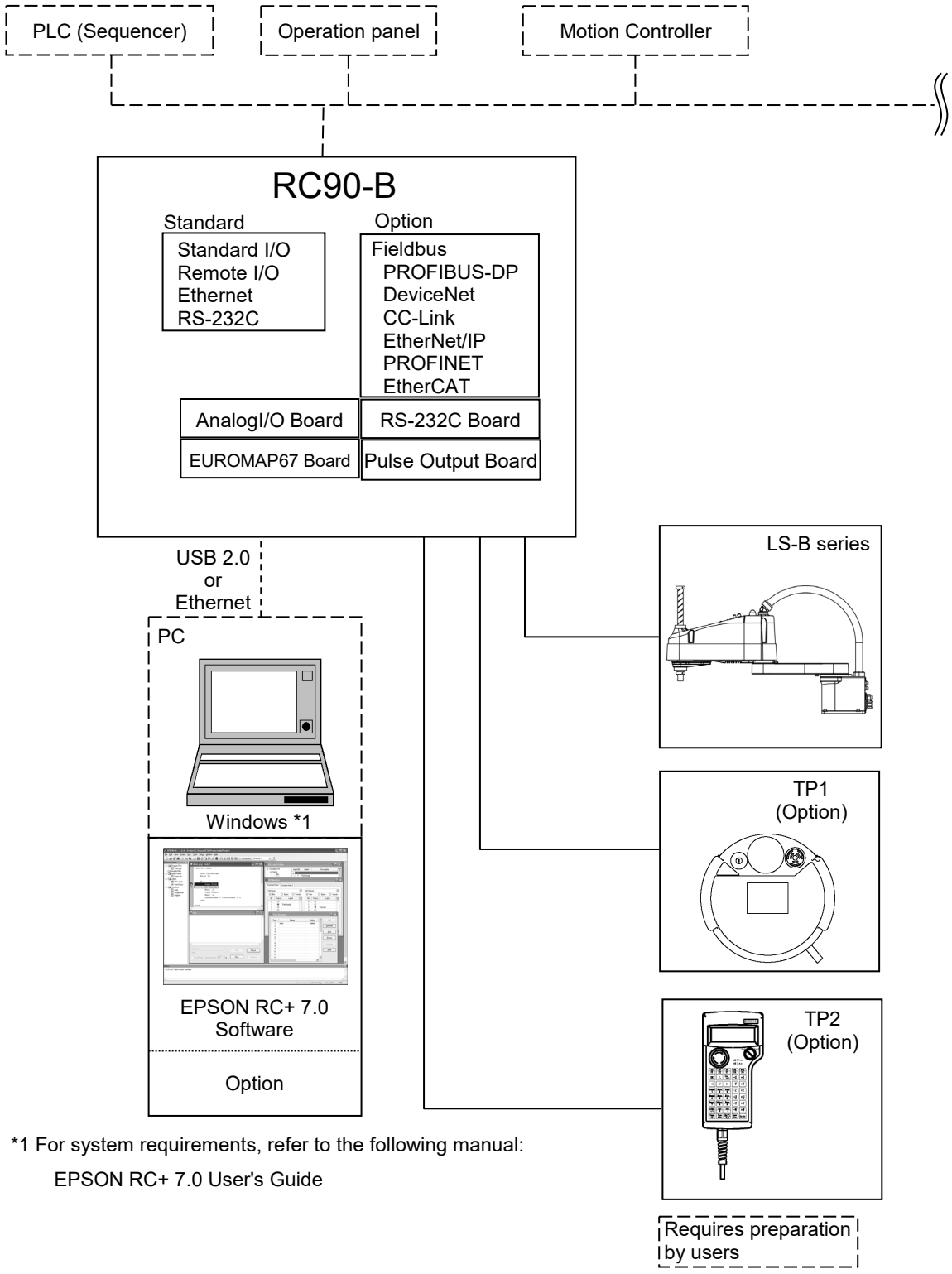
1.1 Conventions

Important safety considerations are indicated throughout the manual by the following symbols. Be sure to read the descriptions shown with each symbol.

 WARNING	This symbol indicates that a danger of possible serious injury or death exists if the associated instructions are not followed properly.
 WARNING	This symbol indicates that a danger of possible harm to people caused by electric shock exists if the associated instructions are not followed properly.
 CAUTION	This symbol indicates that a danger of possible harm to people or physical damage to equipment and facilities exists if the associated instructions are not followed properly.

2. Specifications

2.1 System Example



*1 For system requirements, refer to the following manual:

EPSON RC+ 7.0 User's Guide

2.2 Unpacking

Unpacking of the Manipulators and robotic equipment shall be performed by personnel who has taken robot system training held by us and suppliers and should conform to all national and local codes.

Please read “Safety Manual” and check safety considerations before installing the robot system or before connecting cables.

2.3 Transportation

2.3.1 Transportation precautions

Transporting of the Manipulators and robotic equipment shall be performed by personnel who has taken robot system training held by us and suppliers and should conform to all national and local codes.
Please read “Safety Manual” and check safety considerations before installing the robot system or before connecting cables.
During unpacking and relocation, avoid applying external force to the arms and motors of the Manipulator.
When transporting the Manipulator for a long distance, secure it to the delivery equipment so that the Manipulator cannot fall. If necessary, pack the Manipulator in the same way as it was delivered.

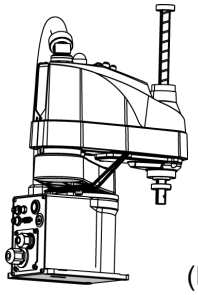
2.3.2 Number of people and position to hold, when transporting manipulator.

LS3-B, LS6-B, LS10-B

Where to hold : Under the arm 1 and bottom of the base *(shade part)
* When hold bottom of the base, be careful not caught your hands or fingers.

Minimum number of people : 2 people

Do not hold : -



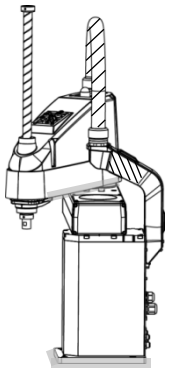
(Figure: LS6-B)

LS20-B

Where to hold : Under the arm 1 and bottom of the base *(shade part)
* When hold bottom of the base, be careful not caught your hands or fingers.

Minimum number of people : 2 people

Do not hold : Metal duct part, resin duct part (slashed part)



(Figure: LS20-B)

2.4 Manipulator Installation

2.4.1 Transportation Precautions

Installing of the Manipulators and robotic equipment shall be performed by personnel who has taken robot system training held by us and suppliers and should conform to all national and local codes.
Please read “Safety Manual” and check safety considerations before installing the robot system or before connecting cables.

2.4.2 Environment

A suitable environment is necessary for the robot system to function properly and safely. Be sure to install the robot system in an environment that meets the following conditions:

Item	Environment
Ambient temperature *	5 ~ 40 °C
Ambient relative humidity	10 ~ 80 % (no condensation)
Fast transient burst noise	1 kV or less(signal wire)
Electrostatic noise	4 kV or less
Elevation	1000 m or less
Environment	- Install indoors. - Keep away from direct sunlight. - Keep away from dust, oily smoke, salinity, metal powder or other contaminants. - Keep away from flammable or corrosive solvents and gases. - Keep away from water. - Keep away from shocks or vibrations. - Keep away from sources of electric noise. - Keep away from explosive area - Keep away from a large quantity of radiation

* When the product is used in a low temperature environment around the minimum temperature of the product specification, or when the product is suspended for a long time on holidays or at night, a collision detection error may occur due to the large resistance of the drive unit immediately after the start of operation. In such a case, it is recommended to warm up for about 10 minutes.

NOTE

Manipulators are not suitable for operation in harsh environments such as painting areas, etc. When using Manipulators in inadequate environments that do not meet the above conditions, please contact the supplier of your region.

Special Environmental Conditions

The surface of the Manipulator has general oil resistance. However, if your requirements specify that the Manipulator must withstand certain kinds of oil, please contact the supplier of your region.

Rapid change in temperature and humidity can cause condensation inside the Manipulator.

If your requirements specify that the Manipulator handles food, please contact the supplier of your region to check whether the Manipulator will damage the food or not.

The Manipulator cannot be used in corrosive environments where acid or alkaline is used. In a salty environment where the rust is likely to gather, the Manipulator is susceptible to rust.

 WARNING	■ Use an earth leakage breaker on the AC power cable of the Controller to avoid the electric shock and circuit breakdown. Prepare the earth leakage breaker that pertains the Controller you are using. For details, refer to the Robot Controller manual.

 CAUTION	■ When cleaning the Manipulator, do not rub it strongly with alcohol or benzene. It may lose luster on the coated face.

2.4.3 Noise level

About noise level due to manipulator operation, refer to the following manual.

Manipulator Manual Appendix A. Specifications Table

2.4.4 Base Table

A base table for anchoring the Manipulator is not supplied. Please make or obtain the base table for your Manipulator. The shape and size of the base table differs depending on the use of the robot system. For your reference, we list some Manipulator table requirements here.

The base table must not only be able to bear the weight of the Manipulator but also be able to withstand the dynamic movement of the Manipulator when it operates at maximum acceleration. Ensure that there is enough strength on the base table by attaching reinforcing materials such as crossbeams.

Use the mounting bolts conforming to the strength of ISO898-1 property class 10.9 or 12.9.

The table must be secured on the floor or wall to prevent it from moving.

The Manipulator installation surface should have a flatness of 0.5 mm or less and an inclination of 0.5 ° or less. If the flatness of the installation surface is improper, the base may be damaged or the robot may not fully show its performance.

When using a leveler to adjust the height of the base table, use a screw with M16 diameter. The torque and reaction force produced by Manipulator motion are as follows:

	LS20-B	LS10-B	LS6-B	LS3-B
Max. Reaction torque on the horizontal plate (N·m)	1,000	550	350	250
Max. Horizontal reaction force (N)	7,500	3,200	1,700	1,000
Max. Vertical reaction force (N)	2,000	1,500	1,500	1,000
Threaded holes for Mounting screw	M12	M8	M8	M8

The plate for the Manipulator mounting face should be 20 mm thick or more and made of steel to reduce vibration. The surface roughness of the steel plate should be 25 μm or less.

When using the Manipulator in the cleanroom, follow the steps below before the installation.

(1) Unpack the Manipulator outside of the cleanroom.

(2) Secure the Manipulator to delivery equipment such as a pallet with bolts so that the Manipulator does not fall over.

(3) Wipe off the dust on the Manipulator with a little alcohol or distilled water on a lint-free cloth.

(4) Transport the Manipulator into the cleanroom.

(5) Secure the Manipulator to the base table.

2.4.5 Installation Procedure

NOTE

When the Manipulator is cleanroom model, unpack it outside of the cleanroom. Secure the Manipulator not to fall, and then wipe off the dust on the Manipulator with a little alcohol or distilled water on a lint-free cloth. After that, transport the Manipulator into the cleanroom. Connect an exhaust tube to the exhaust port after installation.

LS3-B, LS6-B, LS10-B

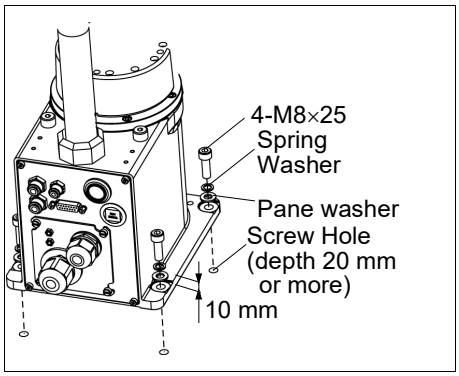
CAUTION	■ Installing or relocate LS3-B, LS6-B, LS10-B with two or more people. The Manipulator weights are as follows. Be careful not to get hands, fingers, or feet caught and/or have equipment damaged by a fall of the Manipulator. LS3-B401* : approx. 14 kg: 31 lbs. LS6-B502* : approx. 17 kg: 37.5 lbs. LS6-B602* : approx. 17 kg: 37.5 lbs. LS6-B602S-V1: approx. 18 kg: 39.7 lbs. LS6-B702* : approx. 18 kg: 39.7 lbs. LS10-B60** : approx. 23 kg: 50.7 lbs. LS10-B70** : approx. 23 kg: 50.7 lbs. LS10-B80** : approx. 24 kg: 52.9 lbs.

(1) Secure the base to the base table with four bolts.

NOTE

Use bolts with specifications conforming to ISO898-1 property Class: 10.9 or 12.9.

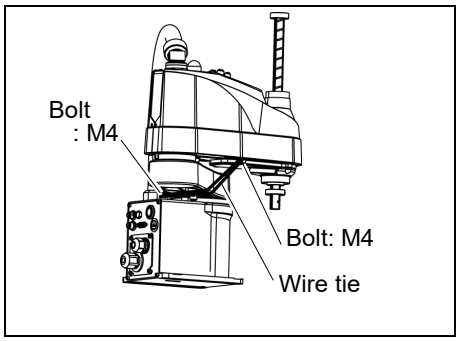
Tightening torque: 32.0 N·m (326 kgf·cm)



(2) Using nippers, cut off the wire tie binding the shaft and arm retaining bracket on the base.

(3) Remove the bolts securing the wire ties removed in step (2).

Remove the fixing sheet for transportation which is attached to the arm.



NOTE

Make sure to remove the wire tie for protection of mechanical stop.

LS20-B

CAUTION	■ Install the LS20-B Manipulator with two or more people. The Manipulator weights are as follows. Be careful not to get hands, fingers, or feet caught and/or have equipment damaged by a fall of the Manipulator. LS20-B804*: approx. 48 kg: 105.8 lbs. LS20-BA04*: approx. 51 kg: 112.5 lbs.

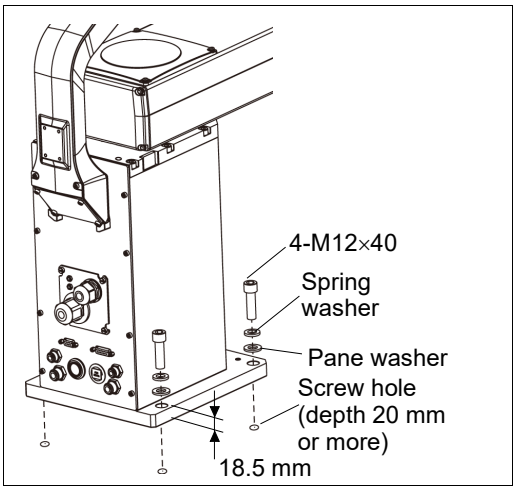
(1)

NOTE

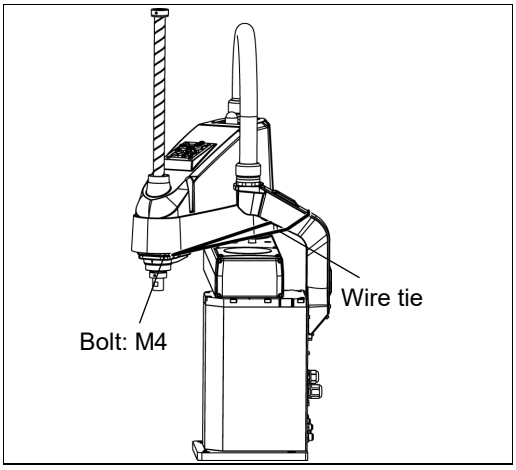
Secure the base to the base table with four bolts.

Use bolts with specifications conforming to ISO898-1 Property Class: 10.9 or 12.9.

Tightening torque:
73.5 N·m (750 kgf·cm)



(2) Using nippers, cut off the wire tie binding the shaft and arm retaining bracket on the base.
Remove the fixing bolt.

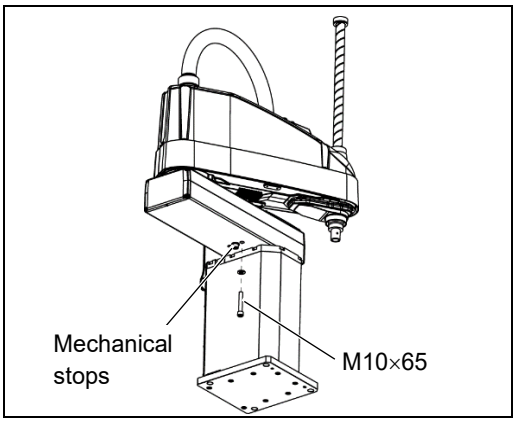


(3) Remove the arm lock.

NOTE

Removed screw will be used when re-locating manipulator.
Keep it with the arm lock that packed at shipment
Do not lose them.

Do not remove the Mechanical Stops.



2.5 Controller Installation

2.5.1 Environment

For using the robot system function properly and safely, a suitable environment is required. Install the control unit in a location meets the following conditions.

CAUTION	■ Install the controller at a height of 0.4 to 2m above the floor to maintain easily. ■ Install a power-off switch at a height of 0.6 to 1.9m.

NOTE

- The Controller are not designed for cleanroom specification. If it must be installed in a cleanroom, be sure to install it in a proper enclosure with adequate ventilation and cooling.
- Install controller in a location that allows easy connection / disconnection of cables.
- Install Controller outside of the safety door.

Item	Condition
Ambient temperature	5 to 40 °C
Ambient relative humidity	20% to 80% (with no condensation)
Fast transient burst noise	2 kV or less (Power supply wire) 1 kV or less (Signal wire)
Electrostatic noise	4 kV or less
Base table	Use a base table that is at least 100 mm off the floor. Placing the Controller directly on the floor could allow dust penetration leading to malfunction.
Installation face	Inclination 0.5 ° or less (When installing vertically, the control unit or drive unit may fall over if you touch them.)
Elevation	1000 m or less

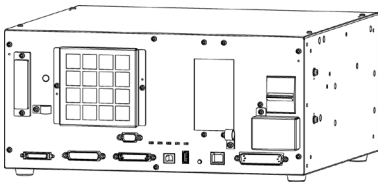
If the controller must be used in an environment that does not fulfill the conditions mentioned above, take adequate countermeasures. For example, the controller may be enclosed in a cabinet with adequate ventilation and cooling.

- Install indoors, and place in a well-ventilated area.
- Keep away from direct sunlight.
- Keep away from radiation heat.
- Keep away from dust, oily mist, oil, salinity, metal powder or other contaminants.
- Keep away from water.
- Keep away from shocks or vibrations.
- Keep away from sources of electronic noise
- Keep away from strong electric or magnetic fields.

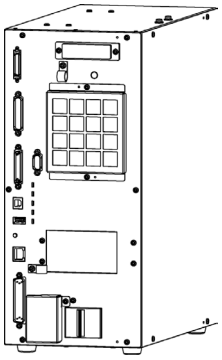
2.5.2 Installation

Install the Controller on a flat surface such as wall, floor, and Controller box in the direction shown in (A) to (C).

(A) Flat mounting

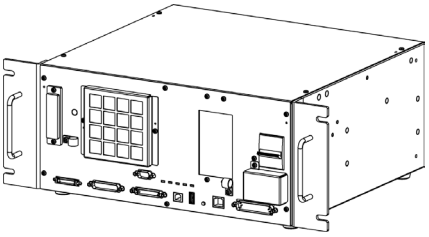


(B) Upright mounting



* The rubber foot needs to be replaced. Attach the rubber foot with the flat side facing the controller.
  The size of the screw that secure the rubber foot is M4×8.
  When replacing the rubber foot, be careful not to lose the screw. Also, do not use a screw of size other than M4×8.

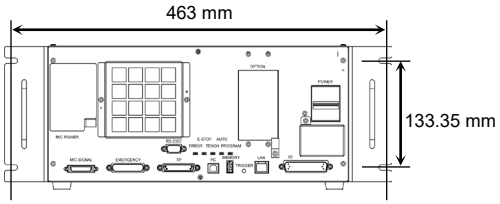
(C) Rack mounting



* A plate for rack mounting is required.

NOTE

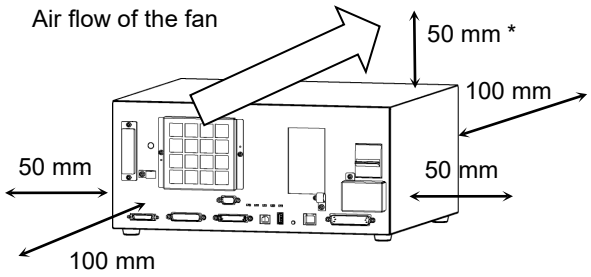
For the Controller installation to the Controller box or the base table, process screw hole drilling as follows.



- Ensure the air flow around the supply and exhaust ports, and install the Controller while leaving space from other equipment or walls as below.

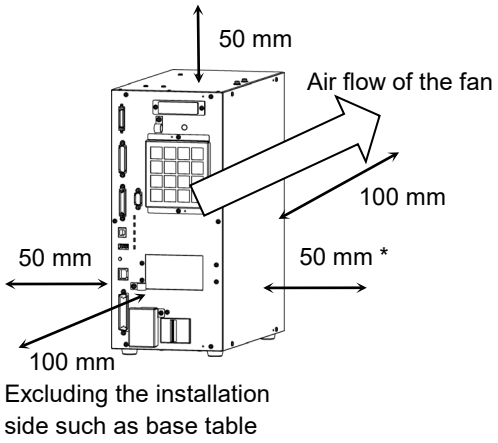
  * Keep the space about 200 mm or more on the top to maintenance.

(A)Flat mounting, (C)Rack mounting



(Figure Flat mounting)

(B) Upright mounting



- Hot air with higher temperature than the ambient temperature (about 10 °C) comes out from the Controller. Make sure that heat sensitive devices are not placed near the outlet.
- Arrange the cables in front of the Controller so that you can pull the Controller forward.

2.6 Designing a Safe Robot System

It is important to operate robots safely. It is also important for robot users to give careful consideration to the safety of the overall robot system design.
This section summarizes the minimum conditions that should be observed when using EPSON robots in your robot systems.
Please design and manufacture robot systems in accordance with the principles described in this and the following sections.

2.6.1 Environmental Conditions

Carefully observe the conditions for installing robots and robot systems that are listed in the "Environmental Conditions" tables included in the manuals for all equipment used in the system.

2.6.2 System Layout

When designing the layout for a robot system, carefully consider the possibility of error between robots and peripheral equipment. Emergency stops require particular attention, since a robot will stop after following a path that is different from its normal movement path. The layout design should provide enough margins for safety. Refer to the manuals for each manipulator, and ensure that the layout secures ample space for maintenance and inspection work.

When designing a robot system to restrict the area of motion of the robots, do so in accordance with the methods described in each manipulator manual. Utilize both software and mechanical stops as measures to restrict motion.

Install the emergency stop switch at a location near the operation unit for the robot system where the operator can easily press and hold it in an emergency.

Do not install the controller at a location where water or other liquids can leak inside the controller. In addition, never use liquids to clean the controller.

2.6.3 Disabling Power to the System using lock out / tag out

The power connection for the robot controller should be such that it can be locked and tagged in the off position to prevent anyone from turning on power while someone else is in the safeguarded area.

For further details, refer to the following section:

RC90 Series Maintenance Manual 1.2 Lockout/Tagout

2.6.4 End Effector Design

Provide wiring and piping that will prevent the robot end effector from releasing the object held (the work piece) when the robot system power is shut off.

Design the robot end effector such that its weight and moment of inertia do not exceed the allowable limits. Use of values that exceed the allowable limits can subject the robot to excessive loads. This will not only shorten the service life of the robot but can lead to unexpectedly dangerous situations due to additional external forces applied to the end effector and the work piece.

Design the size of the end effector with care, since the robot body and robot end effector can interfere with each other.

2.6.5 Peripheral Equipment Design

When designing equipment that removes and supplies parts and materials to the robot system, ensure that the design provides the operator with sufficient safety. If there is a need to remove and supply materials without stopping the robot, install a shuttle device or take other measures to ensure that the operator does not need to enter a potentially dangerous zone.

Ensure that an interruption to the power supply (power shutoff) of peripheral equipment does not lead to a dangerous situation. Take measures that not only prevent a work piece held from being released as mentioned in “End effector Design” but that also ensure peripheral equipment other than the robots can stop safely. Verify equipment safety to ensure that, when the power shuts off, the area is safe.

2.6.6 Remote Control

For the safety of the overall system, however, safety measures are needed to eliminate the risks associated with the start-up and shutdown of peripheral equipment by remote control.

With this product, the robot system can be operated remotely by assigning a remote function to the controller I/O. To prevent danger from unintended remote control, the remote function will not be enabled without proper settings. Also when remote is valid, motion command execution and I/O output are available only from remote.

Pressing the Emergency Stop switch immediately changes the Manipulator to deceleration motion and stops it at the maximum deceleration speed.

2.6.7 Emergency Stop

Each robot system needs equipment that will allow the operator to immediately stop the system's operation. Install an emergency stop device that utilizes emergency stop input from the Controller and all other equipment.

Use the emergency switch which has safety function more than PLd.

The safety function of Emergency Stop Input:	Category 3, PLd (reference ISO13849-1 2015)
The Stop category of Emergency Stop Input:	Category 0, (refer Safety Standard IEC60204-1)

NOTE

Test pulse cannot be used with the emergency stop input of this model.

During the emergency stop, the power supplied to the motor for driving the robot is shut off (non-operating state), and the robot is stopped due to the dynamic brake.

Pressing the Emergency Stop switch immediately changes the Manipulator to deceleration motion and stops it at the maximum deceleration speed.

The controller is equipped with an emergency stop input terminal. If the normally closed emergency stop circuit is broken, the power supplied to all motors will be shut off (and enter servo-free status) and the robot will be stopped by dynamic braking.

The emergency stop circuit should also remove power from all external components that must be turned off during an emergency. Do not assume that the robot controller will turn off all outputs if configured to. For example, if an I/O card is faulty, the controller cannot turn off a component connected to an output. The emergency stop on the controller is hardwired to remove motor power from the robot, but not external power supplies.

Do not press the Emergency Stop switch unnecessarily while the Robot is operating. Pressing the switch during the operation makes the brakes work. This will shorten the life of the brakes due to the worn friction plates.

Normal brake life cycle: About 2 years (when the brakes are used 100 times/day)

However, the rough normal relay life is approximately 20,000 times. If you press the emergency stop switch unnecessarily, the life of the relay will be shortened.

Also, the Emergency Stop during operation applies impact on the reduction gear unit, and it may result in the short life of the reduction gear unit.
Following may result in the short life when stopping the operation by pressing emergency switch in the manipulator working properly.
Brakes
Reduction gear unit

To place the robot system in emergency mode during normal operation, press the Emergency Stop switch while the Manipulator is not moving.
Refer to the Controller manual for instructions on how to wire the Emergency Stop switch circuit.

Do not turn OFF the Controller while the Manipulator is operating.

If the Manipulator is stopped by turning OFF the Controller while it is operating, the following problems may occur.
Reduction of the life and damage of the reduction gear unit
Position gap at the joints

In addition, if the Controller was forced to be turned OFF by blackouts and the like while the Manipulator is operating, make sure to check the following points after power restoration.
Whether or not the reduction gear is damaged
Whether or not the joints are in their proper positions
If there is a position gap, perform calibration.
For the calibration, refer to the *"Manipulator Maintenance Manual"*.

Before using the Emergency Stop switch, be aware of the followings.
- The Emergency Stop (E-STOP) switch should be used to stop the Manipulator only in case of emergencies.
- To stop the Manipulator operating the program except in emergency, use Pause (halt) or STOP (program stop) commands.
  Pause and STOP commands do not turn OFF the motors. Therefore, the brake does not function.
- For the Safeguard system, do not use the circuit for E-STOP.

To check brake problems, refer to the following manuals.
Manipulator Manual Inspection Point 1.1.2 Inspection Item

Stopping distance in emergency

The operating Manipulator cannot stop immediately after the Emergency Stop switch is pressed. In addition, stopping time and stopping distance vary by following factors:

Hand weight	WEIGHT Setting	ACCEL Setting
Workpiece weight	SPEED Setting	Posture etc.

For stopping time and stopping distance, refer to the following manual:
Manipulator Manual
Appendix B. Stopping Time and Stopping Distance in Emergency

2.6.8 Safeguard

To ensure safe operation, install a safety system using safety doors, light curtains, safety floor mats, etc.
When installing the safeguard system, strictly observe the following points:
Install the safeguard which has safety function more than PLd.

The safety features of Safety Door Input:	Category 3, PLd (reference ISO13849-1 2015)
The Stop category of Safety Door Input:	Category 1, (refer Safety Standard IEC60204-1)

NOTE

Test pulse cannot be used with the safeguard input of this model.

Refer to each manipulator manual and install the safeguard system outside the maximum space. Carefully consider the size of the end effector and the work pieces to be held so that there will be no error between the moving parts and the safeguard system.

Manufacture the safeguard system to withstand calculated external forces (forces that will be added during operation and forces from the surrounding environment).

When designing the safeguard system, make sure that it is free of sharp corners and projections, and that the safeguard system itself is not a hazard.

Make sure that the safeguard system can only be removed by using a tool.

There are several types of safeguard devices, including safety doors, safety barriers, light curtains, safety gates, and safety floor mats. Install the interlocking function in the safeguard device. The safeguard interlock must be installed so that the safeguard interlock is forced to work in case of a device failure or other unexpected accident. For example, when using a door with a switch as the interlock, do not rely on the switch's own spring force to open the contact. The contact mechanism must open immediately in case of an accident.

Connect the interlock switch to the safeguard input of the drive unit's EMERGENCY connector. The safeguard input informs the robot controller that an operator may be inside the safeguard area. When the safeguard input is activated, the robot stops immediately and enters pause status, as well as either operation-prohibited status or restricted status (low power status).

It is dangerous to allow someone else to release the safeguard interlock by mistake while the operator is working inside the safeguarded area. To protect the operator working inside the safeguarded area, take measures to lock out and tag out the latch-release switch.

WARNING	■ The EMERGENCY connector on the controller has a safeguard input circuit to connect the safety device interlock switch. To protect operators working near the robot, be sure to connect the interlock switch and make sure that it works properly.

When a closed safeguard is open during robot motion, the safeguard interlock function operates. The robot stops immediately and enters into pause state. Then, all robot motors are turned OFF. The descriptions below explain how the safeguard input works.

Safeguard open : The robot stops immediately, motors are turned OFF, and further operation is impossible until either the safeguard is closed or TEACH or TEST mode is turned ON and the enable circuit is engaged.
Safeguard closed : The robot can automatically operate in unrestricted (high power) state.

Do not open the safeguard unnecessarily while motor is ON. Frequent safeguard inputs affect the life of the relay.
Rough normal relay life: Approximately 20,000 times

For the safeguard, do not use the E-STOP circuit.
For details of wiring instructions, refer to the "*9. Emergency*".

For details of the safeguard, refer to the following section as well.

2.7.1 Connection to EMERGENCY Connector

Stopping distance when the safeguard is opened

The Manipulator in operation cannot stop immediately after the safeguard is opened. In addition, stopping time and stopping distance vary by following factors:

Hand weight	WEIGHT Setting	ACCEL Setting
Workpiece weight	SPEED Setting	Posture etc.

For stopping time and stopping distance of the Manipulator refer to the following manual.

Manipulator Manual
Appendix C: Stopping Time and Stopping Distance When the Safeguard is Opened.

2.6.9 Presence Sensing Device

The above mentioned safeguard interlock is a type of presence sensing device, since it indicates the possibility of somebody being inside the safeguard system. When separately installing a presence sensing device, however, perform a satisfactory risk assessment and pay thorough attention to its dependability.

Here are precautions that should be noted:

- Design the system so that when the presence sensing device is not activated or a dangerous situation still exists that no personnel can go inside the safeguard area or place their hands inside it.
- Design the presence sensing device so that regardless of the situation the system operates safely.
- If the robot stops operating when the presence sensing device is activated, it is necessary to ensure that it does not start again until the detected object has been removed. Make sure that the robot cannot automatically restart.

2.6.10 Resetting the Safeguard

Ensure that the robot system can only be restarted through careful operation from outside the safeguarded system. The robot will never restart simply by resetting the safeguard interlock switch. Apply this concept to the interlock gates and presence sensing devices for the entire system.

2.6.11 Robot Operation Panel

The robot operation panel must not be located inside of the robot work envelope / work cell. Ensure that the robot system can be operated from outside of the safeguard.

2.7 Connecting

2.7.1 Connection to EMERGENCY Connector

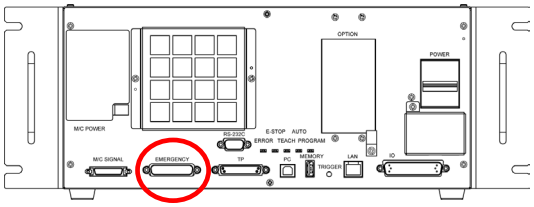
NOTE

The details of safety requirements for this section are described in Safety Manual. Please refer to them to keep the robot system safe.

WARNING	■ Not only when turning ON the device, but also changing use environment such as add options or replace parts for maintenance, make sure that the emergency stop or safety door work properly.

Connect a safeguard switch or Emergency Stop switch to the Control unit and Drive unit EMERGENCY connector for safety.
When nothing is connected to the EMERGENCY connector, Control unit and Drive unit does not operate normally.

WARNING	■ Before connecting the connector, make sure that the pins are not bent. Connecting with the pins bent may damage the connector and result in malfunction of the robot system.



EMERGENCY Connector

Safety Door Switch and Latch Release Switch

The EMERGENCY connector has input terminals for the Safety Door switch and the Emergency Stop switch. Be sure to use these input terminals to keep the system safe.

Connector	Standard
EMERGENCY connector (Controller side)	D-sub25 Pin (male) Screwlock #4-40

Safety Door Switch

WARNING	■ The interlock of the Safety Door must be functioning when the robot system is operated. Do not operate the system under the condition that the switch cannot be turned ON/OFF (e.g. covering the switch with tape). Operating the robot system when the switch is not functioning properly is extremely hazardous and may cause serious safety problems as the Safety Door input cannot fulfill its intended function.

In order to maintain a safe working zone, a safeguard must be set up around the Manipulator. The safeguard must have an interlock switch at the entrance to the working zoon. The Safety Door that is described in this manual is one of the safeguards and an interlock of the Safety Door is called a Safety Door switch. Connect the Safety Door switch to the Safety Door input terminal on the EMERGENCY connector.

The Safety Door switch has safety features such as temporary hold-up of the program or the operation-prohibited status that are activated whenever the Safety Door is opened.

Observe the following in designing the Safety Door switch and the Safety Door.

- For the Safety Door switch, select a switch that opens as the Safety Door opens, and not by the spring of the switch itself.
- The signal from the Safety Door (Safety Door input) is designed to input to two redundant signals. If the signals at the two inputs differ by two seconds or more, the system regards it as a critical error. Therefore, make sure that the Safety Door switch has two separate redundant circuits and that each connects to the specified pins at the EMERGENCY connector on the Controller.
- The Safety Door must be designed and installed so that it does not close accidentally.

Latch Release Switch

The controller software latches the following conditions:

- The safety door is open.
- The operation mode is “TEACH”.

The EMERGENCY connector has an input terminal for a latch release switch that cancels the latched conditions.

Open :The latch release switch latches conditions that the safety door is open or the operation mode is “TEACH”.

Closed :The latch release switch releases the latched conditions.

NOTE When the latched TEACH mode is released while the safety door is open, the status of Manipulator power is operation-prohibited because the safety door is open at that time.
To execute a Manipulator operation, close the safety door again, and then close the latch release input.

Checking Latch Release Switch Operation

Refer to 4.2 Development PC and Controller Connection and connect the development PC and Controller before checking the function.
After connecting the safety door switch and latch release switch to the EMERGENCY connector, be sure to check the switch operation for safety by following the procedures described below before operating the Manipulator.

(1) Turn ON the Controller while the safety door is open in order to boot the Controller software.
(2) Make sure that “Safety” is displayed on the EPSON RC+ 7.0 status bar.
(3) Close the safety door, and turn ON the switch connecting to the latch release input. Make sure that the “Safety” is dimmed on the status bar.

The information that the safety door is open can be latched by software based on the latch release input condition. To cancel the condition, close the safety door, and then close the safety door latch release input.

Open : The latch release switch latches condition that the safety door is open.

Closed : The latch release switch does not latch the condition that the safety door is open.

NOTE
The latch release input also functions to acknowledge the change of TEACH mode.
In order to change the latched condition of the TEACH mode, turn the mode selector key switch on the Teach Pendant to “Auto”. Then, close the latch release input.

Emergency Stop Switch

If it is desired to create an external Emergency Stop switch in addition to the Emergency Stop on the Teach Pendant, make sure to connect the Emergency Stop switch to the Emergency Stop input terminal on the EMERGENCY connector.
The Emergency Stop switch connected must comply with the related safety standards (such as IEC60947-5-5) and the following:

- It must be a push button switch that is “normally closed”.
- A button that does not automatically return or resume.
- The button must be mushroom-shaped and red.
- The button must have a double contact that is “normally closed”.

NOTE
The signal from the Emergency Stop switch is designed to use two redundant circuits.
If the signals at the two circuits differ by two seconds or more, the system recognizes it as a critical error. Therefore, make sure that the Emergency Stop switch has double contacts and refer to the following to connect.

Circuit Diagrams

Checking Emergency Stop Switch Operation

NOTE

Refer to *4.2 Development PC and Controller Connection* and connect the development PC and Controller before checking the function.

Once the Emergency Stop switch is connected to the EMERGENCY connector, continue the following procedure to make sure that the switch functions properly.

(1) Turn ON the Controller to boot the controller software while pressing the Emergency Stop switch.

(2) Make sure that E-STOP LED of the controller is lighting.

(3) Make sure that “E.Stop” is displayed on the status bar.

(4) Release the Emergency Stop Switch.

(5) Execute the RESET command.

(6) Make sure that E-STOP LED is turned OFF and that “E-Stop” is dimmed on the main window status bar.

Pin Assignments

The EMERGENCY connector pin assignments are as follows:

Pin No.	Signal	Function	Pin No.	Signal	Function
1	ESW11	Emergency Stop switch	14	ESW21	Emergency Stop switch
2	ESW12	contact (1) *3	15	ESW22	contact (2) *3
3	ESTOP1+	Emergency Stop switch	16	ESTOP2+	Emergency Stop switch
4	ESTOP1–	contact (1) *3	17	ESTOP2-	contact (2) *3
5	Not used	Emergency Stop circuit 1 (+)*4	18	SDLATCH1	Emergency Stop circuit 2 (+)*4
6	Not used	Emergency Stop circuit 1 (–)*4	19	SDLATCH2	Emergency Stop circuit 2 (–)*4
7	SD11	*1	20	SD21	Safety Door Latch Release
8	SD12	*1	21	SD22	Safety Door Latch Release
9	24V	Safety Door input (1) *2	22	24V	Safety Door input (2) *2
10	24V	Safety Door input (1) *2	23	24V	Safety Door input (2) *2
11	24VGND	+24V output	24	24VGND	+24V output
12	24VGND	+24V output	25	24VGND	+24V output
13	Not used				

*1 Do not connect anything to these pins.

*2 A critical error occurs if the input values from the Safety Door 1 and Safety Door 2 are different for two or more seconds. They must be connected to the same switch with two sets of contacts.

*3 A critical error occurs if the input values from the Emergency Stop switch contact 1 and Emergency Stop switch contact 2 are different for two or more seconds. They must be connected the same switch with two sets of contacts.

*4 Do not apply reverse voltage to the Emergency Stop circuit.

Emergency Stop switch output rated load	+30 V 0.3 A or under	1-2, 14-15 pin
Emergency Stop rated input voltage range Emergency Stop rated input current	+24 V ±10% 37.5 mA ±10% / +24V input	3-4, 16-17 pin
Safety Door rated input voltage range Safety Door rated input current	+24 V ±10% 10 mA / +24 V input	7-8, 20-21 pin
Latch Release rated input voltage range Latch Release rated input current	+24 V ±10% 10 mA / +24 V input	18-19 pin

NOTE

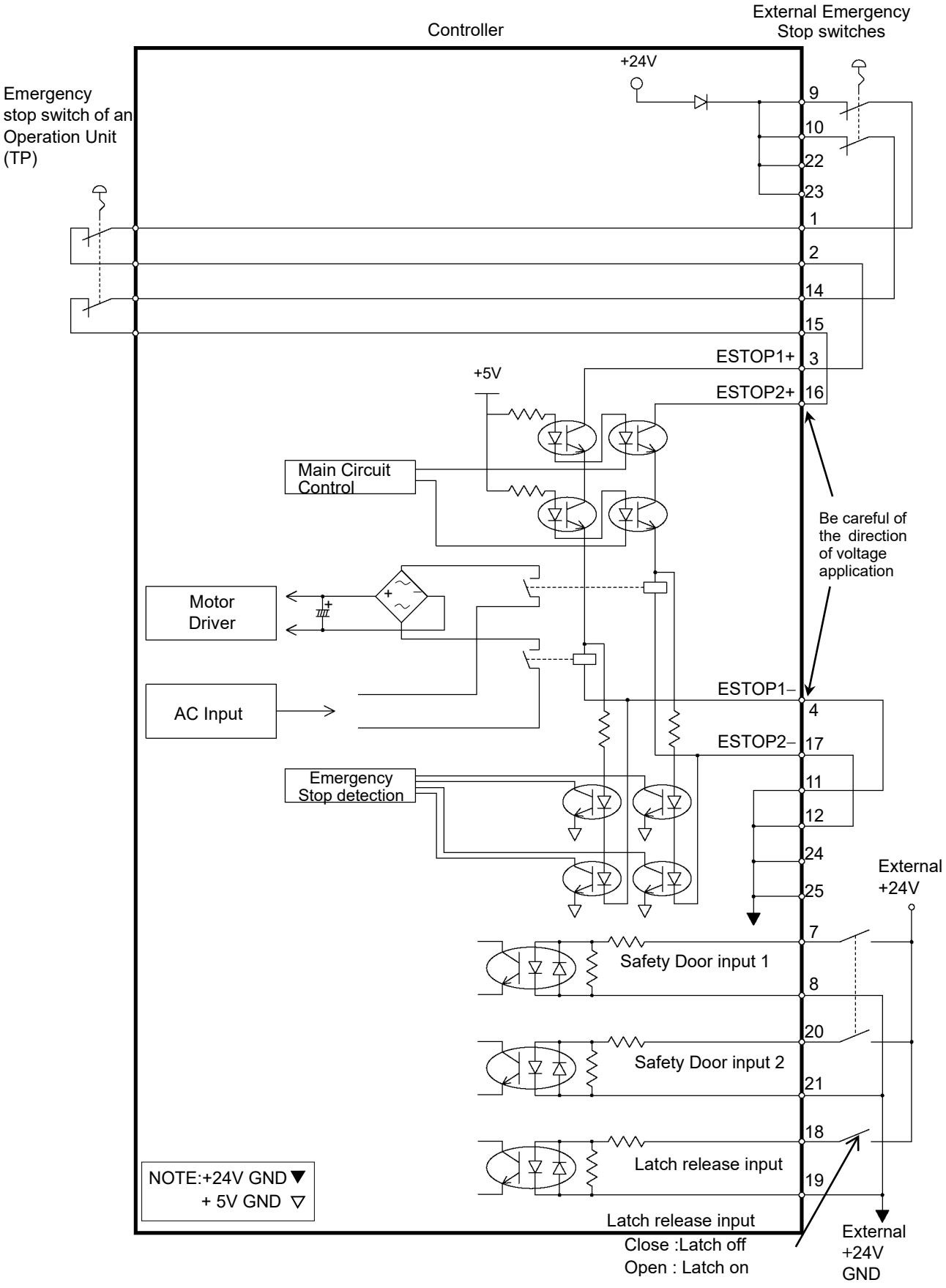
The total electrical resistance of the Emergency Stop switches and their circuit should be 1 Ω or less.

CAUTION

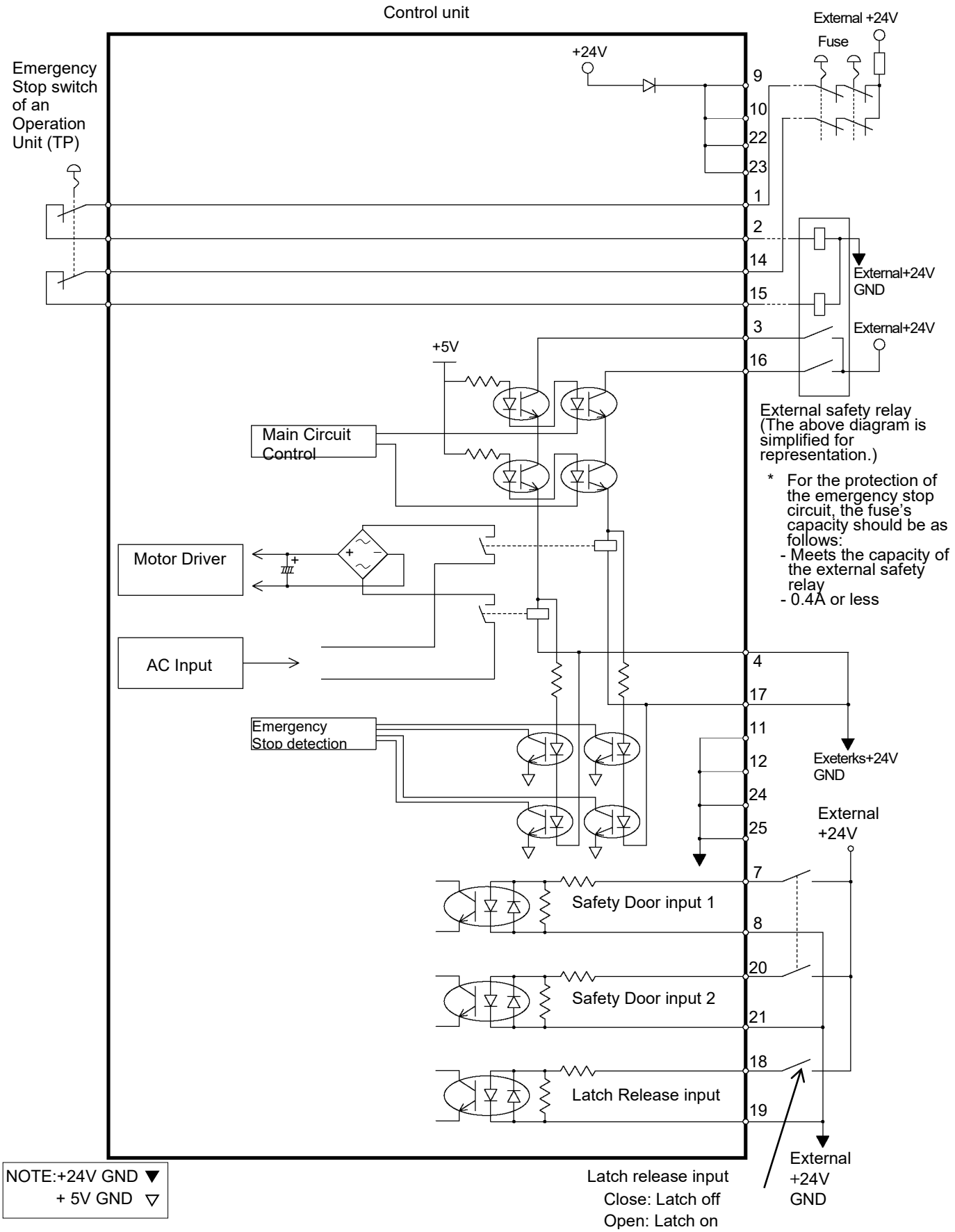
- The 24 V output is for emergency stop. Do not use it for other purposes. Doing so may result in system malfunction.
- Do not apply reverse voltage to the Emergency Stop circuit. Doing so may result in system malfunction.

Circuit Diagrams

Example 1: External emergency stop switch typical application



Example 2: External safety relay typical application



2.7.2 Power supply / AC power cable / Breaker

Power Supply

Ensure that the power meets the following specifications.

Item	Specification
Voltage	200 to 240 VAC Input voltage should be with in ±10 % of the rated voltage.
Phase	Single phase
Frequency	50/60 Hz
Momentary Power Interrupt	10 ms Or less
Power Consumption	Max. 2.5 kVA Actual consumption depends on the model, motion, and load of the Manipulator. See the following for approximate rated consumption LS20-B : 2.4 kVA LS10-B, LS6-B602S-V1 : 1.8 kVA LS6-B : 1.1 kVA LS3-B : 1.0 kVA Refer to the Manipulator manual for rated consumption of the Manipulator motor.
Max. load current	12.5A (Depends to manipulator type.)
Short-circuit current rating	2.5 kA
Maximum Supply Source Impedance	0.61 Ω
Inrush Current	When power is turned ON: approximately 70 A (2 ms) When motor is ON: approximately 50 A (2 ms)
Leakage Current	Max. 10 mA
Ground Resistance	100 Ω or less TN Ground

Install an earth leakage circuit breaker or a circuit breaker in the AC power cable line at 15 A or less rated electric current. Both should be a two-pole disconnect type. If you install an earth leakage circuit breaker, make sure to use an inverter type that does not operate by induction of a 10 kHz or more leakage current. If you install a circuit breaker, please select one that will handle the above mentioned “inrush current”.

The power receptacle should be installed near the equipment and should be easily accessible.

AC Power Cable

WARNING	■ Make sure that operations are done by a qualified personal. ■ Be sure to connect the earth wire (green/yellow) of the AC power cable to the earth terminal of the factory power supply. The equipment must be grounded properly at all times to avoid the risk of electric shock. ■ Always use a power plug or a disconnecting device for power connecting cable. Never connect the Controller directly to the factory power supply. ■ Select the plug or a disconnecting device which conform safety standards for nations.

Following shows specification of power connection side.
Mount to the plug like following below.

Item	Specification
AC power wire (2 cables)	Black, Black
Ground wire	Green / Yellow
Cable length	3 m
Terminal	M4 round solderless terminal



M/C Power Cable Connection

(1) Mount the M/C Power Cable as shown in the picture and form the cables.



(2) Set the M/C Power Cable in the clamp for the M/C Power Cable.

(3) Mount the cover for the M/C Power Connector.



(4) Secure the cover with the screw.

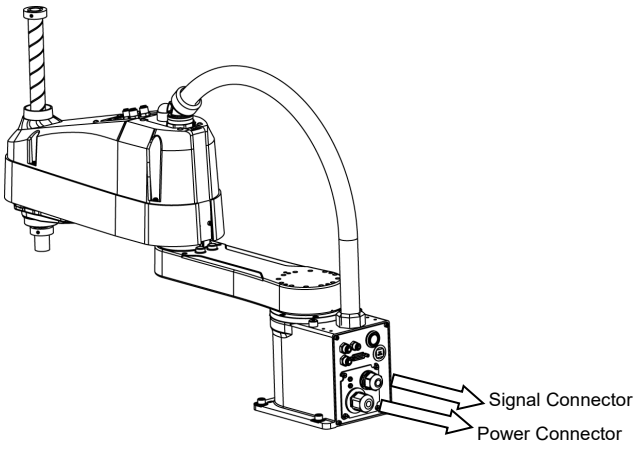


2.7.3 Connecting Manipulator and Controller

Connecting Precautions

Before Connection	:	Before connecting the connector, make sure that the pins are not bent. Connecting with the pins bent may damage the connector and result in malfunction of the robot system.
Connecting procedure	:	Before performing any connecting procedure, turn OFF the Controller and related equipment, and then disconnect the power plug from the power source. Performing any replacement procedure with the power ON is extremely hazardous and may result in electric shock and/or malfunction of the robot system.
Cable	:	Be sure to connect the cables properly. Do not allow unnecessary strain on the cables. (Do not put heavy objects on the cables. Do not bend or pull the cables forcibly.) The unnecessary strain on the cables may result in damage to the cables, disconnection, and/or contact failure. Damaged cables, disconnection, or contact failure is extremely hazardous and may result in electric shock and/or improper function of the robot system.
Connection	:	When connecting the Manipulator and the Controller, make sure that the serial numbers on each equipment match. Improper connection between the Manipulator and Controller may not only cause improper function of the robot system but also safety problems. The serial number of supported Manipulator is labeled on the controller.
Wiring	:	Only authorized or certified personnel should be allowed to perform wiring. Wiring by unauthorized or uncertified personnel may result in bodily injury and/or malfunction of the robot system.
For Clean-model	:	When the Manipulator is a Clean-model, use it with an exhaust system. For details, refer to the Manipulator manual.

Connect the power connector and the signal connector of the M/C cables to the Controller.



2.8 Saving Default Status

The Controller is already setup with the purchased robot(s) at shipment. However, just in case, we recommend saving the default Controller status. A USB memory is necessary to save the Controller status.

For the procedure of Controller status storage, refer to Robot Controller: 6. Memory Port.

3. Power-ON

3.1 Power-ON Precautions

Manipulator check	:	Before operating the Manipulator, make sure that all parts of the Manipulator are in place and have no external defects. Missing or defective parts may cause improper operation of the Manipulator. Improper operation of the Manipulator is extremely hazardous and may result in serious bodily injury and/or severe equipment damage to the robot system.
Shipping bolts and jigs check before turning ON	:	Before turning ON the power after installation, be sure to remove the shipping bolts and jigs from the Manipulator. Turning ON the power while the shipping bolts and jigs are attached may result in equipment damage to the Manipulator.
Power activation	:	Anchor the Manipulator before turning ON the power to or operating the Manipulator. Turning ON the power to or operating the Manipulator that is not anchored is extremely hazardous and may result in serious bodily injury and/or severe equipment damage to the robot system as the Manipulator may fall down
Initial operation	:	Be sure to check at low speed when operating the Manipulator for the first time or executing the operation program for the first time. When it is suddenly operated at high speed, it may not be able to respond to unexpected movements and may cause extremely dangerous, serious injury or serious damage such as collision with the Manipulator.
When supplying the power again	:	When supplying the power to the Controller again, turn OFF the Controller and wait for 5 seconds or more. Then, turn ON the Controller again.

3.2 Power-ON Operations

(1) Check the M/C power cable connection.

(2) Check the M/C signal cable connection.

(3) Check the EMERGENCY connector connection.

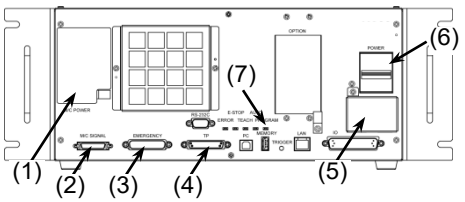
(4) Connect the TP bypass plug to the TP port of the RC90 Controller.

(5) Connect the AC power cable to the power supply socket.

(6) Turn ON the POWER switch of the RC90 Controller.

(7) The PROGRAM LED blinks approximately 30 seconds after Controller starts up normally.

If the ERROR LED is turned ON or blinking, check connection in step (1) to (5) and turn ON the power again.
If the ERROR LED is still ON or blinking, contact the supplier in your region.



4. First Step

This section indicates the procedure to install the development PC EPSON RC+, and execute simple program after connecting the development PC and Controller with a USB. Make sure that the Robot system is installed safely by following the description in *"Safety"* and *"Installation"*. Then, operate the Robot system in the following procedures.

4.1 Installing EPSON RC+ 7.0 Software

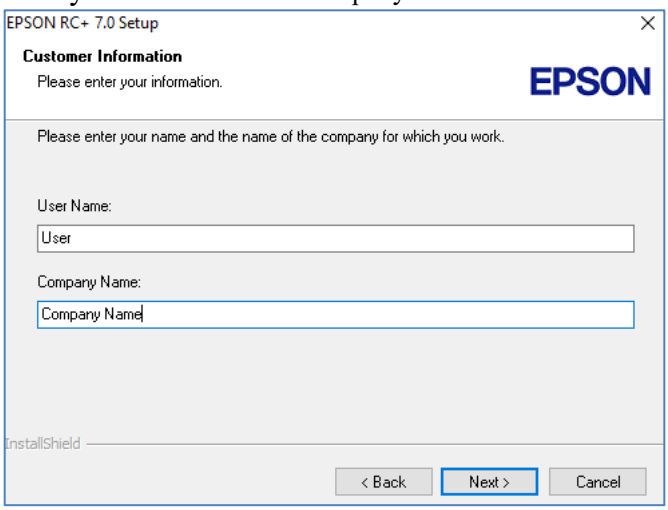
The EPSON RC+ 7.0 software needs to be installed on your development PC.

(1) Insert the EPSON RC+ 7.0 Setup DVD in the DVD drive.

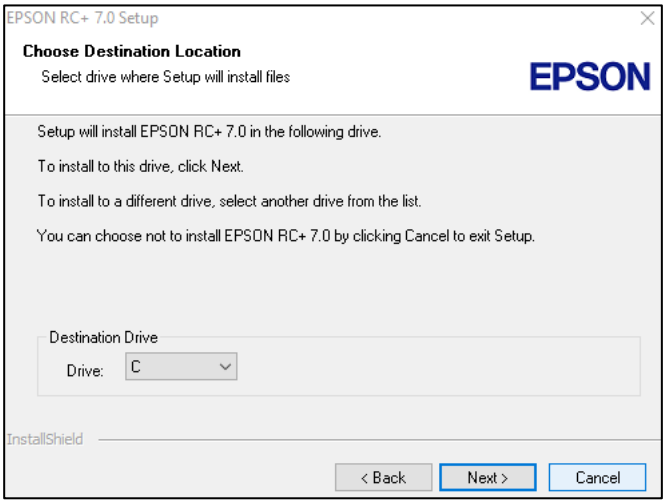
(2) The following dialog box will be displayed. Click <Next>.



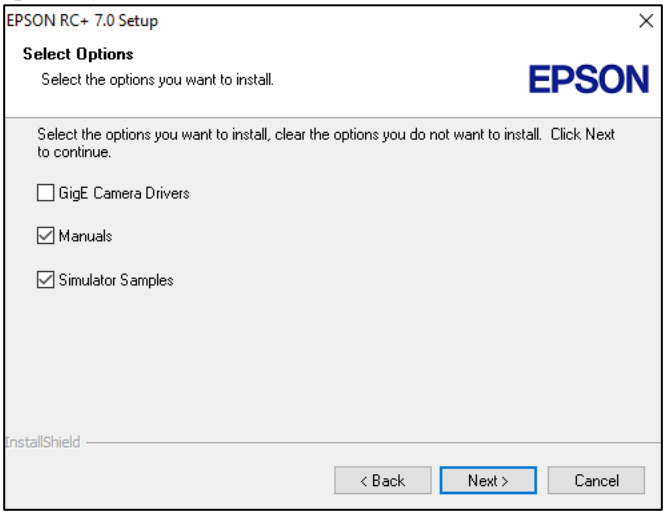
(3) Enter your user name and company name and click <Next>.



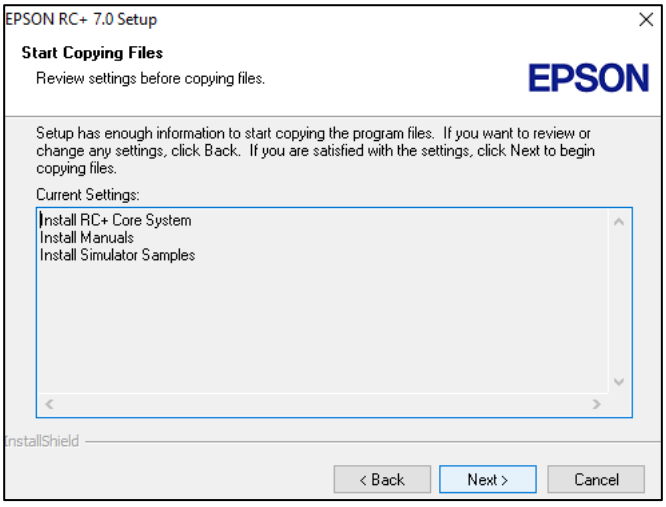
(4) Select the drive where you want to install EPSON RC+ 7.0 and click <Next>.
The installation directory is called EpsonRC70. This cannot be changed.



(5) The dialog box for selecting the options to be installed will be displayed. Check the options you want to install and click <Next>.



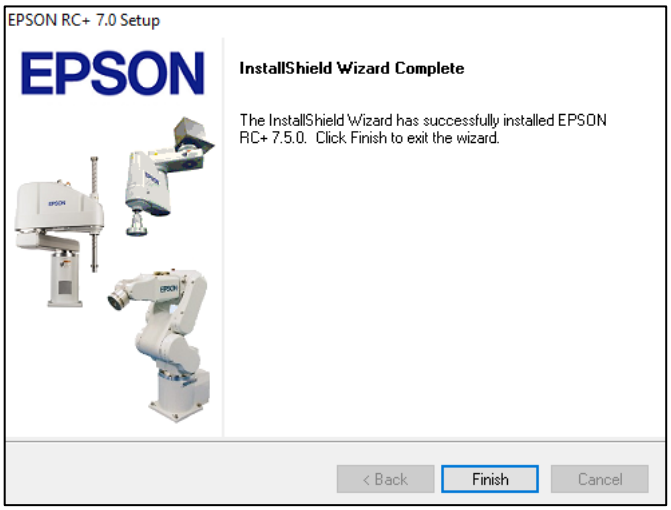
(6) The dialog box to review the settings will be displayed.
If you are satisfied with the settings, click <Next>.



(7) “Windows Installer” will be installed. This may takes several minutes.

NOTE Manuals are provided in PDF. To refer to the manual, use the pre-installed PDF viewer in Windows, or install a PDF viewer such as Adobe Acrobat Reader.

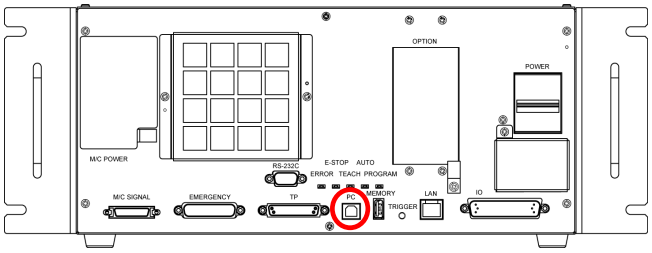
(8) After the installation has completed, restart your computer.
The EPSON RC+ 7.0 software installation is now completed.



NOTE When installing RC; an error “Cannot create parser instance.” may occur.
If the error occurs, uninstall RC+ once and execute Microsoft\VC151719.exe in the DVD and re-install RC+.

4.2 Development PC and Controller Connection

Connect the development PC and the USB port for connection (USB B series connector).



Development PC connection port

NOTE

- For other details of development PC and Controller connection, refer to *EPSON RC+ 7.0 User's Guide: PC to Controller Communications Command*.
- For RC90, be sure to install the EPSON RC+7.0 to the development PC first, then connect the development PC and RC90 with the USB cable.
  If RC90 and the development PC are connected without installing the EPSON RC+7.0 to the development PC, [Add New Hardware Wizard] appears. If this wizard appears, click the <Cancel> button.

4.2.1 About Development PC Connection Port

Development PC connection port supports following USB.

- USB2.0 HighSpeed/FullSpeed (Speed auto selection, or FullSpeed mode)
- USB1.1 FullSpeed

Interface Standard: USB specification Ver.2.0 compliant
(USB Ver.1.1 upward compatible)

Connect the Controller and development PC by a USB cable to develop the robot system or set the Controller configuration with the EPSON RC+ 7.0 software installed in the development PC.

Development PC connection port supports hot plugging. Insertion and removal of the cables while the power of the development PC and the Controller are ON is available. However, the manipulator stops when the USB cable is removed while the Controller and the development PC are connected.

4.2.2 Precaution

When connecting the development PC and the Controller, make sure of the following.

- Connect the development PC and the Controller with 5 m or shorter USB cable. Do not use the USB hub or extension cable.
- Make sure that no other devices except development PC is used for development PC connection port.
- Use PC or USB cable that supports USB2.0 HighSpeed mode to operate in USB2.0 HighSpeed mode.
- Do not pull or bend the cable strongly.
- Do not allow unnecessary strain on the cable.
- When development PC and the Controller is connected, do not insert or remove other USB devices from the development PC. Connection with the Controller may disconnect.

4.2.3 Software Setup and Connection Check

The following is the steps of connecting the development PC and the Controller.

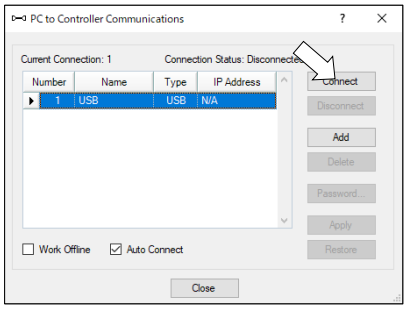
(1) Make sure that EPSON RC+ 7.0 is installed on the Controller connected to the development PC.
(Install the software when it is not installed.)

(2) Connect the development PC and the Controller by the USB cable.

(3) Turn ON the Controller.

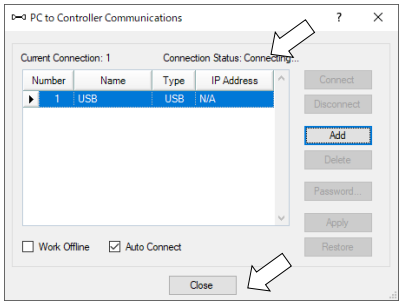
(4) Start the software EPSON RC+ 7.0.

(5) Select the EPSON RC+ 7.0 menu-[Setup]-[PC to Controller Communications] to display the [PC to Controller Communications] dialog box.



(6) Select the “No.1 USB” and click the <Connect> button.

(7) After development PC and the Controller connection is completed, “Connected” is displayed in the [Connection status:]. Make sure that “Connected” is displayed and click the <Close> button to close the [PC to Controller Communications] dialog box.



Connection of the development PC and the Controller is completed. Now robot system can be used from EPSON RC+ 7.0.

4.2.4 Backup the initial condition of the Controller

Backup the Controller data configured before shipment.

Project and system configuration backup procedure:

(1) From the [Project] menu, select [Copy].
(2) Change the [Destination Drive] to an arbitrary drive.
(3) Click <OK>. The project will be copied to the external media.
(4) From the [Tools] menu, select [Controller].
(5) Click on the <Backup Controller> button.
(6) Select the arbitrary drive.
(7) Click <OK>. The system configuration will be backed up on the external media.

4.2.5 Disconnection of Development PC and Controller

Disconnection of the development PC and the Controller is indicated.

(1) Select the EPSON RC+ 7.0 menu-[Setup]-[PC to Controller Communications] to display the [PC to Controller Communications] dialog box.

(2) Click the <Disconnect> button.
Controller and the development PC is disconnected and USB cable can be removed.

NOTE If the USB cable is removed while the Controller and the development PC are connected, the Manipulator stops. Be sure to click the <Disconnect> button in the [PC to Controller Communications] dialog box before removing the USB cable.

4.2.6 Moving the Robot to Initial Position

Robot can be operated by the following methods other than creating and executing the program.

Manual operation
Jog motion by the Teach Pendant
Command execution from the EPSON RC+
Jog motion from the EPSON RC+

This section explains about following methods.

A: Manual operation
B: Command execution from the EPSON RC+
C: Jog motion from the EPSON RC+

A: Manual operation

Move the non-excited robot manually.

Joints with no electromagnetic brake can be moved manually.
Joints with the brakes (Joint #3 only, or Joints #3 and #4) can be moved manually while the break release switch on the manipulator is being pressed with the controller ON.

B: Command execution from the EPSON RC+

Move the robot by exciting the robot motors and executing the command.

Following explains the example of moving all joints to the 0 pulse positions by specifying the pulse for each joint.

(1) Start the EPSON RC+ 7.0.
Double click the <EPSON RC+ 7.0> icon on the desktop.
(2) Open the command window.
EPSON RC+ 7.0 menu-[Tools]-[Command Window]
(3) Execute the following command in [Command Window].

```
>Motor On
>Go Pulse (0,0,0,0)
```

For position and posture of Manipulator at 0 pulse position, refer to *Motion Range* in the Manipulator manuals.

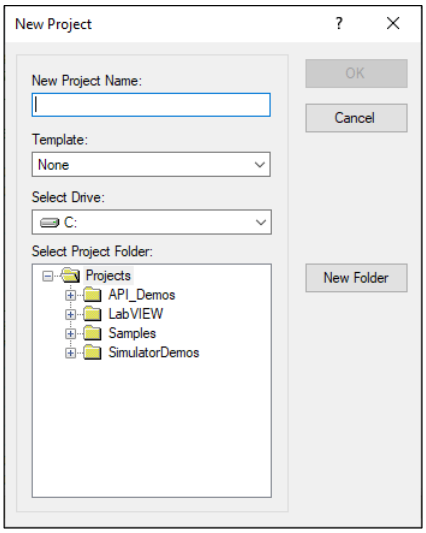
C: Jog motion from the EPSON RC+

Move the robot by exciting the motors and operating from the Jog & Teach window of the EPSON RC+.

(1) Start the EPSON RC+ 7.0.
Double click the <EPSON RC+ 7.0> icon on the desktop.

(2) Create a new project.

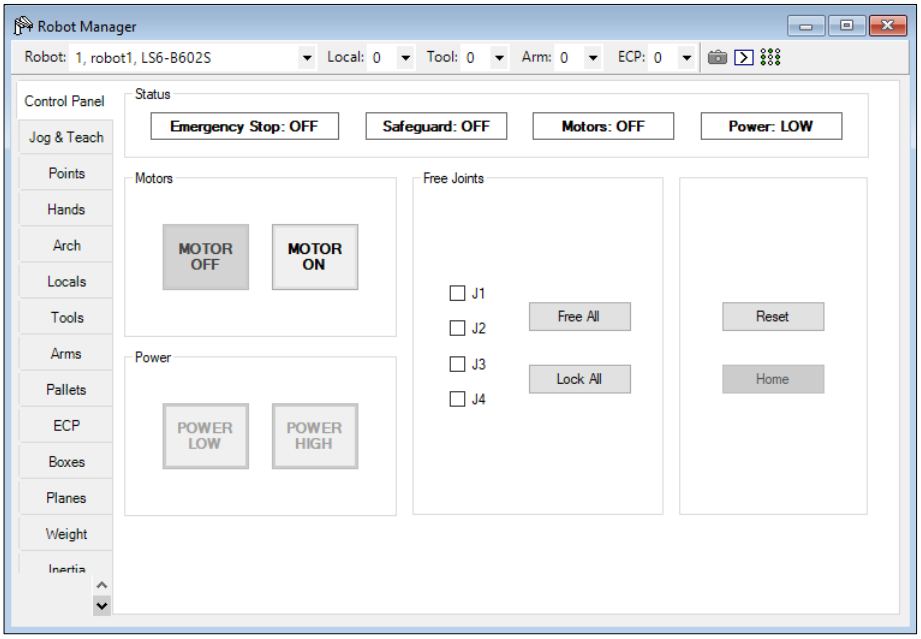
1. Select [New] from the [Project] Menu. The [New Project] dialog box will be displayed.



2. Type a project name in the [New Project Name] box.
(e.g. FirstApp)
3. Click the <OK> button and create the new project.

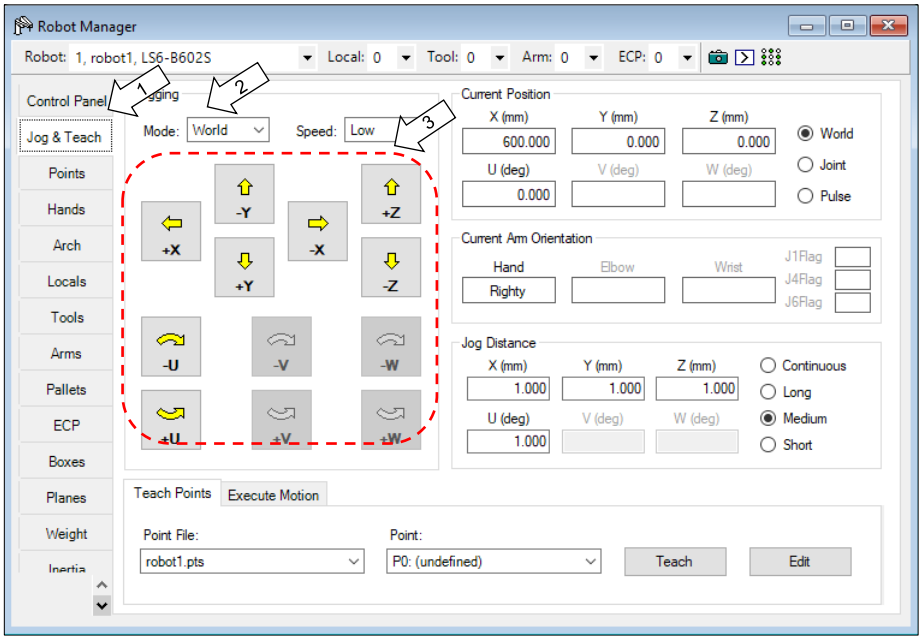
(3) Open the robot manager.
EPSON RC+ 7.0 menu-[Tools]-[Robot Manager].

(4) Turn ON the motors.
Check that the [Control Panel] tab is open. Click the <MOTOR ON> button.



(5) Move the Manipulator with Jog motion.

1. Select [Jog & Teach] tab.



2. Select “Joint” in [Jogging]-<Mode>.
3. Move the Manipulator by joint by clicking J1-J6 jog keys.
   Manipulator can be moved by setting to other modes or setting the jog distance.

4.3 Writing your first program

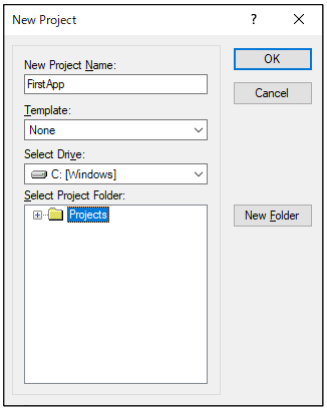
After installing the Controller, Manipulator, and EPSON RC+ 7.0 software on your PC, follow the instructions below to create a simple application program so that you will become more familiar with EPSON RC+ 7.0 development environment.

1. Start EPSON RC+ 7.0

   Double-click the EPSON RC+ 7.0 icon on the desktop.

2. Create a new project

   (1) Select [New] from the [Project] Menu. The [New Project] dialog box will be displayed.



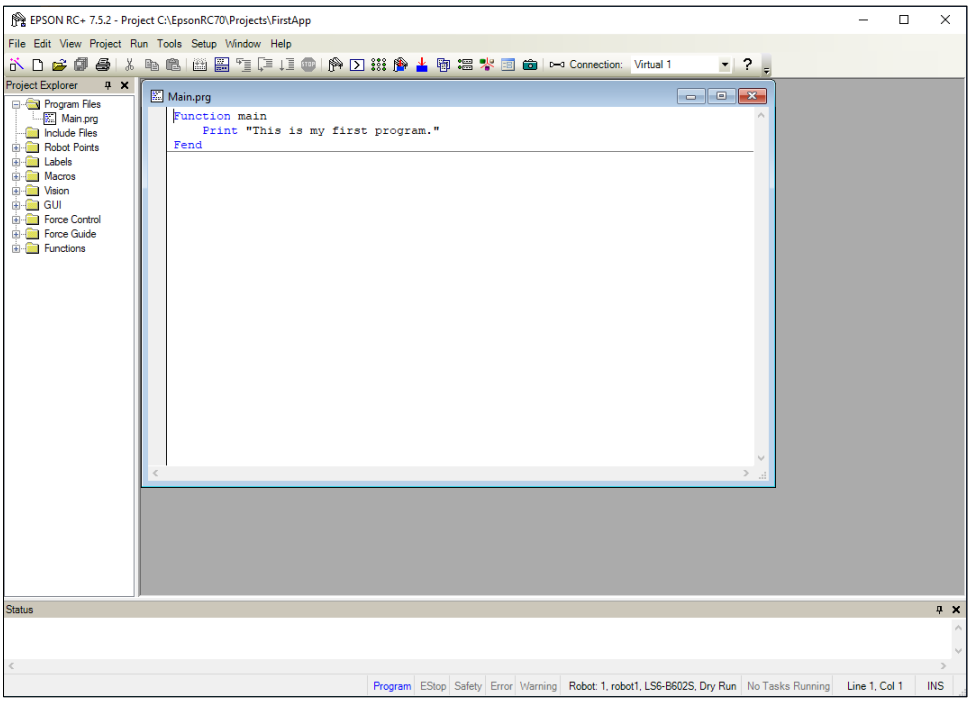
   (2) Type a project name in the [New Project Name] box.
   (e.g. FirstApp)

   (3) Click <OK> to create the new project.
   When the new project is created, a program called Main.prg is created.
   The window titled Main.prg will be displayed with a cursor flashing in the upper left corner.
   Now you are ready to start entering your first program.

3. Edit the program

   Type the following program lines in the Main.prg edit window.

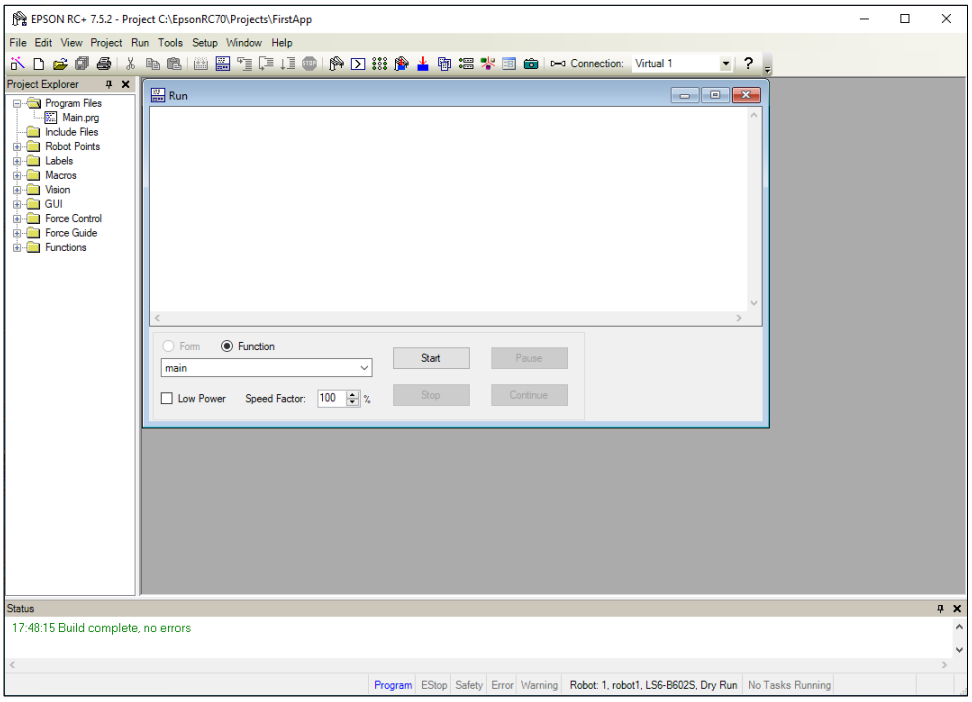
```
Function main
    Print "This is my first program."
Fend
```



4. Run the program

(1) Press F5 to run the program. (F5 is the hot key for the [Run Window] selection of the [Run] Menu). You will see the Status window located at the bottom of the main window showing the build operation status.

(2) During project build, your program is compiled and linked. Then communications are established with the Controller and project files are sent to the Controller. If there are no errors during build, the Run window will appear.



(3) Click the Start button on the Run window to run the program.

(4) The texts like the following will be displayed in the status window.

```
19:32:45 Task main started
19:32:45 All tasks stopped
```

On the Run window, you will see the output of the print statement.

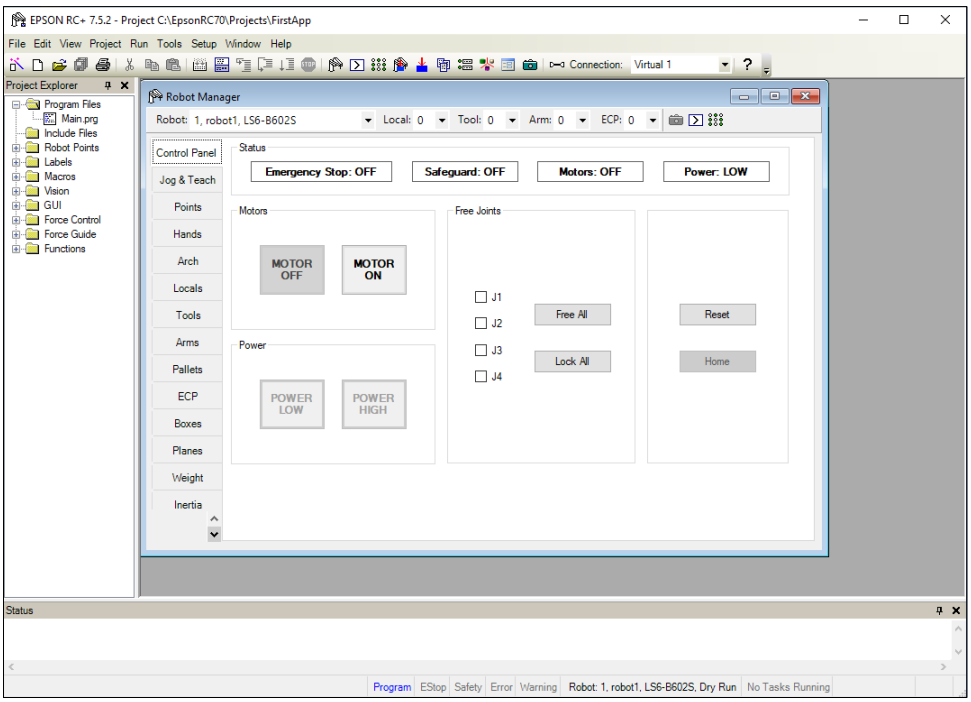
Now let's teach some robot points and create a program to move the robot.

NOTE

Teaching described in this procedure should be done outside the safeguard.

5. Teach robot points

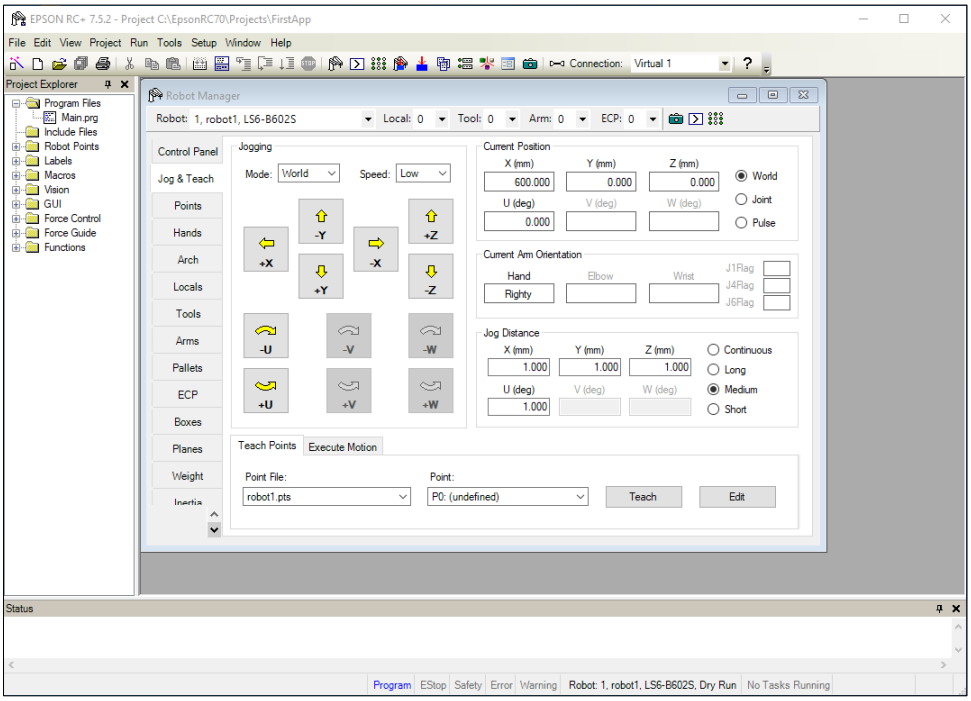
(1) Make sure that it is safe to operate the robot. Click the <Robot Manager> button on the toolbar to display the [Robot Manager] window.



(2) Click the [Control Panel] tab.
Then, click the <Motor On> button to turn on the robot motors. You will be prompted to confirm the operation.

(3) Click <Yes> to continue.

(4) Click the [Jog & Teach] tab.



(5) Click the <Teach> button in the lower right corner to teach point P0. You will be prompted for a point label and description.
(6) Click the <+Y> jog button to jog the robot in +Y direction. Hold the button down to keep jogging. Move the robot until it reaches the middle of the work envelope.
(7) Click the <-Z> button to lower the Z axis of the robot.
(8) Select "P1" in the [Point (P)] dropdown list which is next to the <Teach> button. The current point is set to P1.
(9) Click the <Teach> button. You will see a confirmation message to teach the point.
(10) Click the <Yes> button.
(11) Click the <+X> button to jog the robot in the +X direction.
(12) Select "P2" in the [Point (P)] dropdown list which is next to the <Teach> button. The current point is set to P2.
(13) Click the <Teach> button. You will see a confirmation message to teach the point.
(14) Click the <Yes> button.
(15) Click the <Save Project> toolbar button to save the changes.

6. Modify the program to include robot motion commands

(1) Insert three new Go statements into the Main.prg program as shown below:

```
Function main
   Print "This is my first program."
   Go P1
   Go P2
   Go P0
Fend
```

(2) Press F5 to display the Run window.
(3) Click the <Start> button to execute the program.
The robot will move to the points you taught.

7. Modify the program to change speed of robot motion commands

(1) Insert the Power, Speed, and Accel commands as shown in the program below:

```
Function main

   Print "This is my first program."
   Power High
   Speed 20
   Accel 20, 20
   Go P1
   Go P2
   Go P0
Fend
```

(2) Press F5 to display the Run window.
(3) Click the <Start> button to execute the program.
(4) The robot will move to each of the points you taught at 20% speed, acceleration, and deceleration. The Power High statement executes the program to operate the robot with increased speed and acceleration.

8. Backup the project and system configuration

Even though this is only a sample project, we will backup the project and Controller configuration. Backup can be done easily with EPSON RC+ 7.0. It is important that you keep regular backups of your applications on an external media such as a USB memory key.

Follow the steps below to back up the project and system configuration:

(1) Select the EPSON RC+ 7.0 menu - [Project] - [Copy].

(2) Change the [Destination Drive] to an arbitrary drive.

(3) Click <OK>. The project will be copied to the external media.

(4) Select the EPSON RC+ 7.0 menu - [Tools] - [Controller].

(5) Click on the <Backup Controller> button.

(6) Select the arbitrary drive.

(7) Click <OK>. The system configuration will be backed up on the external media.

5. Second Step

After operating the robot system as instructed in *4. First Step,* setup other functions as necessary.
This section shows the manuals which contains information of necessary settings and setting procedure.

5.1 Connection with External Equipment

5.1.1 Remote Control

EPSON RC+ 7.0 User's Guide *12. Remote Control*
Functions
"12.1 Remote I/O"

I/O
EPSON RC+ 7.0 User's Guide *11. Inputs and Outputs*
Functions
"11. I/O Connector", "13.2 Expansion I/O Board"

Fieldbus I/O (Option)
Robot Controller *Option Fieldbus I/O Board*

5.1.2 Ethernet

EPSON RC+ 7.0 User's Guide
"1.9 Security for Controller Ethernet Connection"
"1.10 Security for Compact Vision CV2-A Ethernet Connection"
"1.11 Security for Feeder Ethernet Connection"
"4.3.3 Ethernet Communication"
Functions
"7. LAN (Ethernet Communication) Port"

5.1.3 RS-232C (Option)

EPSON RC+ 7.0 User's Guide *"RS-232C Communication"*
Functions
"13.4 RS-232C Board"

5.1.4 Analog I/O board (Option)

Functions
"13.5 Analog I/O Board"

5.2 Ethernet Connection of Development PC and Controller

EPSON RC+ 7.0 User's Guide

"1.9 Security for Controller Ethernet Connection"
"1.10 Security for Compact Vision CV2-A Ethernet Connection"
"1.11 Security for Feeder Ethernet Connection"
"4.3.3 Ethernet Communication"

Functions

"7. LAN (Ethernet Communication) Port"

5.3 Connection of Teach Pendant (Option)

Functions

"8. TP Port"

Robot Controller Option TP1 manual

"Functions & Installation: Installation"

Robot Controller Option TP2 manual

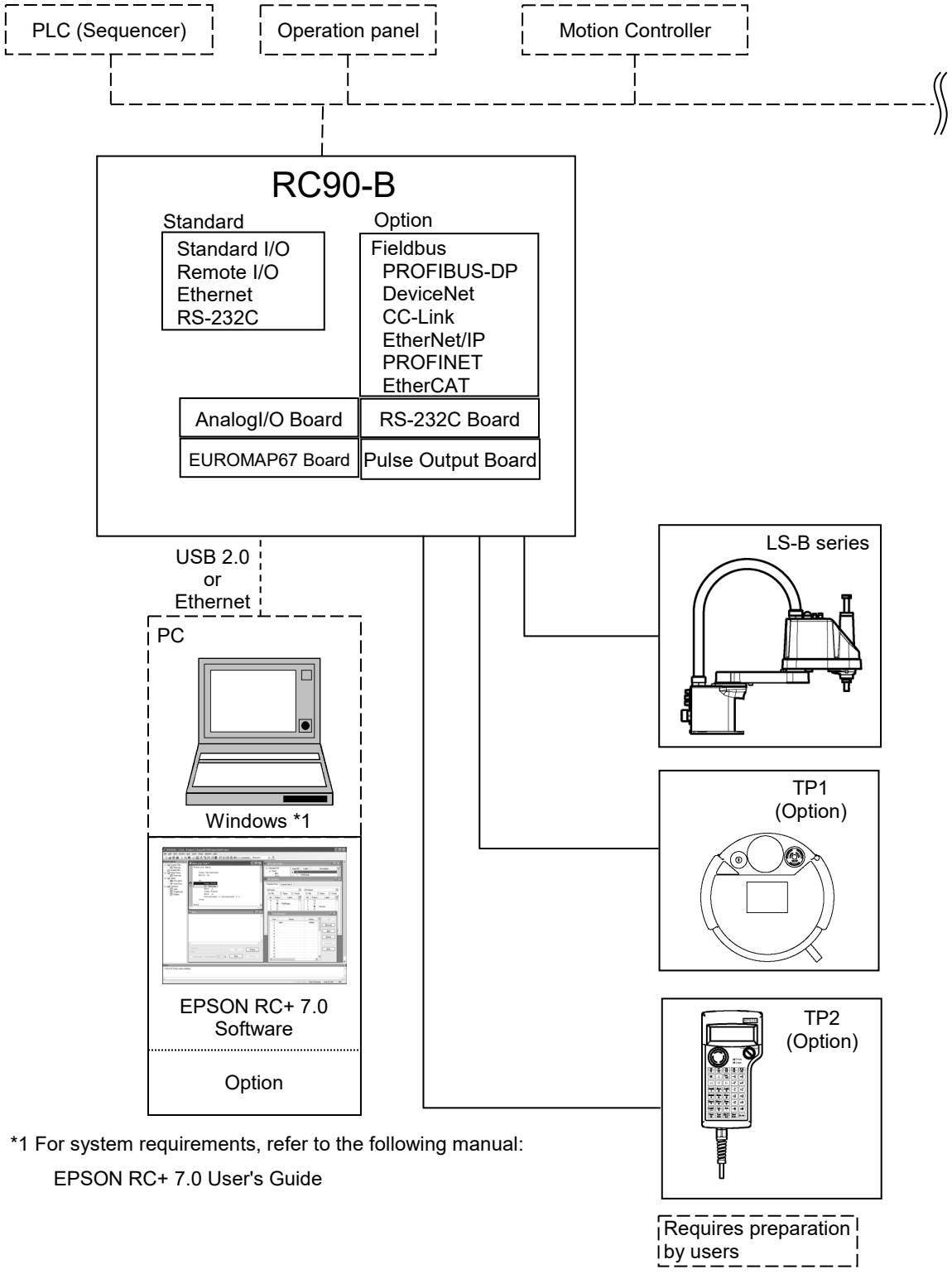
"Functions & Installation: Installation"

Functions required for designing

This section contains information for functions of the Robot Controller.

1. Specifications

1.1 System Example



*1 For system requirements, refer to the following manual:

EPSON RC+ 7.0 User's Guide

1.2 Standard Specifications

Item	Specification		
Model	Robot Controller RC90-B		
CPU	32 bits Micro Processor		
Controllable axes	4 AC servo motors		
Robot manipulator control	Programming language and Robot control software	EPSON RC+ 7.0 (multi-tasking robot language) Ver.7.0.2 or later is recommended.	
	Joint Control	Up to 4 joints simultaneous control Software AC servo control	
	Speed Control	PTP motion : Programmable in the range of 1 to 100% CP motion : Programmable (Actual value to be manually entered.)	
	Acceleration/ deceleration control	PTP motion : Programmable in the range of 1 to 100%; Automatic CP motion : Programmable (Actual value to be manually entered.)	
Positioning control	PTP (Point-To-Point control) CP (Continuous Path control)		
Memory capacity	Maximum Object Size : 4 MB Point data area : 1000 points (per file) Backup variable area : Max. 100 KB (Includes the memory area for the management table.) Approx. 1000 variables (Depends on the size of array variables.)		
External input/output signals (standard)	Standard I/O	Input : 24 Output : 16	Including 8 inputs, 8 outputs with remote function assigned Assignment change allowed
Communication interface (standard)	Ethernet	1 channel	
RS-232C port	1 port		

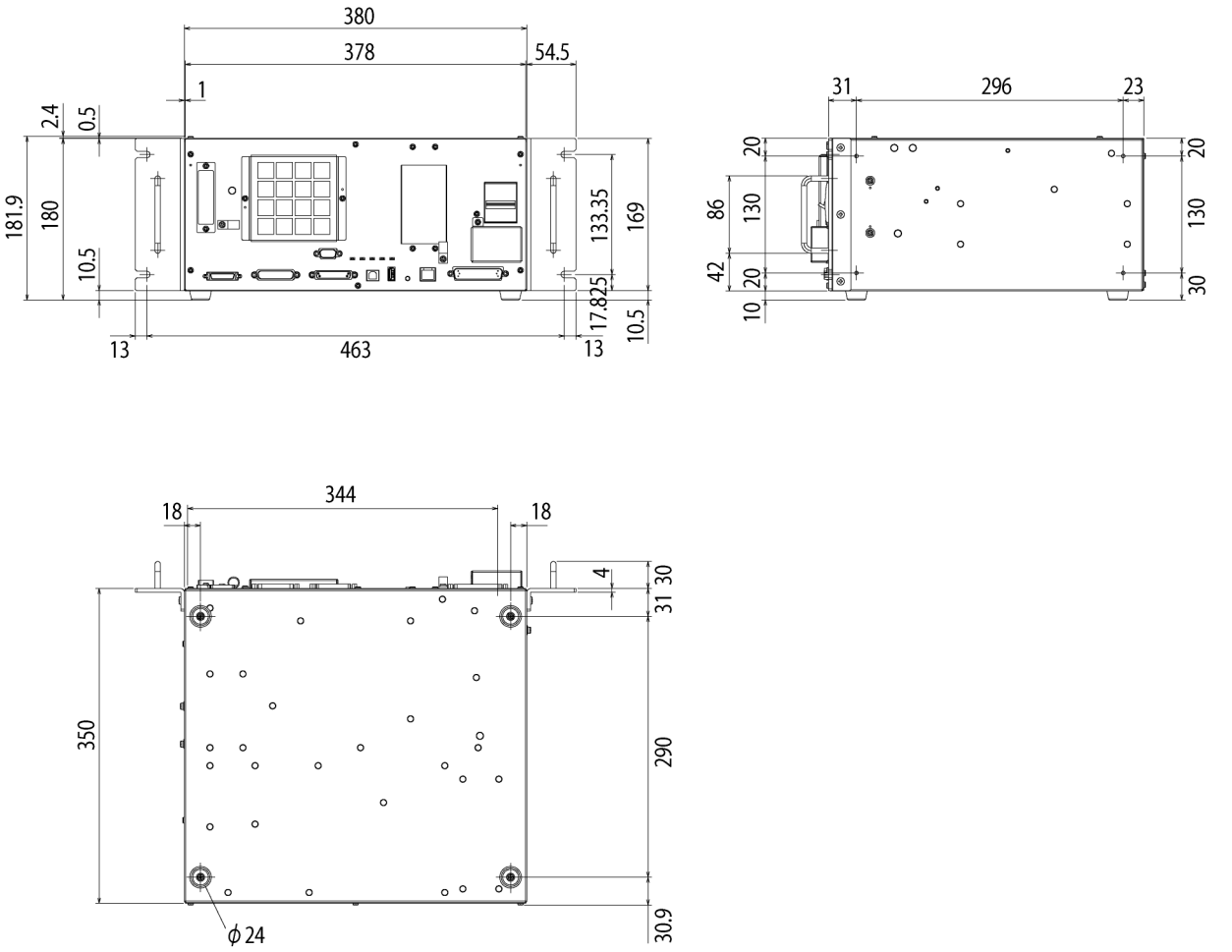
Item	Specification		
Options (Max. 2 slots)	Expansion I/O	Input : 24 per board Output : 16 per board	Addition of 2 boards allowed
	Communication interface	RS-232C : 2ch per board	Addition of 2 boards allowed
		Fieldbus I/O : 1ch per board PROFIBUS-DP, DeviceNet, CC-Link, EtherNet/IP, PROFINET, EtherCAT	Addition of 1 board from the left allowed
	PG	Controllable joints 4 joints/board	Addition of 2 board allowed
	Analog I/O	SKU1 Output: 1ch	Addition of 2 board from the left allowed
		SKU2 Output: 2ch Input: 2ch	
	EUROMAP67	Input : 15 Output : 16	Addition of 1 board allowed
Safety features	- Emergency stop switch - Safety door input - Low power mode - Dynamic brake - Motor overload detection - Irregular motor torque (out-of-control Manipulator) detection - Motor speed error detection - Positioning overflow - servo error - detection - Speed overflow - servo error - detection - CPU irregularity detection - Memory check-sum error detection - Overheat detection at the Motor Driver Module - Relay welding detection - Over-voltage detection - AC power supply voltage reduction detection - Temperature error detection - Fan error detection		
Power Source	200 VAC to 240 VAC Single phase 50/60 Hz		
Maximum Rated Capacity	2.5 kVA (Depending on the Manipulator model)		
Maximum Load Current	12.5A (Depends to the manipulator type.)		
Short-circuit current rating	2.5kA		
Maximum Supply Source Impedance	0.61 Ω		
Insulation Resistance	100 MΩ or more		
Rated Ambient Temperature	Installation: 5 to 40 °C Transportation or Storage: -20 to 60 °C		
Rated Relative Humidity	Installation: 20 to 80 % (with no condensation) Transportation or Storage: 10 to 90% (with no condensation)		
Weight *1	7.5 kg or 10 kg (Depending on the Manipulator model)		
Degree of protection	IP20		

*1 Weight of the unit is indicated on the controller itself.
Make sure to check the weight before units transfer or relocation and prevent throwing out your back at holding the unit.
Also, make sure to keep your hands, fingers, and feet safe from being caught or serious injury.

NOTE
Use this product under the condition of overvoltage category 2, degree of stain category 2.

1.3 Outer Dimensions

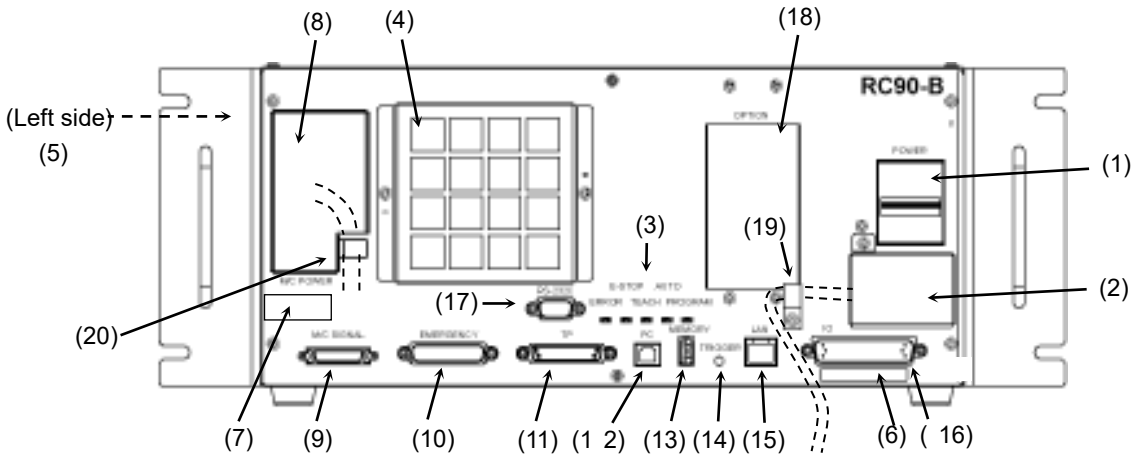
[Unit: mm]



(Figure: RC700)

2. Part Names and Functions

2.1 Part Names



2.2 Functions

(1) POWER switch

Turns ON or OFF the Controller.

(2) AC IN

The terminal block for 200VAC power input.
For details, refer to *3.3.2 AC Power Cable*.

(3) LED

The LED indicates current operation mode (ERROR, E-STOP, TEACH, AUTO, or PROGRAM mode). For details, refer to *2.3 LED*.

(4) Fan Filter

A protective filter is installed in front of the fan to filter out dust.
Check the condition of the filter regularly and clean it when necessary. A dirty filter may result in malfunction of the robot system due to temperature rise of the Controller.

(5) Signature label

The serial number of the Controller and other information are shown.

(6) Controller Number label

The serial number of the Controller is indicated.

(7) Connection Check label

The details of the Manipulator to be connected are recorded on the label as shown in the right. The label indicates the Manipulator model and Manipulator serial number.

MANIPULATOR	
LS3-B401S	00002

(8) M/C POWER connector

A connector for the Manipulator power source.
Connect the dedicated power cable attached to the Manipulator.

(9) M/C SIGNAL connector

This connector is used for signals such as the manipulator's motor position detector, etc.
Connect the Manipulator's dedicated signal cable.

(10) EMERGENCY connector

This connector is used for input/output from/to Emergency Stop and Safety Door switches. For details, refer to the *9. EMERGENCY*.

(11) TP port

Connects Teach Pendant TP1 (Option), TP2 (Option) and TP bypass plug. For details, refer to *8. TP Port*.

NOTE ☞ Do not connect the following to the TP port of RC90-B. Connecting to the followings may result in malfunction of the device since the pin assignments are different.

OPTIONAL DEVICE dummy plug
Operation Pendant OP500
Operator Pendant OP500RC
Jog Pad JP500
Teaching Pendant TP-3**
Operator Panel OP1

(12) Development PC connection port

This port connects the Controller and the Development PC using a USB cable.
Do not connect other devices except the Development PC.
For details, refer to *5. Development PC Connection Port*.

(13) Memory port

This port connects the common USB memory for Controller status storage function.
Do not connect other USB devices except the USB memory.
For details, refer to *6. Memory Port*.

(14) Trigger Switch

This switch is for Controller status storage function using the USB memory.
For details, refer to *6. Memory Port.*

(15) LAN (Ethernet communication) port

This port connects the Controller and the Development PC using an Ethernet cable.
100BASE-TX / 10BASE-T communication are available.
For details, refer to *7. LAN (Ethernet communication) Port*.

(16) I/O connector

This connector is used for input/output device. There are 24 inputs and 16 outputs.
For details, refer to *11. I/O Connector*.

(17) Standard RS-232C port

This port is used for the RS-232C communication with external devices.
For details, refer to *10. Standard RS-232C Port.*

(18) Option slot

Option boards such as expansion I/O board, Fieldbus I/O board, RS-232C board, PG board, Analog I/O board can be installed. Two slots are available.
For details, refer to *13.Option Slots*.

(19) Cable Clamp

This can be used to secure the M/C Power Cable and the AC Power cable.

(20) Battery (Mounted inside the controller)

This is a lithium battery for data backup.

2.3 LED

Five LEDs are located on the front panel of the Controller.

LEDs (ERROR, E-STOP, TEACH, AUTO, PROGRAM) turn ON according to the controller status (error, Emergency Stop, TEACH mode, Auto mode, Program mode).

From turning ON the Controller to completing startup

Three LEDs (TEACH, AUTO, PROGRAM) blink.

After Controller Startup

Controller status	LED display
Execute Controller status storage function to the USB memory	TEACH, AUTO, PROGRAM blink.
Complete Controller status storage to USB memory	TEACH, AUTO, PROGRAM turn ON (for 2 seconds). ERROR turns OFF even if an error occurs.
Failure of Controller status storage to USB memory	ERROR, TEACH, AUTO, PROGRAM turn ON (for 2 seconds).
Error	ERROR turns ON.
Warning	ERROR blinks.
Emergency Stop	E-STOP turns ON.
TEACH mode	TEACH blinks.
Auto mode (AUTO mode)	AUTO blinks.
Program mode (AUTO mode)	PROGRAM blinks.
Recovery mode	ERROR, TEACH, PROGRAM turn ON.
AC power supply drop	TEACH, AUTO turn ON.
Test mode	TEACH turns ON.

2.4 Safety Features

The robot control system supports safety features described below. However, it is recommended to strictly follow the proper usage of the robot system by thoroughly reading the attached manuals before using the system. Failure to read and understand the proper usage of the safety functions is highly dangerous.

Among the following safety features, the Emergency Stop Switch and Safety Door Input are particularly important. Make sure that these and other features function properly before operating the robot system.

For details, refer to the *9. EMERGENCY.*

Emergency Stop Switch

The EMERGENCY connector on the Controller has expansion Emergency Stop input terminals used for connecting the Emergency Stop switches.

Pressing any Emergency Stop switch can shut off the motor power immediately and the robot system will enter the Emergency Stop condition.

Use the emergency switch which has safety function more than PLd.

The safety function of Emergency Stop Input: PLd

Safety Door Input

In order to activate this feature, make sure that the Safety Door Input switch is connected to the EMERGENCY connector at the Controller.

When the safety door is opened, normally the Manipulator immediately stops the current operation, and the status of Manipulator power is operation-prohibited until the safety door is closed and the latched condition is released. In order to execute the Manipulator operation while the safety door is open, you must change the mode selector key switch on the Teach Pendant to the “Teach” mode. Manipulator motor can be turned on only when the enable switch is on. In this case, the Manipulator is operated in low power status.

Use the Safety Door Input which has safety function more than PLd.

The safety function of Safety Door Input: PLd

Low Power Mode

The motor power is reduced in this mode.

Executing a power status change instruction will change to the restricted (low power) status regardless of conditions of the safety door or operation mode. The restricted (low power) status ensures the safety of the operator and reduces the possibility of peripheral equipment destruction or damage caused by careless operation.

Dynamic Brake

The dynamic brake circuit includes relays that short the motor armatures. The dynamic brake circuit is activated when there is an Emergency Stop input or when any of the following errors is detected: encoder cable disconnection, motor overload, irregular motor torque, motor speed error, servo error (positioning or speed overflow), irregular CPU, memory check-sum error and overheat condition inside the Motor Driver Module.

Motor Overload Detection

The dynamic brake circuit is activated when the system detects that the load on the motor has exceeded its capacity.

Irregular Motor Torque (out-of-control manipulator) Detection

The dynamic brake circuit is activated when irregularity with motor torque (motor output) is detected (in which case the Manipulator is out of control).

Motor Speed Error Detection

The dynamic brake circuit is activated when the system detects that the motor is running at incorrect speed.

Positioning Overflow –Servo Error- Detection

The dynamic brake circuit is activated when the system detects that the difference between the Manipulator's actual position and commanded position exceeds the margin of error allowed.

Speed Overflow –Servo Error- Detection

The dynamic brake circuit is activated when the Manipulator's actual speed is detected to mark an overflow (the actual speed is outside the nominal range) error.

CPU Irregularity Detection

Irregularity of CPU that controls the motor is detected by the watchdog timer. The system CPU and the motor controlling CPU inside the Controller are also designed to constantly check each other for any discrepancies. If a discrepancy is detected, the dynamic brake circuit is activated.

Memory Check-sum Error Detection

The dynamic brake circuit is activated when a memory check-sum error is detected.

Overheat Detection at the Motor Driver Module

The dynamic brake circuit is activated when the temperature of the power device inside the Motor Driver module is above the nominal limit.

Relay Deposition Detection

The dynamic brake circuit is activated when relay deposition, junction error, or open fault is detected.

Over-Voltage Detection

The dynamic brake circuit is activated when the voltage of the Controller is above the normal limit.

AC Power Supply Voltage Drop Detection

The dynamic brake circuit is activated when the drop of the power supply voltage is detected.

Temperature Anomaly Detection

The temperature anomaly is detected.

Fan Malfunction Detection

Malfunction of the fan rotation speed is detected.

3. Installation

3.1 Unpacking

TP/OP Bypass Plug	1 unit
EMERGENCY Port Connector	1 set
I/O Connector	1 set
Rack-Mount Plate	1 set

3.2 Environmental Requirements

WARNING

■ The Manipulator and the Controller must be used within the environmental conditions described in their manuals. This product has been designed and manufactured strictly for use in a normal indoor environment. Using the product in the environment that exceeds the conditions may not only shorten the life cycle of the product but also cause serious safety problems.

3.2.1 Environment

In order to optimize the robot system's performance for safety, the Controller must be placed in an environment that satisfies the following conditions:

NOTE

- The Controller is not designed for clean-room specification. If it must be installed in a clean room, be sure to install it in a proper enclosure with adequate ventilation and cooling.
- Install Controller in a location that allows easy connection / disconnection of cables.
- Install the controller outside of the safety guard.

Item	Condition
Ambient temperature	5 to 40 °C
Ambient relative humidity	20% to 80% (with no condensation)
Fast transient burst noise	2 kV or less (Power supply wire) 1 kV or less (Signal wire)
Electrostatic noise	4 kV or less
Base table	Use a base table that is at least 100 mm off the floor. Placing the Controller directly on the floor could allow dust penetration leading to malfunction.
Install surface	Tilt 0.5° or less (When installing upright, you put your hand on it, it may fall over.)
Elevation	1000 m or less

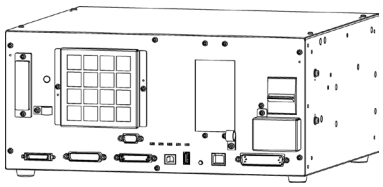
If the Controller must be used in an environment that does not fulfill the conditions mentioned above, take adequate countermeasures. For example, the Controller may be enclosed in a cabinet with adequate ventilation and cooling.

- Install indoors only.
- Place in a well-ventilated area.
- Keep away from direct sunlight and radiation heat.
- Keep away from dust, oily mist, oil, salinity, metal powder or other contaminants.
- Keep away from water.
- Keep away from shocks or vibrations.
- Keep away from sources of electronic noise
- Keep away from strong electric or magnetic fields.

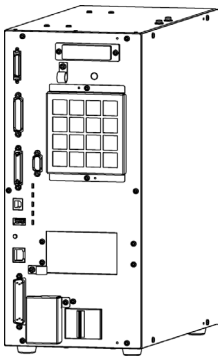
3.2.2 Installation

Install the controller on a flat surface such as wall, floor, and controller box in the direction shown from (A) to (C).

(A) Flat mounting

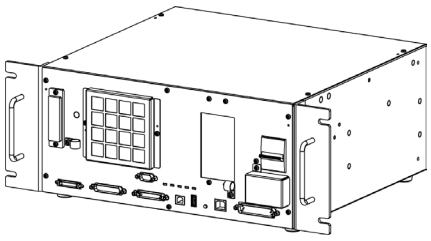


(B) Upright mounting



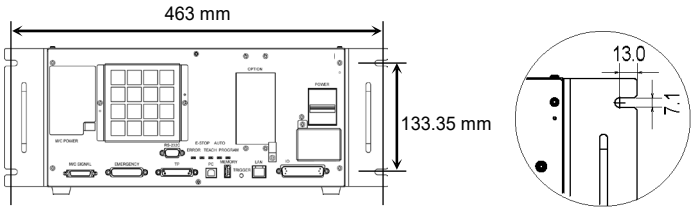
* The rubber foot needs to be replaced. Attach the rubber foot with the flat side facing the controller.
The size of the screw that secure the rubber foot is M4×8.
When replacing the rubber foot, be careful not to lose the screw. Also, do not use a screw of size other than M4×8.

(C) Rack mounting



* A plate for rack mounting is required.

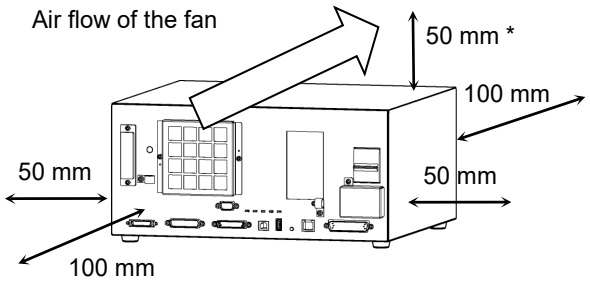
NOTE For installing the Controller to the Controller box or the base table, process screw holes as follows.



- Ensure the air flow around the supply and exhaust ports, and install the Controller while leaving space from other equipment or walls as below.

  * Keep the space about 200 mm or more on the top to maintenance.

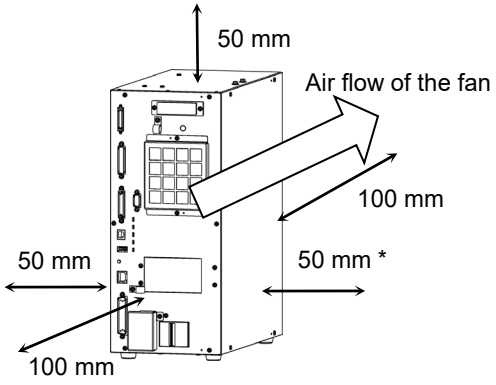
(A) Flat mounting, (C)Rack mounting



Excluding the installation side such as base table.

(Figure Flat mounting)

(B) Upright mounting



Exclude the installation side such as base table.

- Hot air with higher temperature than the ambient temperature (about 10 °C) comes out from the Controller. Make sure that heat sensitive devices are not placed near the outlet.
- Arrange the cables in front of the Controller so that you can pull the Controller forward.

3.3 Power Supply

3.3.1 Specifications

Ensure that the available power meets following specifications.

Item	Specification
Voltage	200 to 240 VAC Input voltage should be with in ±10 % of the rated voltage.
Phase	Single phase
Frequency	50/60 Hz
Momentary Power Interrupt	10 ms or less
Rated Capacity	Max. 2.5 kVA Actual power consumption depends on the model, motion, and load of the Manipulator. For approximate power consumption of each model, refer to the following values LS3-B: 1.0 kVA LS6-B : 1.1 kVA LS20-B : 2.4 kVA LS10-B, LS6-B602S-V1: 1.8 kVA Refer to the Manipulator manual for rated capacity of the Manipulator motor.
Max. load current	Max.12.5A Actual power consumption depends on the model, motion, and load of the Manipulator. For approximate power consumption of each model, refer to the following values. LS3-B: 5.0A LS6-B: 5.5A LS10-B, LS6-B602-V1: 9A LS20-B: 12A
Short-circuit current rating	2.5 kA
Maximum Supply Source Impedance	0.61 Ω
Inrush Current	When power is turned ON : approximately 70 A (2 ms.) When motor is ON : approximately 50 A (2 ms.)
Leakage Current	Max. 10 mA
Ground Resistance	100 Ω or less

Install an earth leakage circuit breaker or a circuit breaker in the AC power cable line at 15 A or less rated electric current. Both should be a two-pole disconnect type. If you install an earth leakage circuit breaker, make sure to use an inverter type that does not operate by induction of a 10 kHz or more leakage current. If you install a circuit breaker, please select one that will handle the above mentioned “inrush current”.

The power receptacle shall be installed near the equipment and shall be easily accessible.

3.3.2 AC Power Cable

WARNING	■ Make sure that operations are done by a qualified personal. ■ Be sure to connect the earth wire (green/yellow) of the AC power cable to the earth terminal of the factory power supply. The equipment must be grounded properly at all times to avoid the risk of electric shock. ■ Always use a power plug or a disconnecting device for power connecting cable. Never connect the Controller directly to the factory power supply. ■ Select the plug or a disconnecting device which conform safety standards for nations.

Following shows specification of power connection side.
Mount to the plug like following below.

Item	Specification
AC power wire (2 cables)	Black, Black
Ground wire	Green / Yellow
Cable length	3 m
Terminal	M4 round solderless terminal



3.3.3 M/C Power Cable

(1) Mount the M/C Power Cable as shown in the picture and form the cables.



(2) Set the M/C Power Cable in the clamp for the M/C Power Cable.

(3) Mount the cover for the M/C Power Connector.



(4) Secure the cover with the screw.

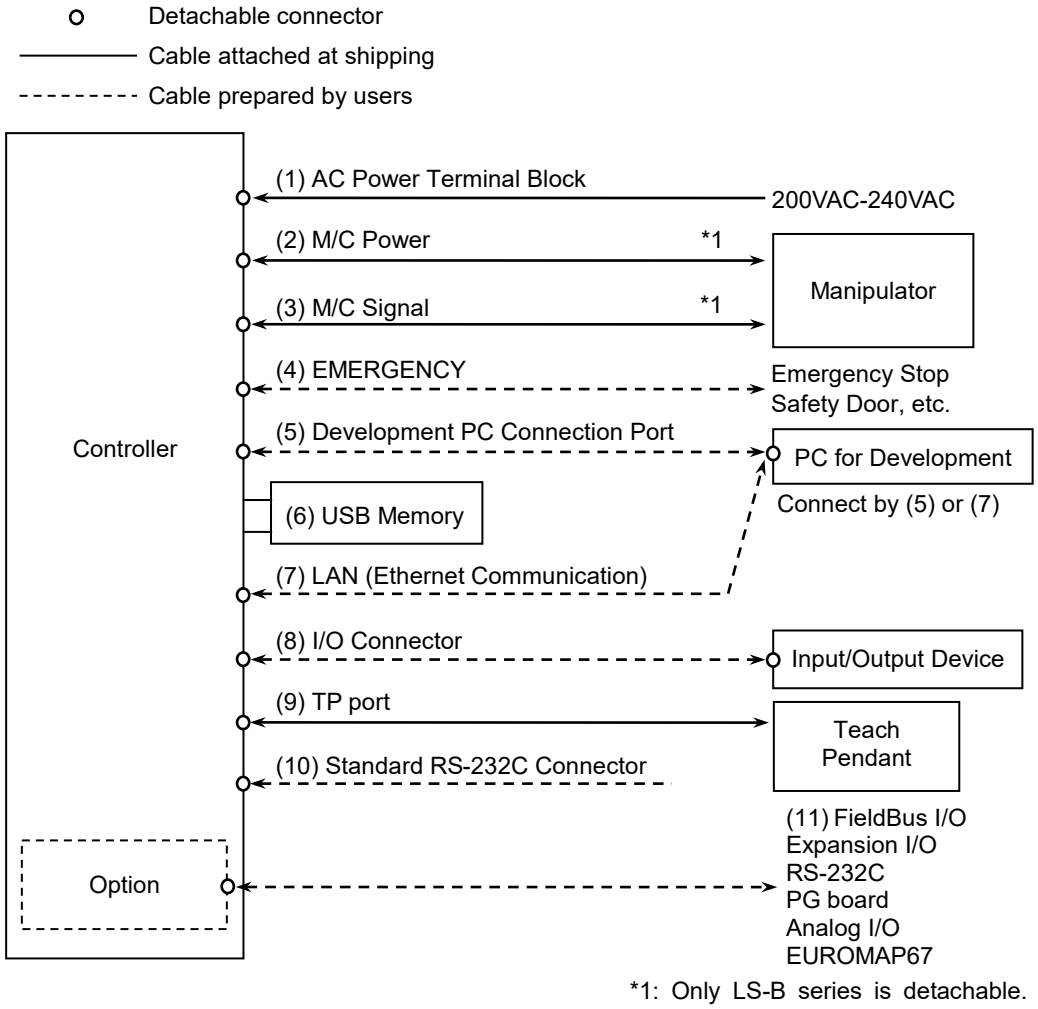


3.4 Cable Connection

 WARNING	■ Make sure that the power to the Controller is turned OFF and the power plug is disconnected before connecting or disconnecting any cables. Connecting or disconnecting any cables with the power ON is extremely hazardous and may result in electric shock and malfunction of the Controller. ■ Be sure to connect the cables properly. Protect it with the strong cable cover and do not allow unnecessary strain on the cables. (Do not put heavy objects on the cables. Do not bend or pull the cables forcibly.) The unnecessary strain on the cables may result in damage to the cables, disconnection, and/or contact failure. Damaged cables, disconnection, or contact failure is extremely hazardous and may result in electric shock and/or improper function of the system.

 CAUTION	■ The serial number of the Manipulator that should be connected is indicated on the Connection Check Label on the Controller. Connect the Controller and the Manipulator correctly. Improper connection between the Controller and the Manipulator may cause not only improper function of the robot system but also safety problems. ■ Before connecting the connector, make sure that the pins are not bent. Connecting with the pins bent may damage the connector and result in malfunction of the robot system.

3.4.1 Typical Cable Connection



(1) AC Power terminal block

Terminal block for 200VAC power input to the Controller.

(2) M/C Power cable

The cable with 20-pin connector on the Controller side.
Connect the Manipulator and the M/C POWER connector on the Controller. Insert the connectors until you hear a “click”.

(3) M/C Signal cable

The cable with 36-pin connector on the Controller side.
Connect the Manipulator and the M/C SIGNAL connector on the Controller.

(4) EMERGENCY

The EMERGENCY connector has inputs to connect the Emergency Stop switch and the Safety Door switch. For safety reasons, connect proper switches for these input devices.
For details, refer to the *9. EMERGENCY*.

(5) PC for development

Connect the PC for development.
For details, refer to the *5. Development PC Port*.

(6) USB memory

Connect the USB memory.
For details, refer to the *6. Memory Port*.

(7) LAN (EtherNet Communication)

Connect the EtherNet cable.
For details, refer to the *7. LAN (Ethernet Communication) Port*.

(8) I/O connector

This connector is used for input/output devices of the user.
When there are input/output devices, use this connector.
There are I/O cable (option) and terminal block (option) for the I/O connector.
For details, refer to the *11. I/O Connector*.

(9) TP port

Connect the option Teach Pendant.
For details, refer to the *8.TP Port*.

(10) Standard RS-232C port

This port is used for the RS-232C communication with external devices.
For details, refer to *10. Standard RS-232C Port*.

(11) Fieldbus I/O

If necessary, take EMC countermeasures for the Fieldbus I/O cable.
For details, refer to *3.5 Noise Countermeasures*.

3.4.2 Connecting Manipulator to Controller

Connect the Manipulator to the Controller by using the Power cable and the Signal cable.

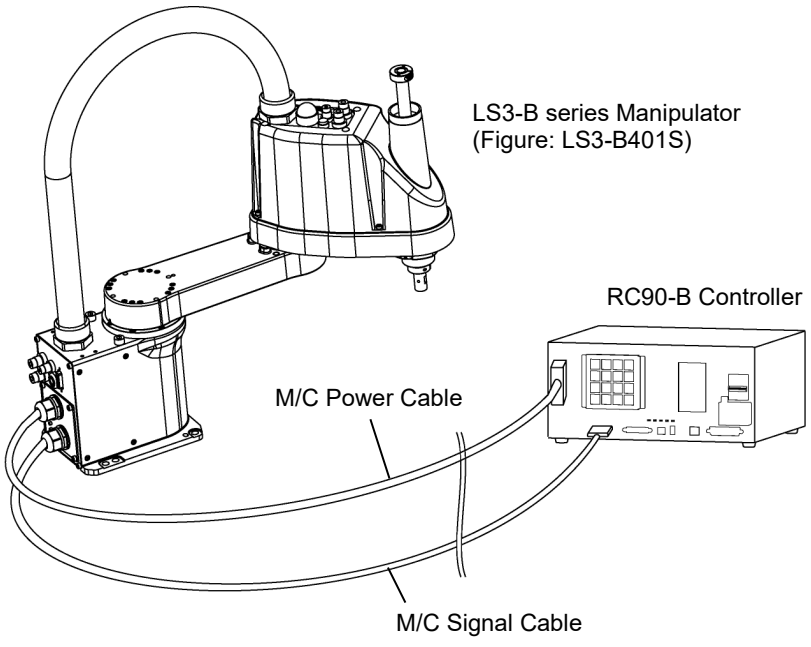
WARNING	■ Make sure that the power to the Controller is turned OFF before connecting or disconnecting any cables. Connecting or disconnecting any cables with the power ON is extremely hazardous and may result in electric shock and malfunction of the Controller. ■ Be sure to connect the cables properly. Protect it with the strong cable cover and do not allow unnecessary strain on the cables. (Do not put heavy objects on the cables. Do not bend or pull the cables forcibly.) The unnecessary strain on the cables may result in damage to the cables, disconnection, and/or contact failure. Damaged cables, disconnection, or contact failure is extremely hazardous and may result in electric shock and/or improper function of the system.

CAUTION	■ The serial number of the Manipulator that should be connected is indicated on the Connection Check Label on the Controller. Connect the Controller and the Manipulator correctly. Improper connection between the Controller and the Manipulator may cause not only improper function of the robot system but also safety problems. ■ When connecting the Manipulator to the Controller, make sure that the serial numbers on each equipment match. Improper connection between the Manipulator and Controller may not only cause improper function of the robot system but also serious safety problems. The connection method varies with the Controller used.

The configuration data for the Manipulator and Manipulator model are stored in the Controller. Therefore the Controller should be connected to the Manipulator whose serial number is specified in the Connection Check label attached on the front of the Controller.

NOTE

The Manipulator's serial number is indicated on the signature label of the Manipulator.

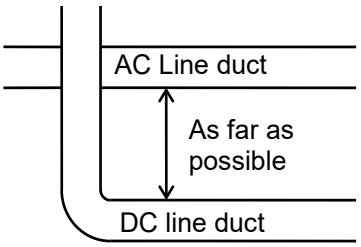


3.5 Noise Countermeasures

To minimize electrical noise conditions, the following items must be observed in the system's cable wiring:

To minimize electrical noise condition, be sure of followings for wiring.

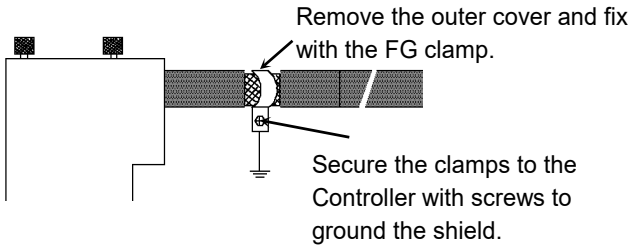
- The earth wire of the power supply should be grounded. (Ground resistance: 100 Ω or less) It is important to ground the frame of Controller not only for prevention from electric shock, but also for reducing the influence of electric noise around the Controller. Therefore, be sure to connect the earth wire (yellow/green) of the Controller's power cable to the ground terminal of the factory power supply. For details about the plug and AC power cable, refer to the *3.3 Power Supply*.
- Do not tap power from a power line that connects to any equipment which may cause noise.
- When you tap power for the Controller and the single-phase AC motor from the same power line, change the phase of one or the other. Ensure that they will not be the same phase.
- Use a twisted pair motor power line.
- Do not run AC power lines and DC power lines in the same wiring duct, and separate them as far as possible. For example, separate the AC motor power line and the Controller power line as far as possible from the sensor or valve I/O lines; and do not bundle both sets of wiring with the same cable tie. If more than one duct/cable must cross each other, they should cross perpendicularly. The preferable example is shown in the right figure.



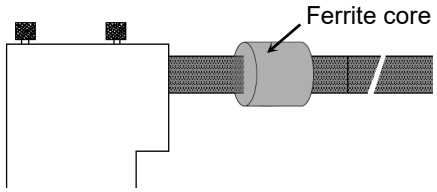
- Wire as short as possible to the I/O connector and EMERGENCY connector. Use a shielded cable and clamp the shield to the attached connector interior. Make sure to keep away from the peripheral noise source as far as possible.
- Make sure that the induction elements used to connect to the Controller's I/O (such as relays and solenoid valves) are noise suppression parts. If an induction element without protection against noise is used, make sure to connect a noise suppression part such as a diode located at the induction element in parallel with it. In selecting noise suppression parts, make sure that they can handle the voltage and current incurred by the induction load.
- To start and change revolutions of the conveyer's (or the like's) AC motor (ex: an induction motor or three-phase induction motor) regularly or abruptly, make sure to install a spark suppressor between the wires. The spark suppressor is more effective when placed closer to the motor.
- As they are easily influenced by noise, keep cable such as USB, Ethernet, RS-232C, or fieldbus away from peripheral noise sources.

- Perform following EMC countermeasure for the Fieldbus I/O cable as necessary.

  - Put together the cable shield side.



  - Mount Ferrite core to the cable.



4. Operation Mode (TEACH/AUTO/TEST)

4.1 Overview

The Robot system has three operation modes.
When operate teaching, place people to check operation outside the safeguard so that the manipulator can be stopped immediately in the event of an anomaly.

TEACH mode	This mode enables point data teaching and checking close to the Robot using the Teach Pendant. In this mode the Robot operates in Low power status.
AUTO mode	This mode enables automatic operation (program execution) of the Robot system for the manufacturing operation, and also programming, debug, adjustment, and maintenance of the Robot system. This mode cannot operate the Robots or run programs with the Safety Door open.
TEST mode	This mode enables program verification while the Enable Switch is held down and the safeguard is open. This is a low speed program verification function (T1: manual deceleration mode) which is defined in Safety Standards. This mode can operate the specified Function with multi-task / single-task, multi-manipulator / single-manipulator at low speed.

NOTE

To use a low speed program verification function (T1: manual deceleration mode) which is defined in Safety Standards, you will need a Teach Pendant that supported for the function.

4.2 Switch Operation Mode

Change the operation mode using the mode selector key switch on the Teach Pendant.

When changing operation mode, operate it outside the safeguard to avoid danger.

TEACH mode	Turn the mode selector key switch to “Teach” for TEACH mode. Switching to TEACH mode pauses the program if it was running. The operating Robot stops by Quick Pause.
AUTO mode	Turn the mode selector key switch to “Auto” and turn on the latch release input signal for AUTO mode.
TEST mode	Turn the mode selector key switch to “Teach” for “TEACH” mode. Push <F1> key-[Test Mode] in [Jog & Teach] dialog of TEACH mode. The mode will be changed to TEST

NOTE

The controller software latches the condition of the operation mode “TEACH”.
To change the mode “TEACH” to “AUTO”, latch release input is required.
The message “Workers must leave the safeguard area.” displayed in the Teach Pendant. Ensure safety when working.
For the detail of latch release, refer to *9.1 Safety Door Switch and Latch Release Switch.*

NOTE

When switching the mode by mode selector key switch, the motor turns off.

NOTE

When switching the mode by mode selector key switch, turn off the Enable Switch. If you switch the mode with the mode selector key switch and turn on the motor while the enable switch is ON, an error will occur. Turn the Enable Switch off and then on again, and then turn on the motor.

4.3 Program Mode (AUTO)

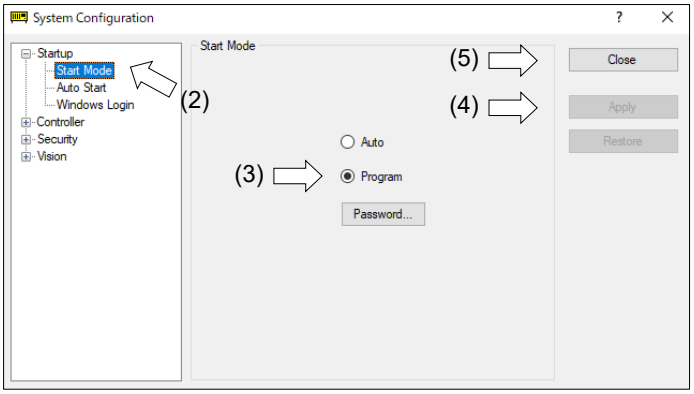
4.3.1 What is Program Mode (AUTO)?

Program mode is for programming, debug, adjustment, and maintenance of the Robot system.
Follow the procedures below to switch to the Program mode.

4.3.2 Setup from EPSON RC+ 7.0

Switch the mode to Program mode from the EPSON RC+ 7.0.

(1) Select EPSON RC+ 7.0 menu-[Setup]-[System Configuration] to display the [System Configuration] dialog.



(2) Select [Startup].

(3) Select [Start mode]-<Program> button.

(4) Click the <Apply> button.

(5) Click the <Close> button.

4.4 Auto Mode (AUTO)

4.4.1 What is Auto mode (AUTO)?

Auto mode (AUTO) is for automatic operation of the Robot system.
Procedures for switching to the Auto mode (AUTO) are the followings.

A : Set the start mode of the EPSON RC+ 7.0 to “Auto” and start the EPSON RC+ 7.0.
(Refer to *4.4.2 Setup from EPSON RC+ 7.0.*)

B : Offline the EPSON RC+ 7.0.

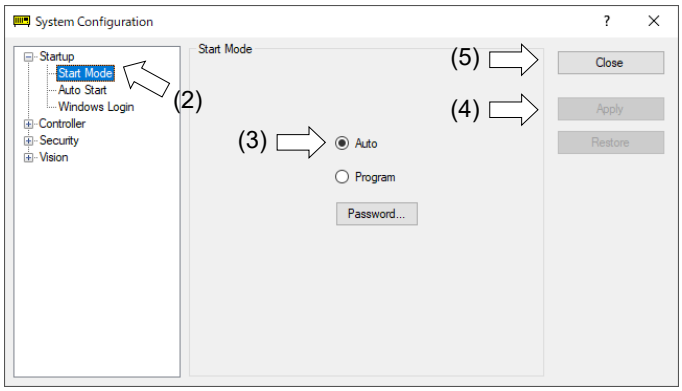
NOTE

Execute and stop the program from the control device specified by the EPSON RC+ 7.0.
(Refer to *4.4.3 Setup Control Device*.)

4.4.2 Setup from EPSON RC+ 7.0

Switch the mode to Auto mode (AUTO) from the EPSON RC+ 7.0.

(1) Select EPSON RC+ 7.0 menu-[Setup]-[System Configuration] to display the [System Configuration] dialog.



(2) Select [Startup].

(3) Select [Start Mode]-<Auto> button.

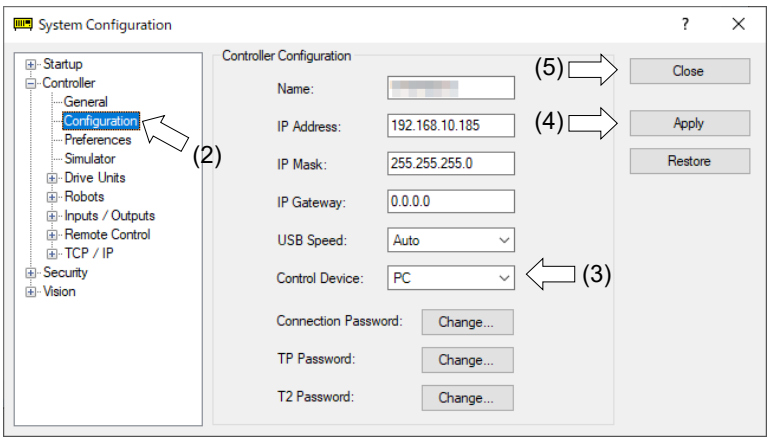
(4) Click the <Apply> button.

(5) Click the <Close> button.

4.4.3 Setup from Control Device

Set the control device from EPSON RC+ 7.0.

(1) Select EPSON RC+ 7.0 menu-[Setup]-[System Configuration] to display the [System Configuration] dialog.



(2) Select [Controller]-[Configuration].

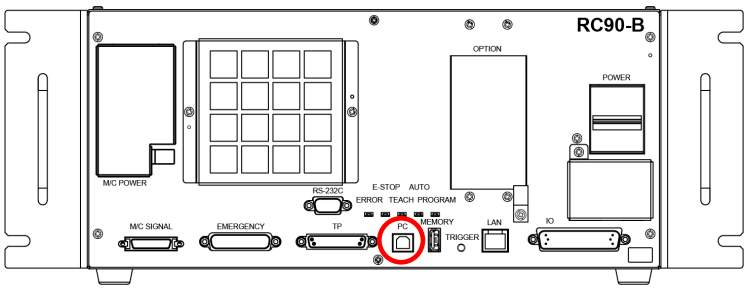
(3) Select [Control Device] to select the control device from the following two types.
- PC
- Remote (I/O)

(4) Click the <Apply> button.

(5) Click the <Close> button.

5. Development PC Connection Port

Development PC connection USB port (USB B series connector)



Development PC connection port

NOTE

For other details of development PC and Controller connection, refer to *EPSON RC+ 7.0 User's Guide 5.13.1 [PC to Controller Communications] Command (Setup Menu)*.

For RC90-B, be sure to install the EPSON RC+ 7.0 to the development PC first, then connect the development PC and RC90-B with the USB cable.
If RC90-B and the development PC are connected without installing the EPSON RC+ 7.0 to the development PC, [Add New Hardware Wizard] appears. If this wizard appears, click the <Cancel> button.

5.1 About Development PC Connection Port

The development PC connection port supports the following USB types.

- USB2.0 HighSpeed/FullSpeed (Speed auto selection, or FullSpeed mode)
- USB1.1 FullSpeed

Interface Standard : USB specification Ver.2.0 compliant
(USB Ver.1.1 upward compatible)

Connect the Controller and development PC by a USB cable to develop the robot system or set the Controller configuration with the EPSON RC+ 7.0 software installed in the development PC.

Development PC connection port supports hot plug feature. Cables insert and remove from the development PC and the Controller is available when the power is ON.
However, stop occurs when USB cable is removed from the Controller or the development PC during connection.

5.2 Precaution

When connecting the development PC and the Controller, make sure of the following:

- Connect the development PC and the Controller with a 5 m or less USB cable.
  Do not use the USB hub or extension cable.
- Make sure that no other devices except the development PC are used for development PC connection port.
- Use a PC and USB cable that supports USB2.0 HighSpeed mode to operate in USB2.0 HighSpeed mode.
- Do not pull or bend the cable strongly.
- Do not allow unnecessary strain on the cable.
- When the development PC and the Controller are connected, do not insert or remove other USB devices from the development PC. Connection with the Controller may be lost.

5.3 Software Setup and Connection Check

Connection of the development PC and the Controller is indicated.

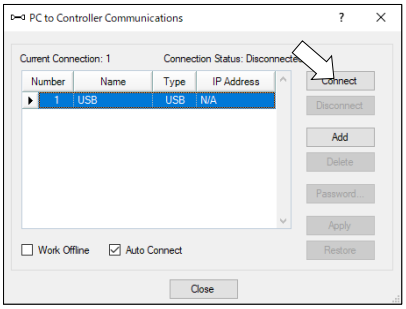
(1) Make sure that software EPSON RC+ 7.0 (Ver.7.0.2 or later is recommended) is installed to the Controller connected to the development PC.
(Install the software when it is not installed. Refer to *Safety Manual* or *EPSON RC+ 7.0 User's Guide.*

(2) Connect the development PC and the Controller using a USB cable.

(3) Turn ON the Controller.

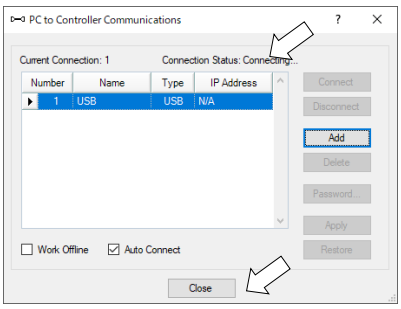
(4) Start EPSON RC+ 7.0.

(5) Select the EPSON RC+ 7.0 menu-[Setup]-[PC to Controller Communications] to display the [PC to Controller Communications] dialog.



(6) Select "No.1 USB" and click the <Connect> button.

(7) After the development PC and the Controller connection has completed, “Connected” is displayed at [Connection status]. Make sure that “Connected” is displayed and click the <Close> button to close the [PC to Controller Communications] dialog.



The connection between the development PC and the Controller is completed. Now the robot system can be used from EPSON RC+ 7.0.

5.4 Disconnection of Development PC and Controller

Disconnection of the development PC and the Controller communication.

(1) Select the EPSON RC+ 7.0 menu-[Setup]-[PC to Controller Communications] to display the [PC to Controller Communications] dialog.

(2) Click the <Disconnect> button.
Communication between the Controller and the development PC is disconnected and the USB cable can be removed.

NOTE

If the USB cable is removed when the Controller and the development PC are connected, the Robot will stop. Be sure to click the <Disconnect> button in the [PC to Controller Communications] dialog before USB cable is removed.

6. Memory Port

Connect a commercial USB memory to the Controller memory port to use the Controller status storage function to the USB memory.

6.1 What is Controller Status Storage Function?

This function saves various kinds of Controller data with one push to the USB memory. Data saved in USB memory is loaded to EPSON RC+ 7.0 to get the status of the Controller and the program simply and accurately.
The saved data can also be used for restoring the Controller.

6.2 Before Using Controller Status Storage Function

6.2.1 Precautions

 CAUTION	Controller status storage function is available at any time and in any Controller status after starting the Controller. However, operations form the console including stop and pause are not available while executing this function. Also, this function influences the robot cycle time and the communication with EPSON RC+ 7.0. Other than only when it is necessary, do not execute this function when operating the robot.

- Make sure that the USB port is used only for USB memory even though the port on the Controller is a universal USB port.
- Insert the USB memory directly into the Controller memory port. Connection with cables or hubs between the Controller and the USB memory is not assured.
- Make sure that the USB memory is inserted or removed slowly.
- Do not edit the saved files with an editor. Operation of the robot system after data restoration to the Controller is not assured.

6.2.2 Adoptable USB Memory

Use USB memory that meets following conditions.

- USB2.0 supported
- Without security function
  USB memory with password input function cannot be used.
- No installation of a driver or software is necessary for Windows 8, or Windows 10, or Windows 11.
  (For supported operating systems for the EPSON RC+ 7.0, refer to *1.1 System Example*.)

6.3 Controller Status Storage Function

6.3.1 Controller Status Storage with Trigger Button

CAUTION	Controller status storage function is available at any time and in any Controller status after starting the Controller. However, operations form the console including stop and pause are not available while executing this function. Also, this function influences the robot cycle time and the communication with EPSON RC+ 7.0. Other than only when it is necessary, do not execute this function when operating the robot.

Use this procedure to save the status of the Controller to USB memory.

(1) Insert the USB memory into the memory port.

(2) Wait approximately 10 seconds for USB memory recognition.

(3) Press the trigger button on the Controller.
When the data transfer starts, the LED of TEACH, AUTO, and PROGRAM starts blinking. Wait until the LED status changes.
(The data transfer time varies according to the data size such as of the projects.)

(4) When the controller status storage is completed successfully, the LED of TEACH, AUTO, and PROGRAM are turned ON for two seconds. Note that the LED of ERROR turns OFF even in the error status.
If it ends in failure, the LED of ERROR, TEACH, AUTO, and PROGRAM are turned ON for two seconds.

(5) Remove the USB memory from the Controller.

NOTE

- USB memory with LED is recommended to check the status changes in procedure (2).
- When storage is executed during Motor ON status, it may fail to store the status. Use another USB memory or execute the storage during Motor OFF status.

6.3.2 Load Data with EPSON RC+ 7.0

The following shows the procedure to read the data stored in the USB memory by EPSON RC+ 7.0 and display the Controller status.

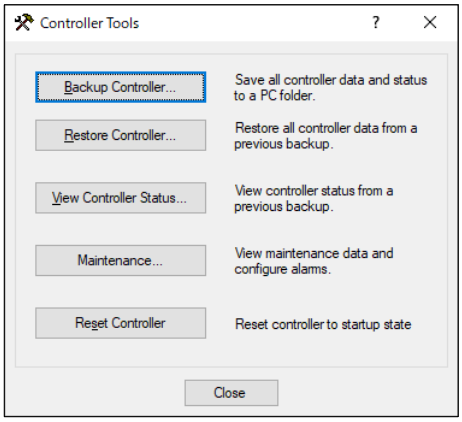
(1) Insert the USB memory into the PC with EPSON RC+ 7.0.

(2) Make sure that the following folder is indicated in the USB memory.

BU_controller model_serial number_data status was saved
→ Example RC90-B : B_RC90-BU_12345_2013-10-29_092951

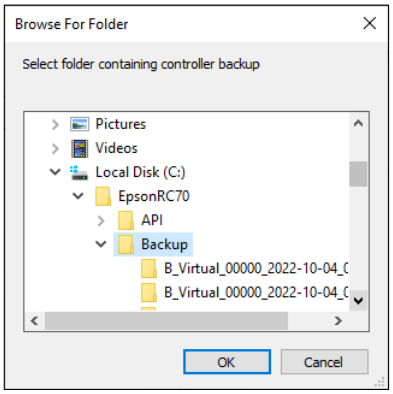
(3) Copy the folder confirmed in procedure (2) to the “\EpsonRC70\Backup” folder.

(4) Select the EPSON RC+ 7.0 menu-[Tools]-[Controller] to display the [Controller Tools] dialog.

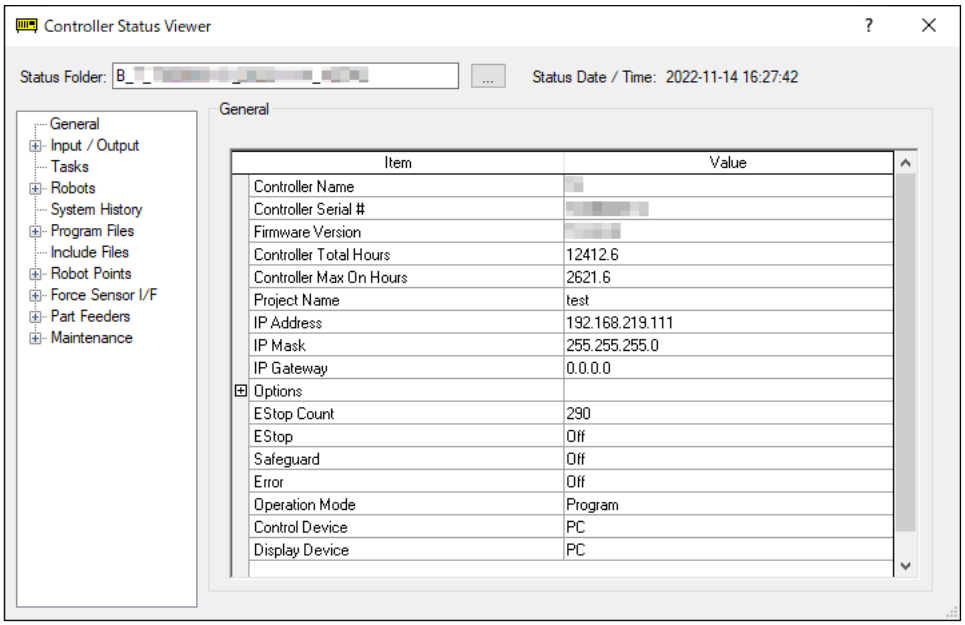


(5) Click the <View Controller Status…> button.

(6) [Browse For Folder] dialog appears. Select the folder copied in procedure (3) and click the <OK> button.



(7) [Controller Status Viewer] dialog appears to confirm the Controller status.
For details, refer to *View Controller Status* in *EPSON RC+ 7.0 User's Guide 5.12.8 [Controller] Command (Tools Menu)*.



6.3.3 Transfer with E-mail

Follow this procedure to transfer the data by e-mail that was saved to the USB memory.

(1) Insert the USB memory to a PC that supports sending of e-mail.

(2) Make sure that the USB memory has following folders.

BU_controller model_serial number_data status was saved
→ Example RC90-B : B_RC90-BU_12345_2013-10-29_092951

(3) Send all the folders by e-mail.

NOTE
- Delete files that do not relate to the project before transfer.
- This function is used to send the data to the system director and EPSON from the end users for problem analysis.

6.4 Details of Data

The following data files are created by the Controller status storage function.

File Name	Outline	
Backup.txt	Information file for restore	File with information for Controller restore.
CurrentMnp01.PRM	Robot parameter	Saves information such as ToolSet.
CurrentStatus.txt	Save status	Saves program and I/O status.
ErrorHistory.csv	Error history	
InitFileSrc.txt	Initial setting	Saves various settings of the Controller.
MCSys01.MCD	Robot setting	Saves information of connected robot.
SrcmcStat.txt	Hardware information	Saves installation information of hardware.
ProjectName.obj	OBJ file	Result of project build. Prg file is not included.
GlobalPreserves.dat *1	Global Preserve variables	Saves values of Global Preserve variables.
WorkQueues.dat	WorkQue information	Saves information of Queues information of the WorkQue.
MCSRAM.bin MCSYSTEMIO.bin MCTABLE.bin MDATA.bin SERVOSRAM.bin VXDWORK.bin	Inner information of Robot operation	
All files related to project except ProjectName.obj *2	Project	When [Include project files when status exported] check box is checked in EPSON RC+ 7.0 menu-[Setup]-[System Configuration]-[Controller]-[Preference], the project file is stored. Includes program files.

1 When the Controller firmware version is Ver.1.0..*, GlobalPreserves.dat is not stored.

*2 Storage of “All files related to project except ProjectName.obj” can be specified by a setting.

7. LAN (Ethernet Communication) Port

NOTE

- Refer to *EPSON RC+ 7.0 User's Guide 5.13.1 [PC to Controller Communications] Command (Setup Menu)* for other details for the development PC and Controller connection.
- For Ethernet (TCP/IP) communication with robot application software, refer to *EPSON RC+ 7.0 Online Help* or *User's Guide 14. TCP/IP Communications*.

7.1 About the LAN (Ethernet Communication) Port

Ethernet communication port supports 100BASE-TX / 10 BASE-T.

This port is used for two different purposes.

Connection with development PC

LAN (Ethernet communication) port is used for connection of the Controller and the development PC.
Equivalent operation is available to connect between the Controller and the development PC with the development PC connection port.
(Refer to *5. Development PC Connection Port*)

Connection with other Controller or PC

The LAN (Ethernet communication) port can be used as an Ethernet (TCP/IP) communication port to communicate between multiple controllers from robot application software.

7.2 IP Address

From the following firmware version password authentication is required when connecting Controllers and PCs to a global accessible network.

F/W : Ver.7.4.8.x

NOTE

For the details of password settings, refer to *User's Guide 1.9.1 Setting Password for PC Ethernet Controller Connection.*

Our robot system is provided on the assumption that customers use it in the closed local area networks. We considered the setting of the global IP address is access to the Internet and changed the specification to perform password authentication in order to secure the connections.
Password authentication is not performed in case of USB connections.

Be sure to use the following private IP addresses.

Private Address List

10.0.0.1	to	10.255.255.254
172.16.0.1	to	172.31.255.254
192.168.0.1	to	192.168.255.254

The following is the configuration of the controller at delivery.

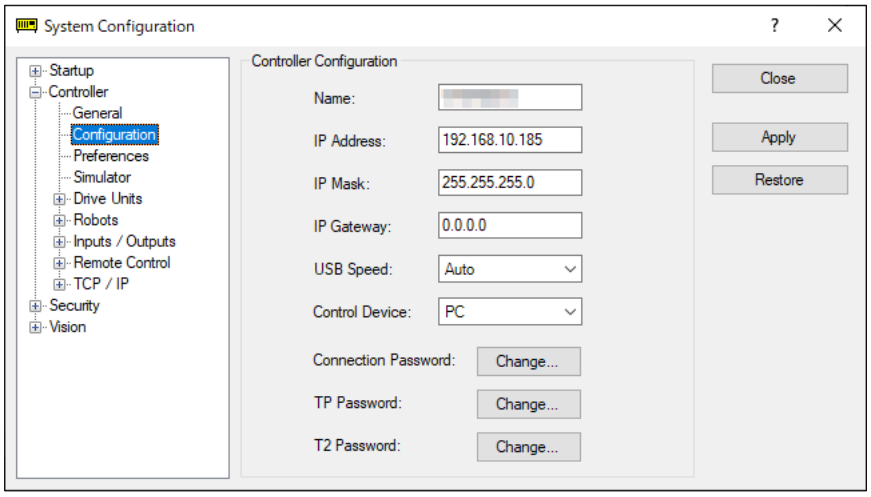
IP Address : 192.168.0.1
IP Mask : 255.255.255.0
IP Gateway : 0.0.0.0

7.3 Changing Controller IP Address

In this section, the procedure to change the Controller IP address when connecting Controller development PC connection port and the development PC by the USB cable is indicated.

(1) For connection between the development PC and the Controller, refer to *5. Development PC Connection Port*.

(2) Select the EPSON RC+ 7.0 menu-[Setup]-[Controller] to display the following dialog.



(3) Select [Controller]-[Configuration].

(4) Enter the proper IP address and subnet mask and click the <Apply> button.

(5) Click the <Close> button. The Controller reboots automatically.
IP address configuration is completed and the Controller reboot dialog disappears.

NOTE

When the Controller and the development PC are connected via the Ethernet, the Controller IP address can also be changed. However, Controller and the development PC do not connect automatically after rebooting the Controller at Ethernet connection.

7.4 Connection of Development PC and Controller with Ethernet

Connection between the development PC and the Controller is shown below.

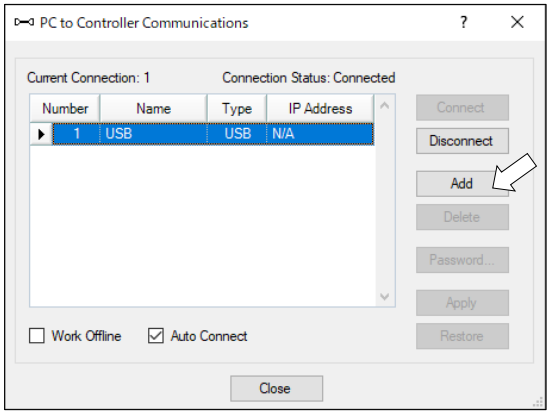
(1) Connect the development PC and the Controller using the Ethernet cable.

(2) Turn on the Controller.

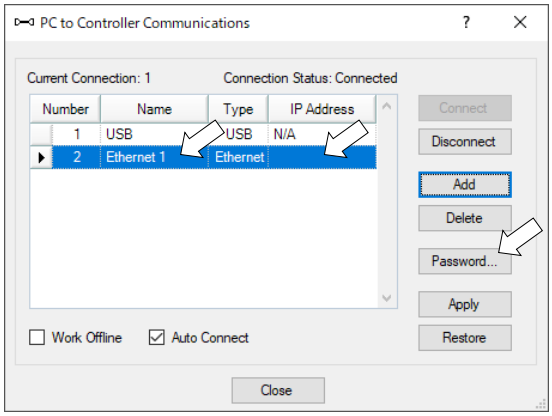
(3) Start EPSON RC+ 7.0.

(4) Display the [PC to Controller Communication] dialog from [Setup] in EPSON RC+ 7.0 menu.

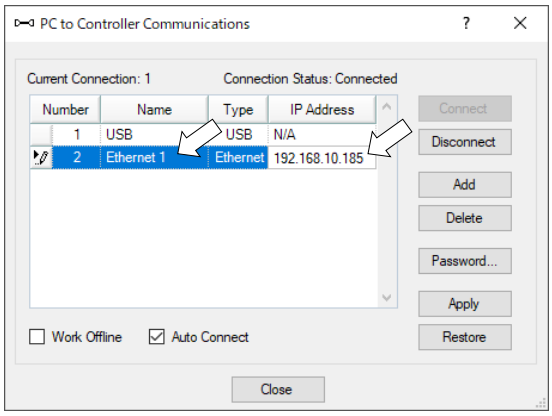
(5) Click the <Add> button.



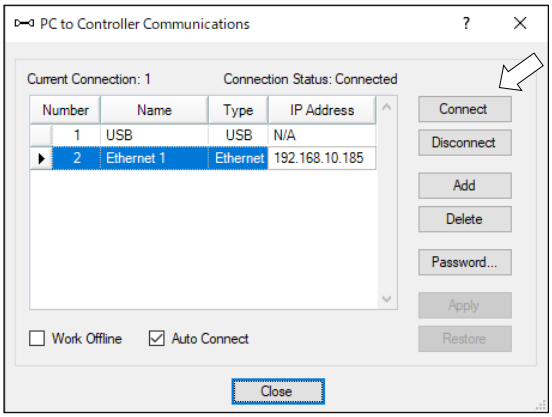
(6) Connection “No.2” is added. Set the following and click the <Apply> button.
Name : Valid value to identify the controller to connect
IP Address : IP address for Controller to connect



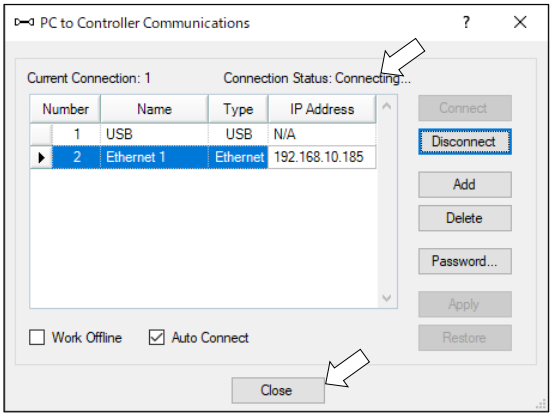
(7) [Name] and [IP Address] specified in procedure (6) is displayed.



(8) Make sure that “No.2” is selected, and click the <Connect> button.



(9) After the development PC and Controller connection is completed, “Connected” is displayed in the [Connection status:]. Make sure that “Connected” is displayed and click the <Close> button to close the [PC to Controller Communications] dialog.



Connection between the development PC and the Controller is complete. Now the robot system can be used via an Ethernet connection from EPSON RC+ 7.0.

7.5 Disconnection of Development PC and Controller with Ethernet

Disconnection of the development PC and the Controller is shown below.

(1) Display [PC-Controller Connection] dialog from [Setup] in EPSON RC+ 7.0 menu.

(2) Click the <Disconnect> button.
Communication between the Controller and the development PC is disconnected and the Ethernet cable can be removed.

NOTE

If the Ethernet cable is removed when the Controller and the development PC is connected, Emergency Stop occurs and the Robot stops. Be sure to click the <Disconnect> button in the [PC to Controller Communications] dialog before the Ethernet cable is removed.

8. TP Port

8.1 What is the TP Port?

The TP port connects the Teach Pendant to the Controller. You can connect the Teach Pendant.

NOTE ☞ When nothing is connected to the TP port, Emergency Stop status occurs in the Controller. When the Teach Pendant is not connected, connect the TP bypass plug.

Save the removed TP in place so that it can be distinguished from the TP connected to the controller.

Do not connect the following devices to the TP port of RC90-B. Connecting these devices may result in malfunction of the device since the pin assignments are different.

OPTIONAL DEVICE dummy plug
Operation Pendant OP500
Operator Pendant OP500RC
Jog Pad JP500
Teaching Pendant TP-3**
Operator Panel OP1

External enable switch cannot be connected to the TP port. Use the enable switch provided with the TP.

8.2 Teach Pendant Connection

A cable for connection to the RC90-B Controller is attached to the Teach Pendant. Connect this cable connector to the TP port.
Communication is set automatically. Enable the Teach Pendant by one of the following procedures.

- Insert the Teach Pendant connector to the Controller and turn ON the Controller.
- Insert the Teach Pendant connector while the Controller is turned ON.

NOTE ☞

- Teach Pendant connection and disconnection from the Controller are allowed when the Controller power is ON.
- When the Teach Pendant connector is removed from the Controller with the mode selector key switch of the Teach Pendant in the “Teach” position, the operation mode will remain in the TEACH mode. The operation mode cannot be switched to AUTO mode. Be sure to remove the Teach Pendant after switching the operation mode to “Auto” mode.
- Manager control taking off or storage of the Teach Pendant. Do not touch it unless the administrator approves it. When storing, make sure that it is not connected to the controller.

For details, refer to manual
Robot Contoroller Option Teach Pendant TP1.
Robot Contoroller Option Teach Pendant TP2.

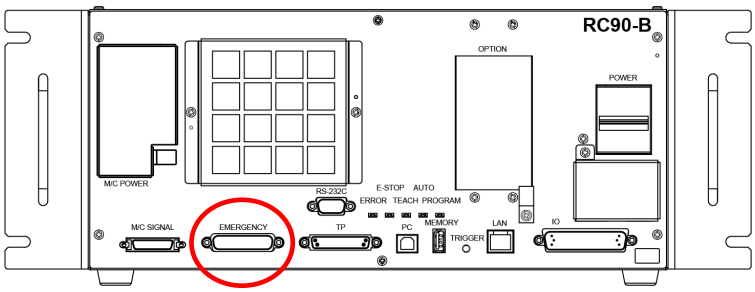
9. EMERGENCY

NOTE

The details of safety requirements for this section are described in *EPSON RC+ 7.0 2. Safety*. Please refer to them to keep the robot system safe.

Connect a safeguard switch or Emergency Stop switch to the Controller EMERGENCY connector for safety.
When nothing is connected to the EMERGENCY connector, the robot system does not operate normally.

WARNING	Before connecting the connector, make sure that the pins are not bent. Connecting with the pins bent may damage the connector and result in malfunction of the robot system.



EMERGENCY Connector

9.1 Safety Door Switch and Latch Release Switch

The EMERGENCY connector has input terminals for the Safety Door switch and the Emergency Stop switch. Be sure to use these input terminals to keep the system safe.

Connector	Standard
EMERGENCY connector (Controller side)	D-sub 25 male pin Screwlock #4 - 40

* The E-STOP BOX, EMERGENCY connector cable, terminal block, and EMERGENCY connector kit are offered as options.

9.1.1 Safety Door Switch

⚠ WARNING	The interlock of the Safety Door must be functioning when the robot system is operated. Do not operate the system under the condition that the switch cannot be turned ON/OFF (e.g. The tape is put around the switch.). Operating the robot system when the switch is not functioning properly is extremely hazardous and may cause serious safety problems as the Safety Door input cannot fulfill its intended function.

In order to maintain a safe working zone, a safeguard must be erected around the Manipulator. The safeguard must have an interlock switch at the entrance to the working zone. The Safety Door that is described in this manual is one of the safeguards and an interlock of the Safety Door is called a Safety Door switch. Connect the Safety Door switch to the Safety Door input terminal on the EMERGENCY connector.
The Safety Door switch has safety features such as temporary hold-up of the program or the operation-prohibited status that are activated whenever the Safety Door is opened.

Observe the followings in designing the Safety Door switch and the Safety Door.

- For the Safety Door switch, select a switch that opens as the Safety Door opens, and not by the spring of the switch itself.
- The signal from the Safety Door (Safety Door input) is designed to input to two redundant signals. If the signals at the two inputs differ by two seconds or more, the system recognizes it to be a critical error. Therefore, make sure that the Safety Door switch has two separate redundant circuits and that each connects to the specified pins at the EMERGENCY connector on the Controller.
- The Safety Door must be designed and installed so that it does not close accidentally.

9.1.2 Latch Release Switch

The controller software latches these conditions:

- The safety door is open.
- The operation mode is set to “TEACH”.

The EMERGENCY connector has an input terminal for a latch release switch that cancels the latched conditions.

Open : The latch release switch latches conditions that the safety door is open or the operation mode is “TEACH”.

Closed : The latch release switch releases the latched conditions.

NOTE ☞ When the latched TEACH mode is released while the safety door is open, the status of Manipulator power is operation-prohibited because the safety door is open at that time.
To execute a Manipulator operation, close the safety door again, and then close the latch release input.

9.1.3 Checking Latch Release Switch Operation

After connecting the safety door switch and latch release switch to the EMERGENCY connector, be sure to check the switch operation for safety by following the procedures described below before operating the Manipulator.

(1) Turn ON the Controller while the safety door is open in order to boot the controller software.

(2) Make sure that “Safety” is displayed on the main window status bar.

(3) Close the safety door, and turn ON the switch connecting to the latch release input. Make sure that the “Safety” is dimmed on the status bar.

The information that the safety door is open can be latched by software based on the latch release input condition. To cancel the condition, close the safety door, and then close the safety door latch release input.

Open : The latch release switch latches the condition that the safety door is open.

Closed : The latch release switch does not latch the condition that the safety door is open.

NOTE ☞ The latch release input also functions to acknowledge the change of to TEACH mode. In order to change the latched condition of TEACH mode, turn the mode selector key switch on the Teach Pendant to “Auto”. Then, close the latch release input.

9.2 Emergency Stop Switch Connection

9.2.1 Emergency Stop Switch

If it is desired to add an external Emergency Stop switch(es) in addition to the Emergency Stop on the Teach Pendant and Operator Panel, be sure to connect such Emergency Stop switch(es) to the Emergency Stop input terminal on the EMERGENCY connector.

The Emergency Stop switch connected must comply with the following:

- It must be a push button switch that is “normally closed”.
- A button that does not automatically return or resume.
- The button must be mushroom-shaped and red.
- The button must have a double contact that is “normally closed”.

NOTE

The signal from the Emergency Stop switch is designed to use two redundant circuits. If the signals at the two circuits differ by two seconds or more, the system recognizes it as a critical error. Therefore, make sure that the Emergency Stop switch has double contacts and that each circuit connects to the specified pins on the EMERGENCY connector at the Controller. Refer to the *9.4 Circuit Diagrams*.
Make sure to give the emergency stop function to all the places where have the activation control.

9.2.2 Checking Emergency Stop Switch Operation

Once the Emergency Stop switch is connected to the EMERGENCY connector, continue the following procedure to make sure that the switch functions properly. For the safety of the operator, the Manipulator must not be powered ON until the following test is completed.

(1) Turn ON the Controller to boot the controller software while pressing the Emergency Stop switch.

(2) Make sure that E-STOP LED of the controller is lighting.

(3) Make sure that “EStop” is displayed on the status bar on the main window.

(4) Release the Emergency Stop Switch.

(5) Execute the RESET command.

(6) Make sure that E-STOP LED is turned OFF and that “EStop” is dimmed on the main window status bar.

9.2.3 Recovery from Emergency Stop

To recover from the emergency stop condition, follow the procedure of safety check as required by the system.

After safety check, the operations below are required to recover from the emergency stop condition.

- Release the Emergency Stop Switch
- Execute the RESET command

9.3 Pin Assignments

The EMERGENCY connector pin assignments are as follows:

Pin No.	Signal	Function	Pin No.	Signal	Function
1	ESW11	Emergency Stop switch contact (1) *3	14	ESW21	Emergency Stop switch contact (2) *3
2	ESW12	Emergency Stop switch contact (1) *3	15	ESW22	Emergency Stop switch contact (2) *3
3	ESTOP1+	Emergency Stop circuit 1 (+) *4	16	ESTOP2+	Emergency Stop circuit 2 (+) *4
4	ESTOP1–	Emergency Stop circuit 1 (–) *4	17	ESTOP2–	Emergency Stop circuit 2 (–) *4
5	Not Used	*1	18	SDLATCH1	Safety Door Latch Release
6	Not Used	*1	19	SDLATCH2	Safety Door Latch Release
7	SD11	Safety Door input (1) *2	20	SD21	Safety Door input (2) *2
8	SD12	Safety Door input (1) *2	21	SD22	Safety Door input (2) *2
9	24V	+24V output	22	24V	+24V output
10	24V	+24V output	23	24V	+24V output
11	24VGND	+24V GND output	24	24VGND	+24V GND output
12	24VGND	+24V GND output	25	24VGND	+24V GND output
13	Not Used				

*1 Do not connect anything to these pins.

*2 A critical error occurs if the input values from the Safety Door 1 and Safety Door 2 are different for two or more seconds. They must be connected to the same switch with two sets of contacts.

*3 A critical error occurs if the input values from the Emergency Stop switch contact 1 and Emergency Stop switch contact 2 are different for two or more seconds. They must be connected the same switch with two sets of contacts.

*4 Do not apply reverse voltage to the Emergency Stop circuit.

Emergency Stop switch output rated load	+30 V 0.3 A or under	1-2, 14-15 pin
Emergency Stop rated input voltage range Emergency Stop rated input current	+24 V ±10% 37.5 mA ±10% /+24 V input	3-4, 16-17 pin
Safety Door rated input voltage range Safety Door rated input current	+24 V ±10% 10 mA/+24 V input	7-8, 20-21 pin
Latch Release rated input voltage range Latch Release rated input current	+24 V ±10% 10 mA/+24 V input	18-19 pin

NOTE

The total electrical resistance of the Emergency Stop switches and their circuit should be 1 Ω or less.

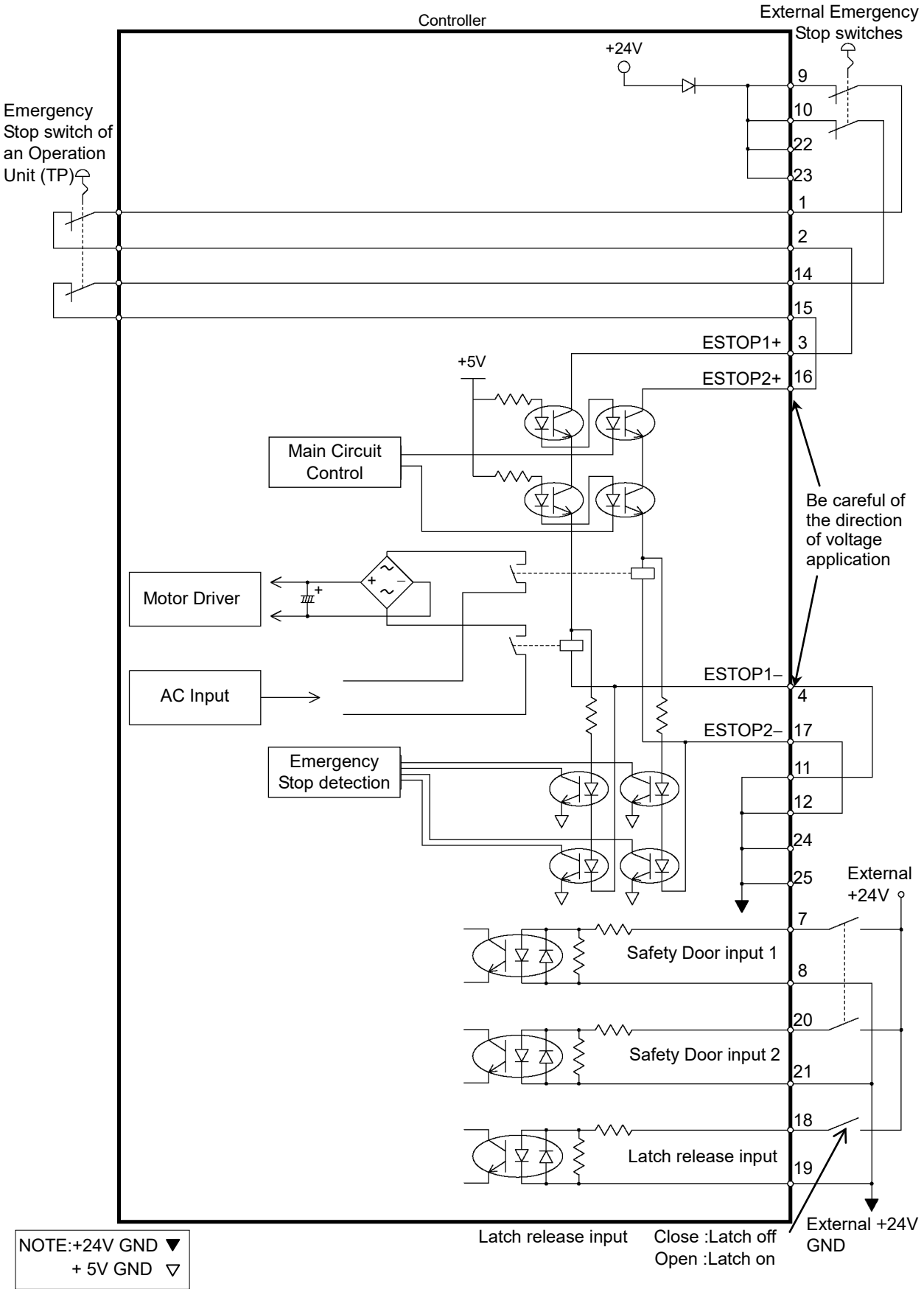
CAUTION

The 24 V output is for emergency stop. Do not use it for other purposes. Doing so may result in system malfunction.

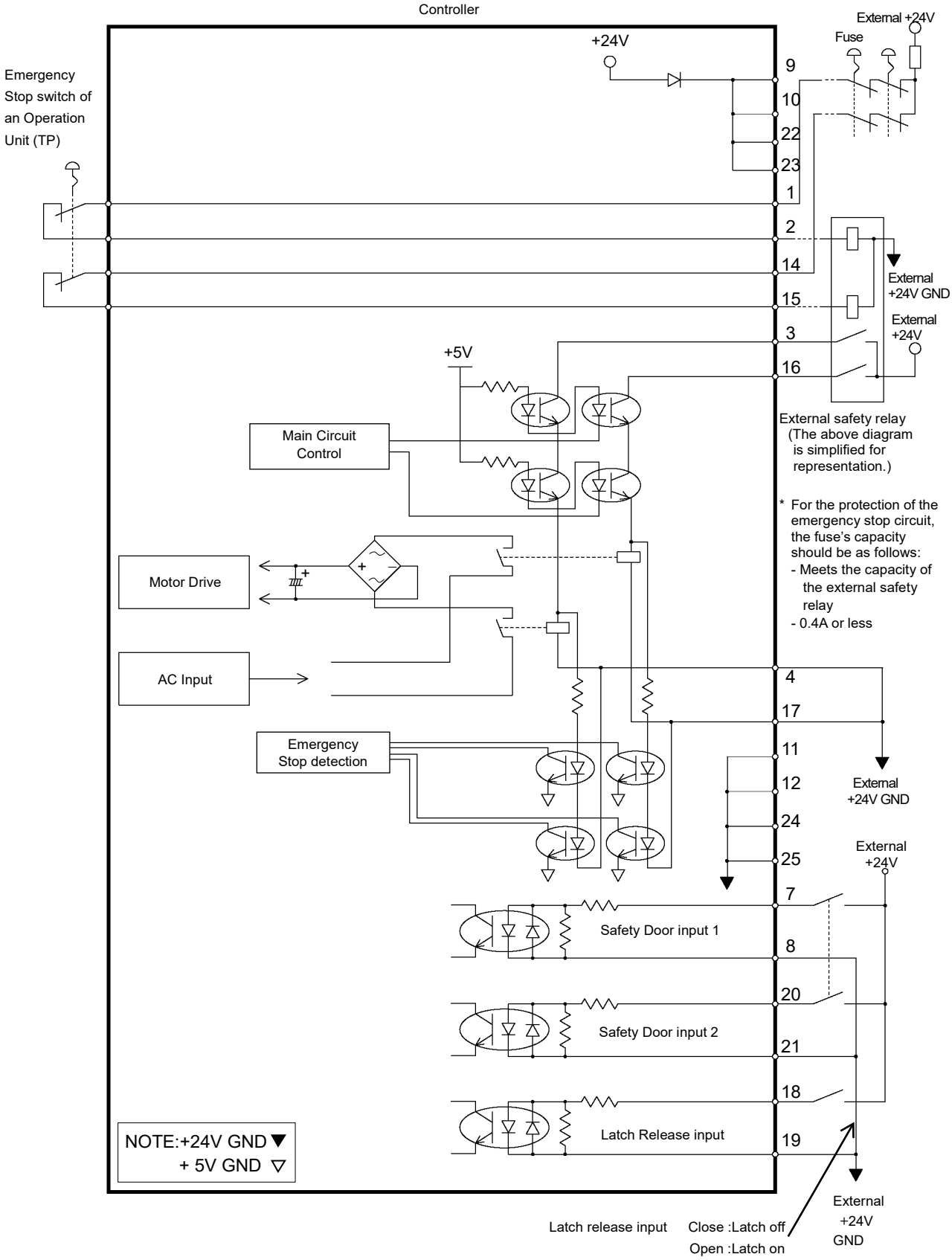
Do not apply reverse voltage to the Emergency Stop circuit. Doing so may result in system malfunction.

9.4 Circuit Diagrams

9.4.1 Example 1: External emergency stop switch typical application



9.4.2 Example 2: External safety relay typical application



10. Standard RS-232C Port

10.1 About the RS-232C Port

A standard RS-232C port is available with the Controller.

Mount the RS-232C board(s) in the option slot to communicate with external equipment with two or more RS-232C ports.

For the details of the expansion port, refer to *13.4 RS-232C Board*.

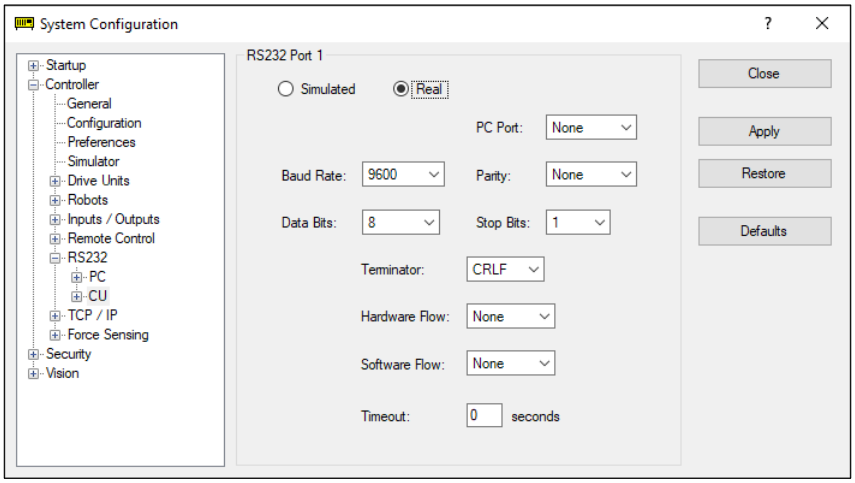
Port numbers are assigned as follows.

Port No.	Supported hardware
#1	Standard RS-232C connector
#2	First expansion RS-232C board CH1
#3	First expansion RS-232C board CH2
#4	Second expansion RS-232C board CH1
#5	Second expansion RS-232C board CH2

10.2 Confirmation with EPSON RC+ 7.0 (RS-232C)

When an RS-232C board is mounted in as option unit, the Controller software automatically identifies the RS-232C board. Therefore, no software configuration is needed. Correct identification can be confirmed from EPSON RC+ 7.0.

(1) Select the EPSON RC+ 7.0 menu-[Setup]-[System Configuration] to display the [System Configuration] dialog.



(2) Select the [RS232]-[CU].

10.3 RS-232C Software Communication Setup (RS-232C)

Available communication settings are as follows.

Item	Specification
Baud Rates	110, 300, 600, 1200, 2400, 4800, 9600, 14400, 19200, 38400, 57600, 115200
Data bit length	7, 8
Stop bit length	1, 2
Parity	Odd, even, NA
Terminator	CR, LF, CRLF

Refer to *EPSON RC+ 7.0 Online Help* or *Users Guide – 13. RS-232C Communications* for RS-232C communication from the Robot application.

10.4 Communication Cable (RS-232C)

Prepare a communication cable as described in this section.

Connector	Standard
RS-232C Connector (Controller side)	D-sub 9 male pin Screwlock #4 - 40

NOTE


Use twisted pair cable for shielded wire.
Clamp the shield to the hood for noise prevention.

Pin assign of the RS-232C connector is as follows.

Pin No	Signal	Function	Signal Direction
1	DCD	Data carrier detect	Input
2	RXD	Receive data	Input
3	TXD	Send data	Output
4	DTR	Terminal ready	Output
5	GND	Signal ground	-
6	DSR	Data set ready	Input
7	RTS	Request to send	Output
8	CTS	Clear to send	Input
9	RI	Ring indicator	Input

11. I/O Connector

The I/O connector is for connecting your input/output equipment to the system.

Non-polar PhotoMOS relay is used. Two types of connection are available.

	Pins	Bit number
Input	24	0 to 23
Output	16	0 to 15

Refer to *13.2. Expansion I/O board*.
For cable wiring, refer to the *3.5 Noise Countermeasures* in order to prevent noise.

Remote function is initially assigned to input from 0 to 7 and output from 0 to 8. For further details, refer to *11. I/O Remote Settings*.

11.1 RC90 (I/O Connector)

11.1.1 Input Circuit (RC90-B)

Input Voltage Range : +12 to 24 V ±10%

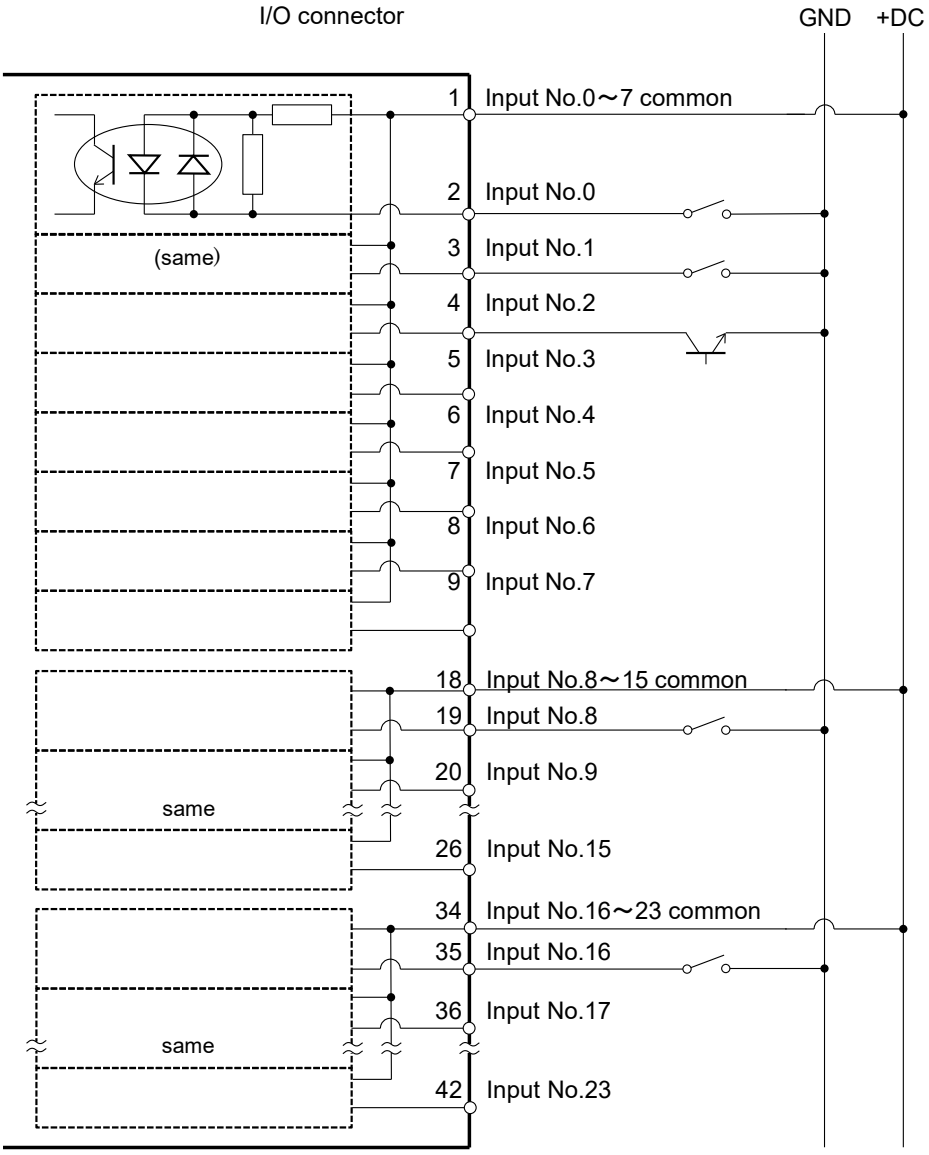
ON Voltage : +10.8 V (min.)

OFF Voltage : +5 V (max.)

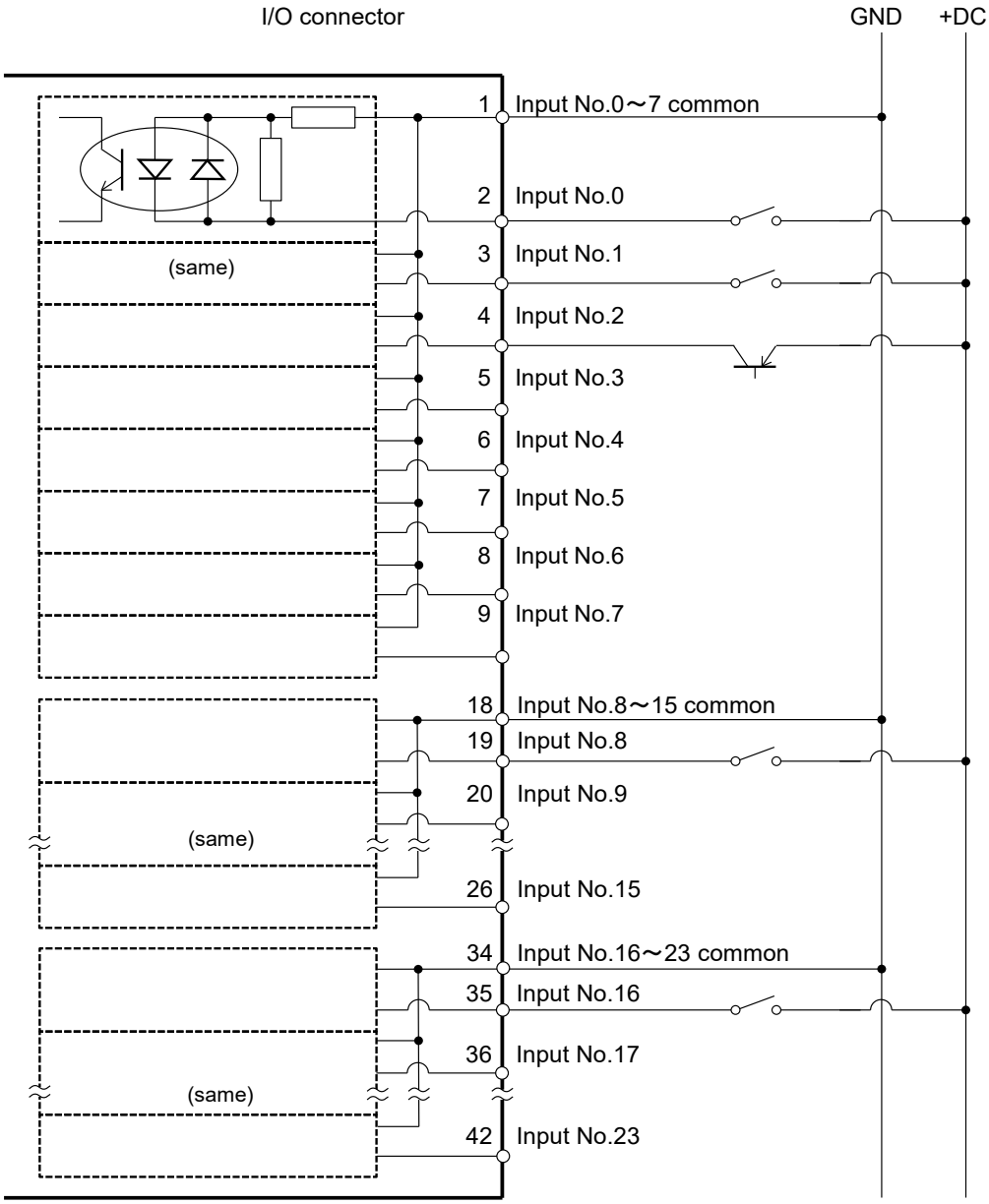
Input Current : 10 mA (TYP) at +24 V input

Two types of wiring are available for use with the two-way photo coupler in the input circuit.

Typical Input Circuit Application 1



Typical Input Circuit Application 2



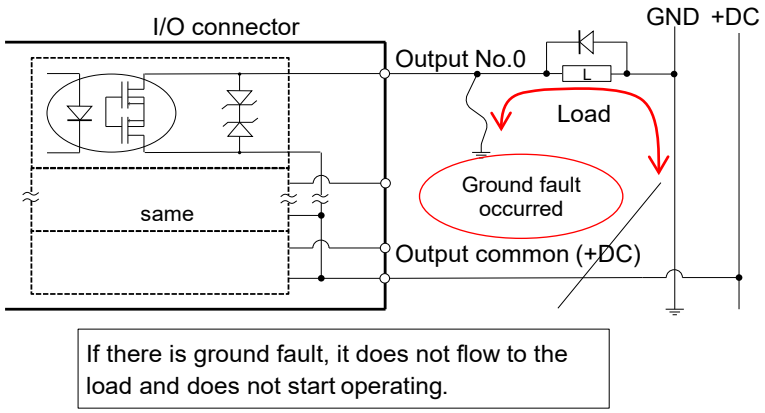
11.1.2 Output Circuit (RC90-B)

Rated Output Voltage : +12V~24V±10%
Maximum Output Current : TYP 100mA/1 output
Output Driver : PhotoMOS relay
On-resistance (average) :0.13Ω or less

Two types of wiring are available for use with the non-polar PhotoMOS relay in the output circuit.

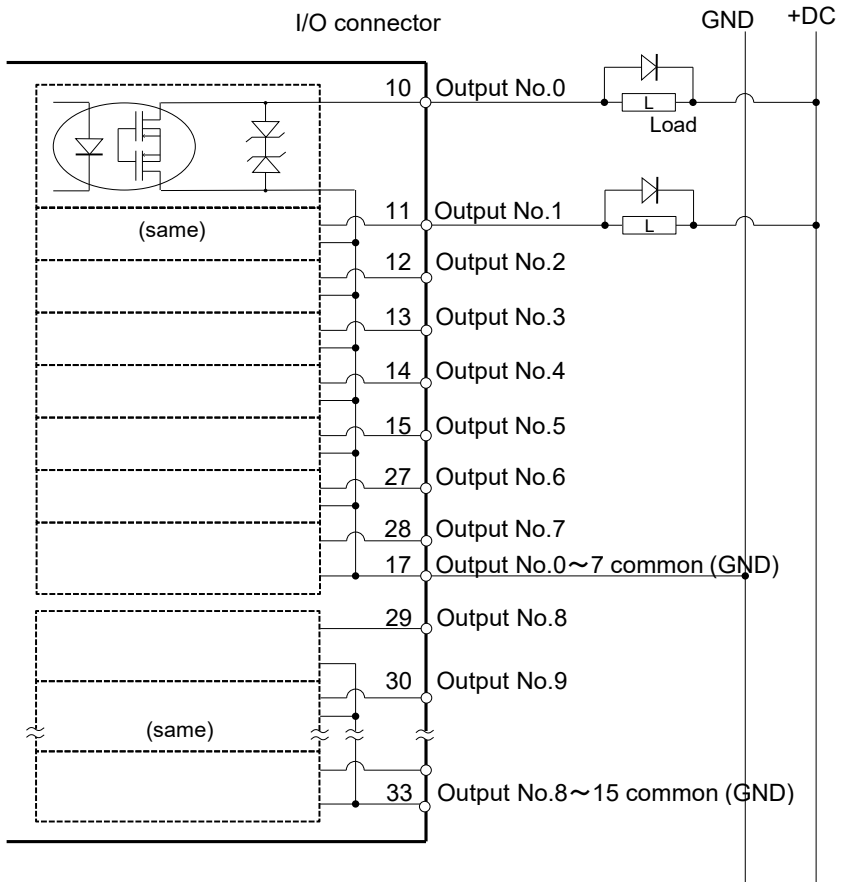
CAUTION

- Use Plus Common (PNP) to prevent the load from operating unintentionally if the wiring between the controller and the load is grounded fault. This conforms to European Machinery Directives.
- Plus Common (PNP) connection

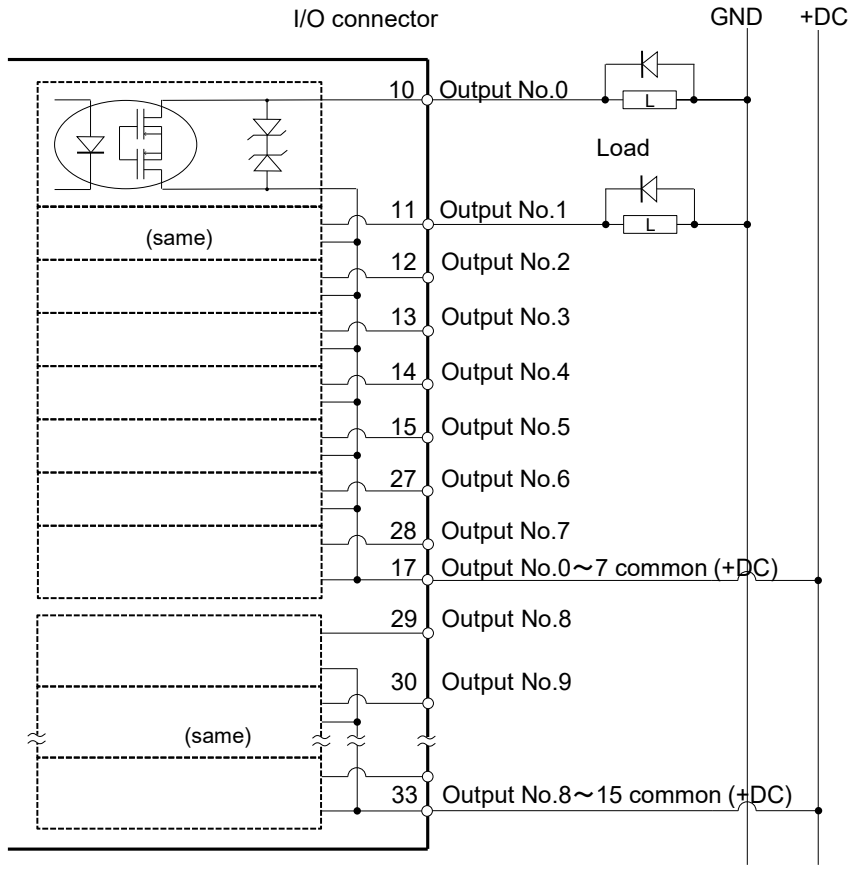


If there is ground fault, it does not flow to the load and does not start operating.

Typical Output Circuit Application 1: Sink Type (NPN) (RC90-B)



Typical Output Circuit Application 2: Source Type (PNP) (RC90-B)



11.2 Pin Assignments

Pin No.	Signal Name	Pin No.	Signal Name	Pin No.	Signal Name
1	Input common No. 0 to 7	18	Input common No. 8 to 15	34	Input common No. 16 to 23
2	Input No. 0 (Start)	19	Input No. 8	35	Input No. 16
3	Input No. 1 (SelProg1)	20	Input No. 9	36	Input No. 17
4	Input No. 2 (SelProg2)	21	Input No. 10	37	Input No. 18
5	Input No. 3 (SelProg4)	22	Input No. 11	38	Input No. 19
6	Input No. 4 (Stop)	23	Input No. 12	39	Input No. 20
7	Input No. 5 (Pause)	24	Input No. 13	40	Input No. 21
8	Input No. 6 (Continue)	25	Input No. 14	41	Input No. 22
9	Input No. 7 (Reset)	26	Input No. 15	42	Input No. 23
10	Output No. 0 (Ready)	27	Output No. 6 (SError)	43	Output No.11
11	Output No. 1 (Running)	28	Output No. 7 (Warning)	44	Output No.12
12	Output No. 2 (Paused)	29	Output No. 8 (EStopOff)	45	Output No.13
13	Output No. 3 (Error)	30	Output No. 9	46	Output No.14
14	Output No. 4	31	Output No.10	47	Output No.15
15	Output No. 5 (SafeguardOn)	32	Not Used	48	Not Used
16	Not Used	33	Output common No. 8 to 15	49	Not Used
17	Output common No. 0 to 7			50	Not Used

Remote function inside ( ) in the table above is initially assigned to input from 0 to 7 and output from 0 to 8. For further details, refer to *12. I/O Remote Settings*.

Connector	Standard
I/O Connector (Controller side)	D-sub 50 male pin Screwlock #4 - 40

* The I/O connector, I/O cable, terminal block, and I/O cable kit are offered as options.

* I/O connector is included with shipment.

12. I/O Remote Settings

This section describes the functions and timings of input and output signals.

The remote functions may be assigned to your standard I/O board(s), expansion I/O board(s), or fieldbus I/O board(s) to enhance robot system control - either from an operational unit of your choice or a sequencer.

Remote function is initially assigned to input from 0 to 7 and output from 0 to 8.

To accept external remote inputs, assign the remote function and the control device is remote. For further details, refer to the section, *Remote Control Software Configuration* in *EPSON RC+ 7.0 User's Guide - Remote Control*.

The user defines the I/O number that a remote function is assigned to using software configuration. For further details, refer to the section, *Remote Control Software Configuration* in *EPSON RC+ 7.0 User's Guide - Remote Control*.

For details about I/O cable connection, refer to sections on *11. I/O Connector* and *13.2. Expansion I/O Board* and *13.3 Fieldbus I/O Board*.

For details about communication with external equipment, refer to *EPSON RC+ 7.0 User's Guide -12. Remote Control*.

 CAUTION	■ When using remote I/O, always make sure of the following. Using the robot system under unsatisfactory conditions may cause malfunction of the system and/or safety problems. - This signal is not safety signal. Do not this use for safety functions. - Assign remote functions to inputs/outputs correctly and wire correctly when setting up remote I/O signals. - Make sure that the functions correspond to the correct input/output signals before turning ON the system. - When verifying the robot system operation, prepare for failures with initial settings or wiring. If the Manipulator functions unusually by the failures with initial settings or wiring, press the Emergency Stop switch immediately to stop the Manipulator.

NOTE

Remote function is available when virtual I/O is enabled.

When you set up a remote I/O signal, please either keep a written record of the settings or store the data in a file for later reference.

When you set up a fieldbus I/O signal to the remote function, response depends on the baud rate of the fieldbus. For details of fieldbus response, refer to *Robot Controller Option Fieldbus I/O*.

12.1 I/O Signal Description

Remote function is initially assigned to input from 0 to 7 and output from 0 to 8.

To change the function assignment from the initial setting, use EPSON RC+ 7.0.

To use all signals, you will need to add Expansion I/O or Fieldbus I/O board(s).

12.1.1 Remote Input Signals

Remote inputs are used to control the Manipulators and start programs. Certain conditions must be met before inputs are enabled, as shown in the table below.
To accept external remote inputs, assign the remote function and set remote to the control device. When external remote input is available, "AutoMode output" turns ON.
Except "SelProg", the signals execute each function when the signal starts in input acceptance condition. The function executes automatically. Therefore, no special programming is needed.

NOTE ☞ When an error occurs, you must execute a "Reset" to clear the error condition before any other remote input commands can be executed. Use the "Error output" and "Reset input" to monitor the error status and clear error conditions from the remote device.

Name	Initial	Description	Input Acceptance Condition (*1)
Start	0	Execute function selected at SelProg. (*2)	Ready output ON Error output OFF EStopOn output OFF SafeguardOn output OFF EStopOff output ON Pause input OFF Stop input OFF
SelProg1 SelProg2 SelProg4 SelProg8 SelProg16 SelProg32	1 2 3 Not Set Not Set Not Set	Specify the executing Main function number. (*2)	
Stop	4	All tasks and commands are stopped.	
Pause	5	All tasks are paused. (*3)	Running output ON
Continue	6	Continue the paused task.	Paused output ON Pause input OFF Stop input OFF
Reset	7	Reset emergency stop and error. (*4)	Ready output ON
Shutdown	Not set	Terminates the system	
ForcePowerLow (*6)	Not Set	Operates as the forced low power function. The robot is operated in the low power mode. Power High control from the command is not accepted. Executes the following according to the controller preferences. Stops or temporarily stops all the tasks and commands. (*12)	Any time This input is acceptable even AutoMode output is OFF.
SelRobot	Not Set	Changes the output condition of MotorsOn, AtHome, PowerHigh, and MCalReqd. (*9)	

Name	Initial	Description	Input Acceptance Condition (*1)
SelRobot1 SelRobot2 SelRobot4 SelRobot8 SelRobot16	Not set	Specify the number of robot which executes a command. (*5)	
SetMotorsOn	Not set	Turn ON robot motors. (*5)	Ready output ON EStopOn output OFF SafeguardOn output OFF EStopOff output ON SetMotorOff input OFF
SetMotorsOff	Not set	Turn OFF robot motors.	Ready output ON
SetPowerHigh	Not set	Set the robot power mode to High (*5)	Ready output ON EStopOn output OFF SafeguardOn output OFF EStopOff output ON SetPowerLow input OFF
SetPowerLow	Not set	Set the robot power mode to Low. (*5)	Ready output ON
Home	Not set	Move the Robot Arm to the home position defined by the user.	Ready output ON Error output OFF EStopOn output OFF SafeguardOn output OFF EStopOff output ON MotorsOn output ON Pause input OFF Stop input OFF
MCal	Not set	Execute MCal (*5) (*7)	Ready output ON Error output OFF EStopOn output OFF SafeguardOn output OFF EStopOff output ON MotorsOn output ON Pause input OFF Stop input OFF
Recover	Not set	After the safeguard is closed, recover to the position where the safeguard is open.	Paused output ON Error output OFF EStopOn output OFF SafeguardOn output OFF EStopOff output ON RecoverReqd output ON Pause input OFF Stop input OFF
ResetAlarm	Not Set	Cancel the alarm (*11)	
SelAlarm1 SelAlarm2 SelAlarm4 SelAlarm8	Not Set	Specify the alarm number to cancel (*10)	
ALIVE	Not Set	Input signal for alive monitoring of the controller. Same signal as the input will be output to ALIVE output. The master equipment can perform alive monitoring of the controller by switching the input periodically and checking the output signal.	

(*1) “AutoMode output” ON is omitted from the table. This is an input acceptance condition for all functions.

(*2) “Start input” executes Function specified by the following six bits: SelProg 1, 2, 4, 8, 16, and 32.

Function name	SelProg1	SelProg2	SelProg4	SelProg8	SelProg16	SelProg32
Main	0	0	0	0	0	0
Main1	1	0	0	0	0	0
Main2	0	1	0	0	0	0
Main3	1	1	0	0	0	0
⋮						
Main60	0	0	1	1	1	1
Main61	1	0	1	1	1	1
Main62	0	1	1	1	1	1
Main63	1	1	1	1	1	1

0=OFF, 1=ON

(*3) “NoPause task” and “NoEmgAbort task” do not pause.
For details, refer to EPSON RC+ 7.0 *Online Help* or *Pause* in *SPEL+ Language Reference*.

(*4) Turns OFF the I/O output and initializes the robot parameter.
For details, refer to EPSON RC+ 7.0 *Online Help* or *Reset* in *SPEL+ Language Reference*.

(*5) The values specified by “SelRobot1, 2, 4, 8, and 16” correspond to the robot numbers, when multiple robots are connected to the controller.

Robot number	SelRobot1	SelRobot2	SelRobot4	SelRobot8	SelRobot16
0 (All)	0	0	0	0	0
1	1	0	0	0	0
2	0	1	0	0	0
3	1	1	0	0	0
⋮					
13	1	0	1	1	0
14	0	1	1	1	0
15	1	1	1	1	0
16	0	0	0	0	1

0=OFF, 1=ON

(*6) Initializes the robot parameter.
For details, refer to EPSON RC+ 7.0 *Online Help* or *Motor* in *SPEL+ Language Reference*.

(*7) For details, refer to *EPSON RC+ 7.0 Online Help* or *MCal* in *SPEL+ Language Reference*.

(*8) This is for experienced users only. Make sure that you fully understand the input specification before using.
CmdRunning output and CmdError output will not change for this input.
“NoEmgAbort task” will not stop by this input.
When the input changes from ON to OFF, all tasks and commands will stop.

(*9) This function changes the output condition of MotorsOn, AtHome, PowerHigh, and MCalReqd.
By setting this signal with the condition selected using SelRobot1 - SelRobot16, you can switch the output condition.
Once you select the condition, it will be kept until you change it or turn off / restart the Controller. All manipulators are selected as default.

(*10) The values specified by “SelAlarm1, 2, 4, and 8” correspond to the alarm numbers.

Alarm #	Target	SelAlarm1	SelAlarm2	SelAlarm4	SelAlarm8
1	Controller battery	1	0	0	0
2	-	0	1	0	0
3	Grease of the Manipulator	1	1	0	0
4	-	0	0	1	0
5	-	1	0	1	0
6	-	0	1	1	0
7	-	1	1	1	0
8	-	0	0	0	1
9	-	1	0	0	1

0=OFF, 1=ON

The following parts are subject to grease up.

Ball screw spline unit on the Joint # 3

(*11) The specified alarm can be canceled by selecting the conditions using SelAlarm1-SelAlarm8 and setting this signal.

(*12) Operation of all tasks and commands, power mode of the robot, and PowerHigh command are executed by the setting value of the controller preferences.

Preferences (1): “Motor power low when ForcePowerLow signal OFF”

Preferences (2): “ForcePowerLow signal change pauses all tasks”

For details of the controller preferences, refer to *EPSON RC+ 7.0 User’s Guide [Setup]-[System Configuration]-[Controller]-[Preferences] in 5.13.2 [System Configuration] Command (Setup Menu)*.

Preferences (1)	Preferences (2)	ForcePowerLow	All tasks and commands	Power mode	PowerHigh
0	0	1→0	Stop	Low only	Accept
0	0	0→1	Stop	Low only	Not accept
0	1	1→0	Continue	High/Low	Accept
0	1	0→1	Temp. stop	Low only	Not accept
1	0	1→0	Stop	Low only	Not accept
1	0	0→1	Stop	Low only	Accept
1	1	1→0	Temp. stop	Low only	Not accept
1	1	0→1	Continue	High/Low	Accept

12.1.2 Remote Output Signals

Remote outputs provide status for the Manipulator and Controller.

Remote outputs provide the assigned function using with any control device The outputs execute automatically. Therefore, no special programming is needed.

Name	Initial	Description
Ready	0	Turns ON when the controller startup completes and no task is running.
Running	1	Turns ON when task is running. However, turns OFF when “Paused output” is ON.
Paused	2	Turns ON when pause task exists.
Error	3	Turns ON when an error occurs. Use “Reset input” to recover from the error. (*13)
EStopOn	Not set	Turns OFF other than at Emergency Stop. Turns ON at Emergency Stop. Turns OFF at controller power OFF (*11)
SafeguardOn	5	Turns ON when the safeguard is open.
SError	6	Turns ON when critical error occurs. When a critical error occurs, “Reset input” does not function. Reboot the controller to recover. (*13)
Warning	7	Turns ON when warning occurs. The task runs as normal with the warning. However, be sure to eliminate the cause of the warning as soon as possible. (*13)
EStopOff	8	Turns OFF other than at Emergency Stop. Turns ON at Emergency Stop. Turns OFF at controller power OFF
MotorsOn	Not set	Turns ON when the motor is ON. (*5)
AtHome	Not set	Turns ON when the robot is in the home position. (*5)
PowerHigh	Not set	Turns ON when the robot’s power mode is High. (*5)
MCalReqd	Not set	Turns ON when the robot hasn’t executed MCal. (*5)
RecoverReqd	Not set	Turns ON when at least one robot is waiting for Recover after the safeguard is closed.
RecoverInCycle	Not set	Turns ON when at least one robot is executing Recover.
WaitingRC	Not set	Turns ON when the controller waits for RC+ to connect.
CmdRunning	Not set	Turns ON when an input command is executing.
CmdError	Not set	Turns ON when an input command cannot be accepted.
CurrProg1 CurrProg2 CurrProg4 CurrProg8 CurrProg16 CurrProg32	Not set	Indicates the running or the last main function number (*1)
AutoMode	Not set	Turns ON in remote input acceptable status. (*2)
TeachMode	Not set	Turns ON in TEACH mode.
TestMode	Not set	Turns ON in TEST mode.
EnableOn	Not set	Turns ON when enable switch is ON.
ErrorCode1 ⋮ ErrorCode8192	Not set	Indicates the error number.
InsideBox1 ⋮ InsideBox15	Not set	Turns ON when the robot is in the approach check area. (*3)
InsidePlane1 ⋮ InsidePlane15	Not set	Turns ON when the robot is in the approach check plane. (*4)
Alarm	Not set	Turns ON when any of the alarms is occurring. (*9) (*13)

Name	Initial	Description
Alarm1	Not set	Turns ON when a battery alarm of the controller is occurring. (*12) (*13)
Alarm2	Not set	Turns ON when a battery alarm of the robot connected to CU is occurring. (*12) (*13)
Alarm3	Not set	Turns ON when a grease alarm of the robot connected to CU is occurring. (*10) (*12) (*13)
Alarm4	Not set	Turns ON when a battery alarm of the robot connected to DU1 is occurring. (*12) (*13)
Alarm5	Not set	Turns ON when a grease alarm of the robot connected to DU1 is occurring. (*10) (*12) (*13)
Alarm6	Not set	Turns ON when a battery alarm of the robot connected to DU2 is occurring. (*12) (*13)
Alarm7	Not set	Turns ON when a grease alarm of the robot connected to DU2 is occurring. (*10) (*12) (*13)
Alarm8	Not set	Turns ON when a battery alarm of the robot connected to DU3 is occurring. (*12) (*13)
Alarm9	Not set	Turns ON when a grease alarm of the robot connected to DU3 is occurring. (*10) (*12) (*13)
PositionX	Not set	Outputs current X coordinate in the World coordinate system (*6) (*7)
PositionY	Not set	Outputs current Y coordinate in the World coordinate system (*6) (*7)
PositionZ	Not set	Outputs current Z coordinate in the World coordinate system (*6) (*7)
PositionU	Not set	Outputs current U coordinate in the World coordinate system (*6) (*7)
PositionV	Not set	Outputs current V coordinate in the World coordinate system (*6) (*7)
PositionW	Not set	Outputs current W coordinate in the World coordinate system (*6) (*7)
Torque1	Not set	Outputs the current torque value of Joint #1 (*6) (*7)
Torque2	Not set	Outputs the current torque value of Joint #2 (*6) (*7)
Torque3	Not set	Outputs the current torque value of Joint #3 (*6) (*7)
Torque4	Not set	Outputs the current torque value of Joint #4 (*6) (*7)
Torque5	Not set	Outputs the current torque value of Joint #5 (*6) (*7)
Torque6	Not set	Outputs the current torque value of Joint #6 (*6) (*7)
CPU	Not set	Outputs the CPU load factor of the user program (*8)
ESTOP	Not set	Outputs how many times emergency stops have been executed.
ALIVE	Not set	Output signal for alive monitoring of the controller. The signal input by ALIVE input will be output. The master equipment can perform alive monitoring of the controller by switching the input periodically and checking the output signal.
ForceControlOn	Not set	Turns ON when the manipulator operating force control function. (*5)
ExtCmdGet	Not set	For extended remote I/O command. For details, refer to the following manual. *Remote Control Reference 4. Remote I/O to Be Used*
ExtRespSet	Not set	
ExtCmdResult	Not set	
ExtError	Not set	
ExtResp_0-15	Not set	
ExtResp_16-31	Not set	
ExtResp_32-47	Not set	
ExtResp_48-63	Not set	
ExtResp_64-79	Not set	
ExtResp_80-95	Not set	
ExtResp_96-111	Not set	
ExtResp_112-127	Not set	

(*1) Outputs the current or the last function number of CurrProg1, 2, 4, 8, 16, or 32.

Function name	CurrProg1	CurrProg2	CurrProg4	CurrProg8	CurrProg16	CurrProg32
Main	0	0	0	0	0	0
Main1	1	0	0	0	0	0
Main2	0	1	0	0	0	0
Main3	1	1	0	0	0	0
⋮						
Main60	0	0	1	1	1	1
Main61	1	0	1	1	1	1
Main62	0	1	1	1	1	1
Main63	1	1	1	1	1	1

0=OFF, 1=ON

(*2) Remote function is available in the followings conditions.
- The setting is Auto mode and the control device is remote.
- The setting is Program mode and Remote I/O is enabled.

(*3) For details, refer to EPSON RC+ 7.0 *Online Help* or *Box* in *SPEL+ Language Reference*.

(*4) For details, refer to EPSON RC+ 7.0 *Online Help* or *Plane* in *SPEL+ Language Reference*.

(*5) Manipulator status is output as follows, according to the condition selected in SelRobot.
Wait at least 40 ms before inputting the signal after changing the condition in SelRobot.

Name	(SelRobot1- SelRobot16) condition when inputting SelRobot	
	0: All robots are selected	1 - 16: Particular robot number is selected
MotorsOn	Turns ON when at least one motor is ON.	Turns ON when the motor of the selected robot is ON.
AtHome	Turns ON when all robots are in the home position.	Turns ON when the selected robot is in the home position.
PowerHigh	Turns ON when at least one robot's power mode is High.	Turns ON when the selected robot's power mode is High.
MCalReqd	Turns ON when at least one robot hasn't executed MCal.	Turns ON when the selected robot hasn't executed MCal.

(*6) Outputs information of the selected robot when SelRobot1, SelRobot2, SelRobot4, SelRobot8, and SelRobot16 are set. If not, information of Robot 1 will be output.

(*7) Outputs information in Real format.

(*8) Outputs the total load factor of the user created tasks. For details on the CPU load factor, refer to the task manager.

(*9) The signal turns on when the alarm occurs either in the controller alarm information or the robot alarm information.

(*10) The following parts are subject to grease up.
Ball screw spline unit on the Joint # 3

(*11) It is not recommended that EStopOn does not match the output with emergency stop state and controller off state. To output the state of emergency stop, assign EStopOff.

(*12) The occurrence of battery alarm and grease alarm is monitored every 5 minutes. The alarm occurrence and output timing of the controller are different. It may be output up to 5 minutes after the controller alarm occurs.

(*13) Following shows status number/error number of output (Error, SError, Warning, Alarm).

Name	Error Number
Error	1000 to 8000s
SError	9000s
Warning	400 to 999
Alarm	(No error codes)

For more information of status number/error number, refer to *"Status Code / Error Code List"* manual.

"Alarm" will on when battery alarm or grease alarm of Controller or Manipulator occurs with "Robot maintenance" enabled. For details of Robot maintenance, refer to "Alarm" section of each Controller maintenance manual.

12.2 Timing Specifications

12.2.1 Design Notes for Remote Input Signals

The following charts indicate the timing sequences for the primary operations of the Controller.

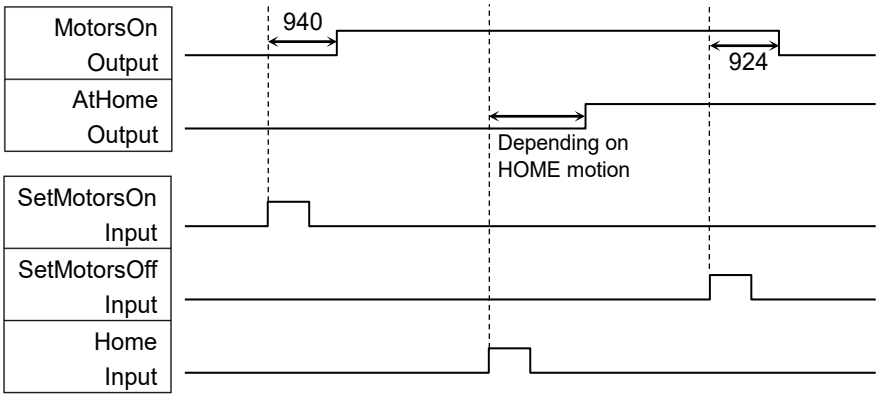
The indicated time lapses (time durations) should be referred to only as reference values since the actual timing values vary depending on the number of tasks running, as well as CPU speed of the Controller. Check carefully and refer to the following charts for the timing interrelation when you enter an input signal.

During system design, make sure that you actuate only one remote input operation at a time, otherwise an error will occur.

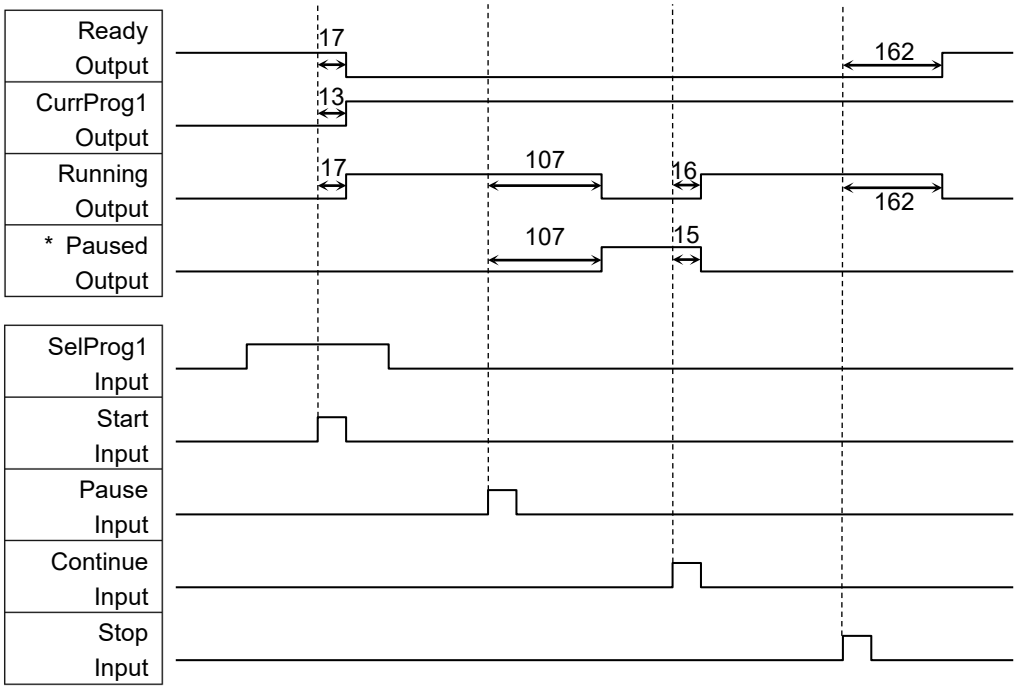
The pulse width of an input signal must be 25 or more milliseconds to be detected.

[Unit: msec]

12.2.2 Timing Diagram for Operation Execution Sequence

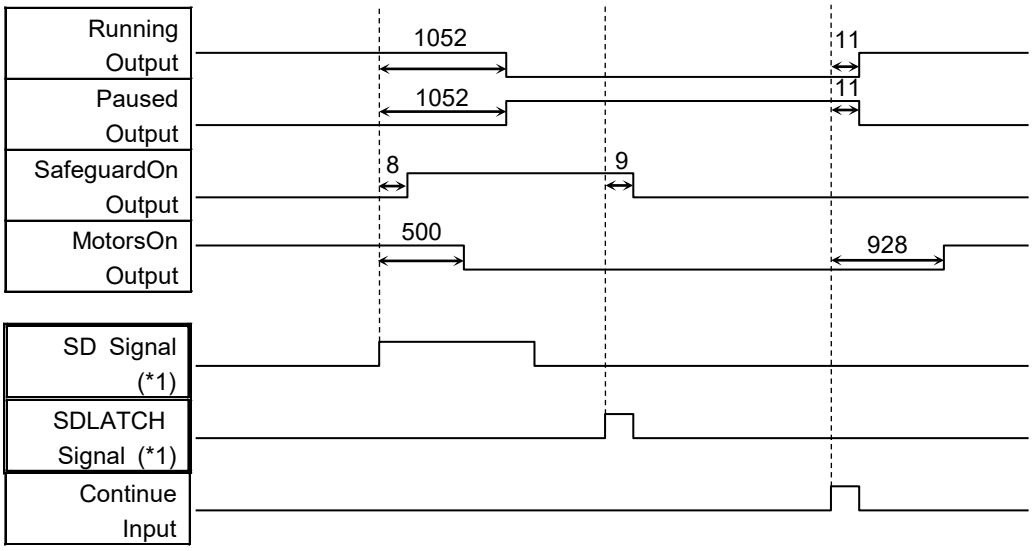


12.2.3 Timing Diagram for Program Execution Sequence



* The duration varies depending on the Quick Pause (QP) setting and the program's perating status at the time of Pause input

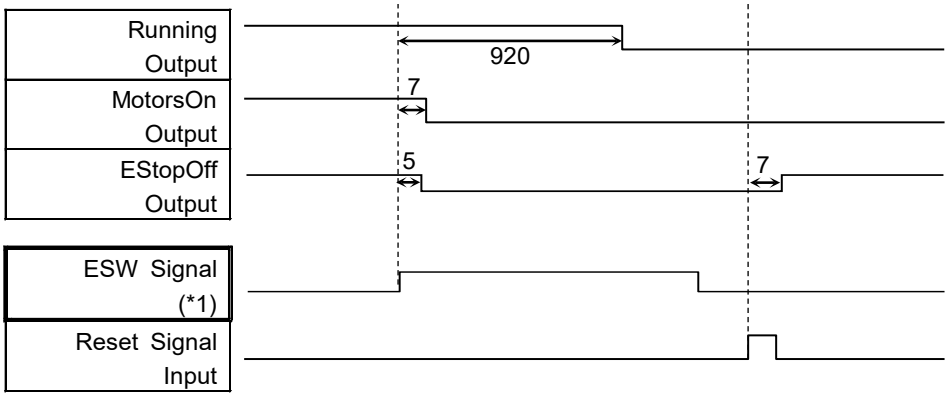
12.2.4 Timing Diagram for Safety Door Input Sequence



[Unit: msec]

(*1) A logical signal to explain the timing of the internal processing of the controller. For the input signal name and operating conditions, refer to the 9.3 Pin Assignments.

12.2.5 Timing Diagram for Emergency Stop Sequence



[Unit: msec]

(*1) A logical signal to explain the timing of the internal processing of the controller. For the input signal name and operating conditions, refer to the 9.3 Pin Assignments.

13. Option Slots

13.1 About Option Slots

Use the Option Slot to install the optional boards of RC90 / RC90-B Controller.
Up to two option boards can be installed in the controller. The types of the option boards are as follows:

13.2 Expansion I/O Board
13.3 Fieldbus I/O Board
13.4 RS-232C Board
13.5 Analog I/O Board
13.6 EUROMAP67 Board

13.2 Expansion I/O Board

13.2.1 About Expansion I/O Board

Each additional expansion I/O board provides 24 inputs and 16 outputs.

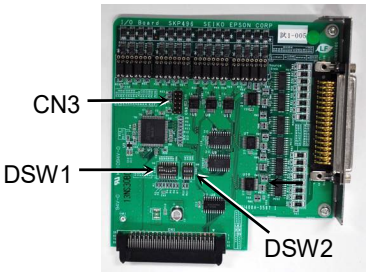
You can install up to two expansion I/O boards in the controller.

The input and output bit numbers are assigned as follows. (Bit number is assigned from CN1.)

Input Bit #	Output Bit #	Applicable Hardware
0 to 23	0 to 15	STANDARD I/O
64 to 87	64 to 79	The 1st Expansion I/O board
96 to 119	96 to 111	The 2nd Expansion I/O board

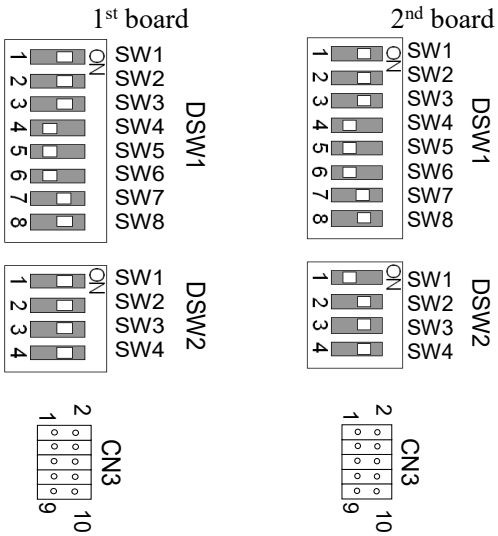
13.2.2 Board Configuration (Expansion I/O)

Board Appearance



Switch and Jumper Configuration

Setup the DSW1 and DSW2. CN3 is all open.

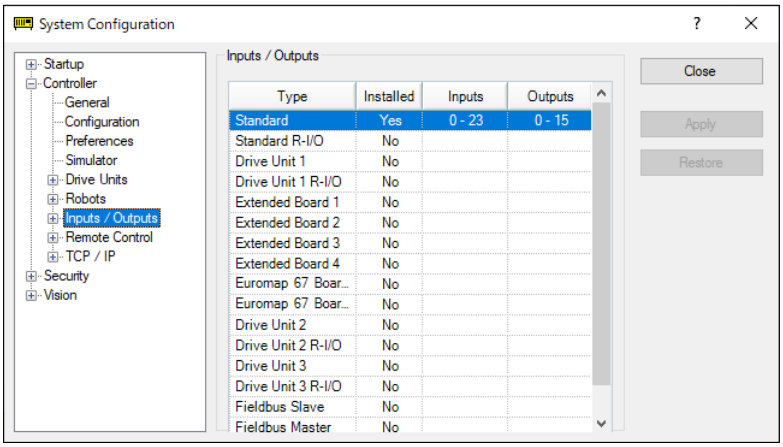


13.2.3 Confirmation with EPSON RC+ 7.0

When an expansion I/O board is mounted to the option unit, the Controller software automatically identifies the expansion I/O board. Therefore, no software configuration is needed.

Correct identification can be confirmed from EPSON RC+ 7.0.

(1) Select the EPSON RC+ 7.0 menu-[Setup]-[System Configuration] to display the [System Configuration] dialog.



(2) Select [Controller]-[Inputs / Outputs].

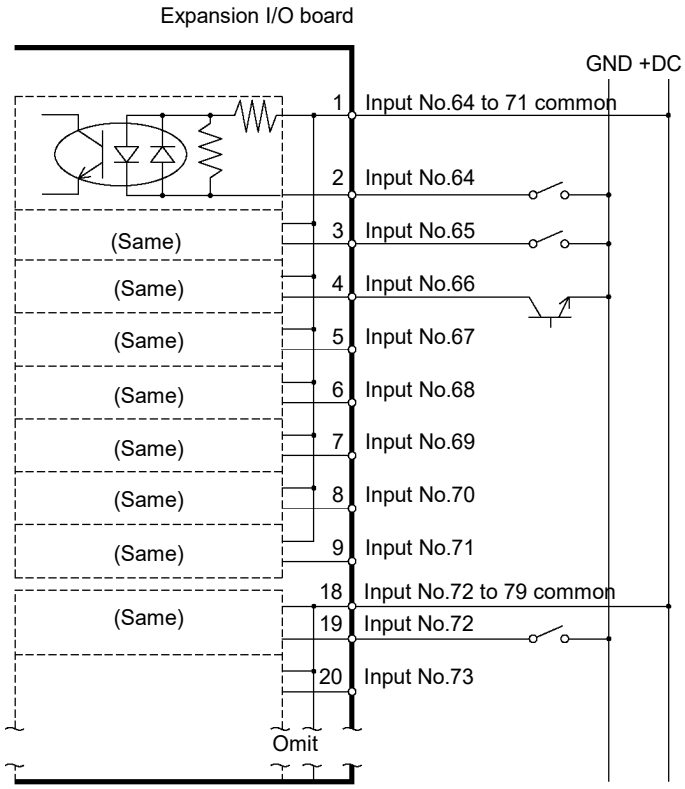
(3) Make sure that “Yes” is displayed in the Installed column.
The expansion I/O board is identified by the Controller software. Corresponding Input and Output is available.

13.2.4 Input Circuit

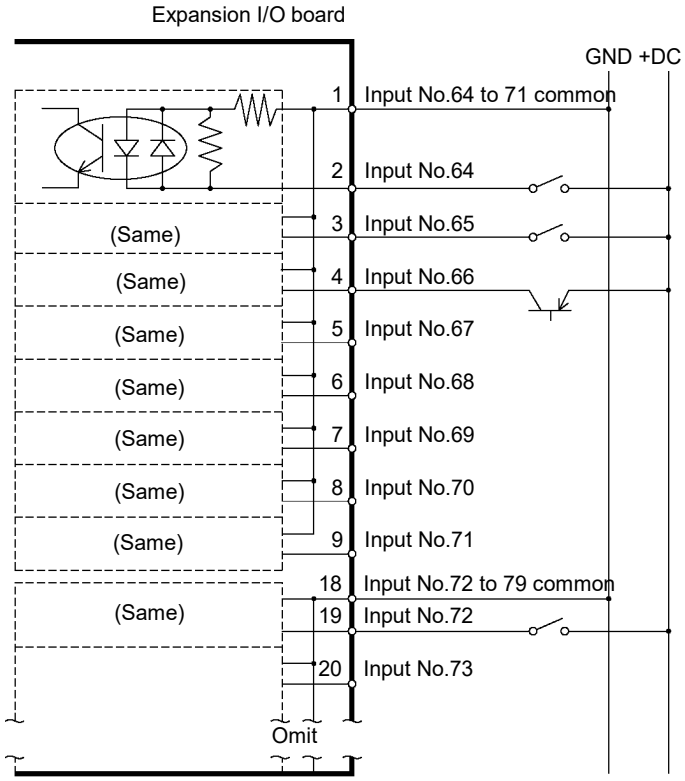
Input Voltage Range	: + 12 V to 24 V ±10%
ON Voltage	: + 10.8 V (Min.)
OFF Voltage	: + 5 V (Max.)
Input Current	: 10 mA (TYP) at + 24 V input

Two types of wiring are available for use with the two-way photo coupler in the input circuit.

Typical Input Circuit Application 1



Typical Input Circuit Application 2



13.2.5 Output Circuit

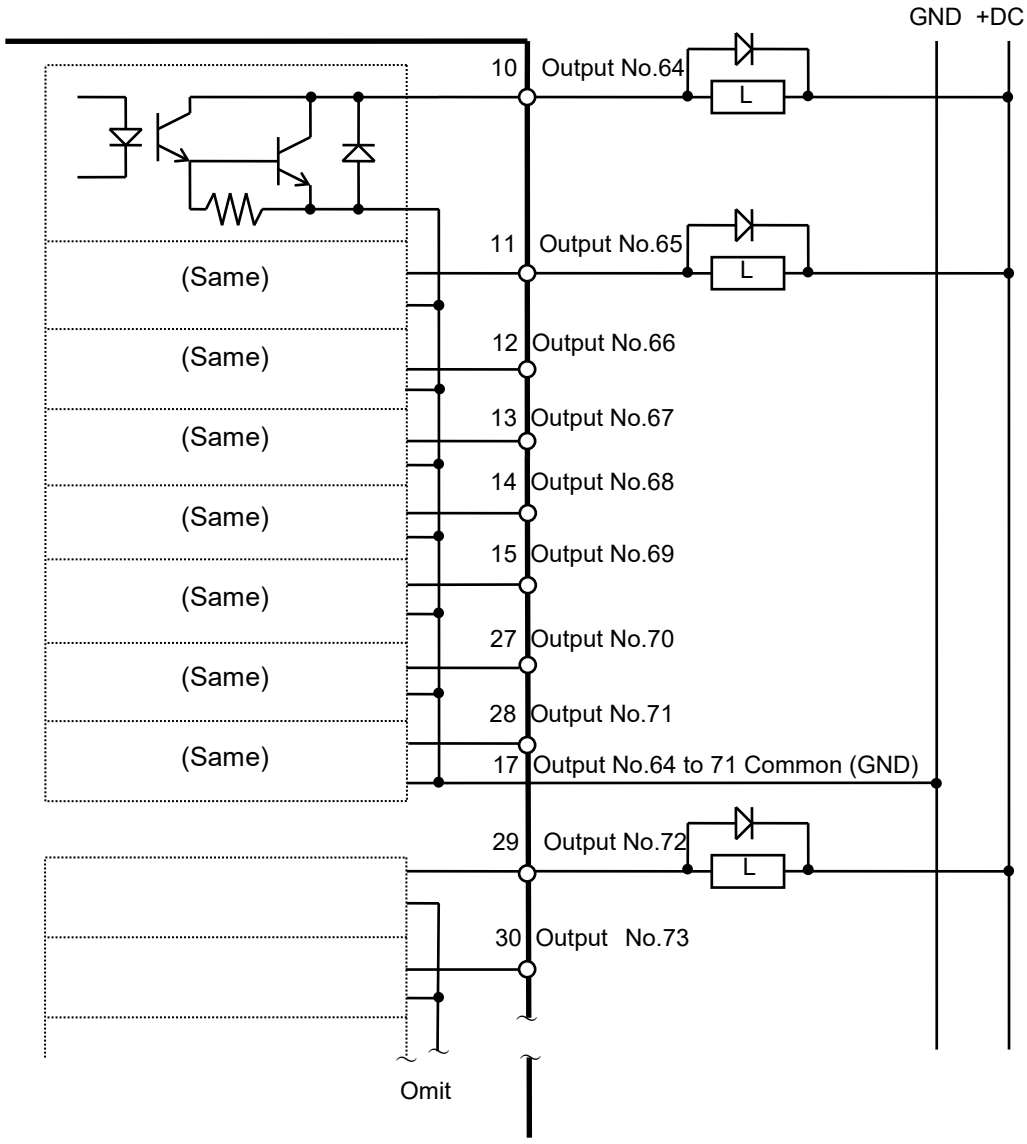
Rated Output Voltage : +12 V to 24 V ±10%

Maximum Output Current : TYP 100 mA/1 output

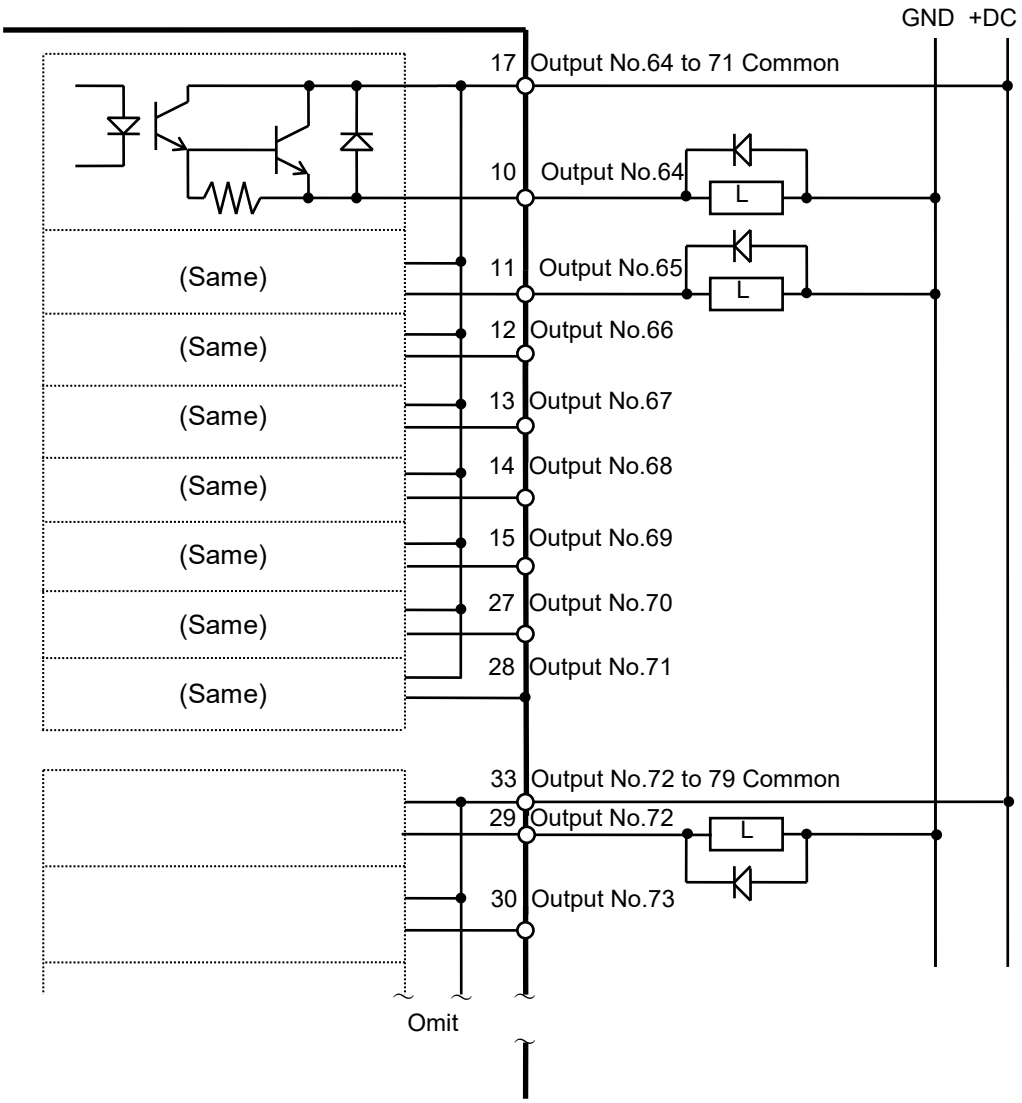
Output Driver : Photo coupler

CAUTION	■ The output circuit has two types: Sink type and Source type. The either type has been configured in the controller before the shipment. Before routing the cables, make sure that the I/O output type of your controller conforms to the external connection devices. If you route the cables with wrong output type, the parts on the board will be broken and the robot system won't operate normally. ■ Use the wiring diagram of *2: Source Type* for CE conformance. Be sure to wire correctly. Improper wiring may cause safety problems as it may make the Manipulator move unusually. ■ Be sure to wire the output circuit properly because it has no protection circuitry for short-circuit and reverse-connection. Improper wiring may cause malfunction of the parts on the board and then improper function of the robot system.

Typical Output Circuit Application 1: Sink Type (NPN)



Typical Output Circuit Application 2: Source Type (PNP)



13.2.6 Pin Assignments

Pin Assignment table of the 1st Expansion I/O board.

Connector 1 Pin Assignments

Pin No.	Signal Name	Pin No.	Signal Name	Pin No.	Signal Name
1	Input common No.64 to 71	18	Input common No.72 to 79	34	Input common No.80 to 87
2	Input No.64	19	Input No.72	35	Input No.80
3	Input No.65	20	Input No.73	36	Input No.81
4	Input No.66	21	Input No.74	37	Input No.82
5	Input No.67	22	Input No.75	38	Input No.83
6	Input No.68	23	Input No.76	39	Input No.84
7	Input No.69	24	Input No.77	40	Input No.85
8	Input No.70	25	Input No.78	41	Input No.86
9	Input No.71	26	Input No.79	42	Input No.87
10	Output No.64	27	Output No.70	43	Output No.75
11	Output No.65	28	Output No.71	44	Output No.76
12	Output No.66	29	Output No.72	45	Output No.77
13	Output No.67	30	Output No.73	46	Output No.78
14	Output No.68	31	Output No.74	47	Output No.79
15	Output No.69	32	Not Used	48	Not Used
16	Not Used	33	Output common No.72 to 79	49	Not Used
17	Output common No.64 to 71			50	Not Used

Connector	Standard
I/O Connector (Controller side)	D-sub 50 male pin Screwlock #4 - 40

* The I/O connector, I/O cable, terminal block, and I/O cable kit are offered as options.

* I/O connector is included with shipment.

13.3 Fieldbus I/O Board

Fieldbus I/O supports the following models.
- DeviceNet™
- PROFIBUS-DP
- PROFINET
- CC-LINK
- EtherNet/IP™
- EtherCAT®
- Modbus (This is not optional but a standard feature.)

Refer to the following manuals.
Robot Controller option Fieldbus I/O
EPSON RC+ 7.0 User's Guide 11.7 Fieldbus Slave I/O

13.4 RS-232C Board

13.4.1 RS-232C Board

One standard RS-232C port is not available with the Controller.

You need to mount the RS-232C board in the Option Slot to communicate with external equipment using two or more port RS-232C.

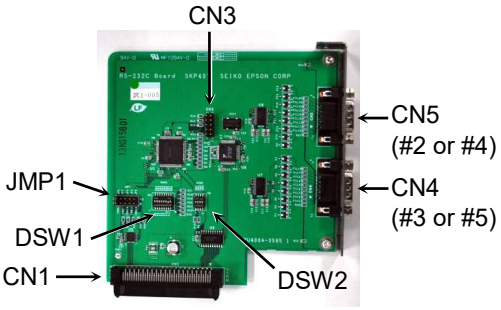
The RS-232C board accepts two ports expansion per board. A maximum of two boards, four ports expansion is available for RS-232C board.

Port numbers are assigned as follows.

Port No.	Supported hardware
#2, #3	First RS-232C board
#4, #5	Second RS-232C board

13.4.2 Board Setup (RS-232C)

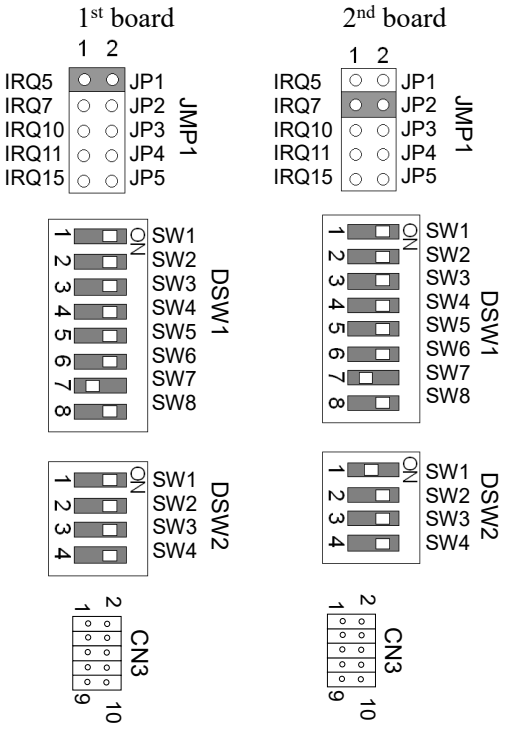
Board Appearance



Switch and Jumper Configuration

Set DSW1, DSW2 and JMP1.

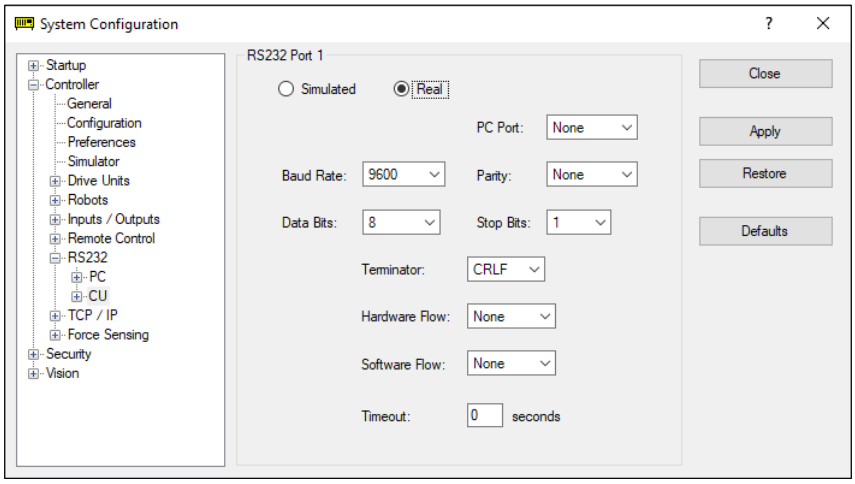
CN3 is all open.



13.4.3 Confirmation with EPSON RC+ 7.0 (RS-232C)

When an RS-232C board is mounted in as option unit, the Controller software automatically identifies the RS-232C board. Therefore, no software configuration is needed. Correct identification can be confirmed from EPSON RC+ 7.0.

(1) Select the EPSON RC+ 7.0 menu-[Setup]-[System Configuration] to display the [System Configuration] dialog.



(2) Select the [RS232]-[CU].

13.4.4 RS-232C Software Communication Setup (RS-232C)

Available communication settings are as follows.

Item	Specification
Baud Rates	110, 300, 600, 1200, 2400, 4800, 9600, 14400, 19200, 38400, 57600, 115200
Data bit length	7, 8
Stop bit length	1, 2
Parity	Odd, even, NA
Terminator	CR, LF, CRLF

Refer to *EPSON RC+ 7.0 Online Help* or *Users Guide – 13. RS-232C Communications* for RS-232C communication from the Robot application.

13.4.5 Communication Cable (RS-232C)

Prepare a communication cable as described in this section.

Connector	Standard
RS-232C Connector (Controller side)	D-sub 9 male pin Screwlock #4 - 40

NOTE ☞ Use twisted pair cable for shielded wire.
Clamp the shield to the hood for noise prevention.

Pin assign of the RS-232C connector is as follows.

Pin No	Signal	Function	Signal Direction
1	DCD	Data carrier detect	Input
2	RXD	Receive data	Input
3	TXD	Send data	Output
4	DTR	Terminal ready	Output
5	GND	Signal ground	-
6	DSR	Data set ready	Input
7	RTS	Request to send	Output
8	CTS	Clear to send	Input
9	RI	Ring indicator	Input

13.5 Analog I/O Board

13.5.1 About Analog I/O Board

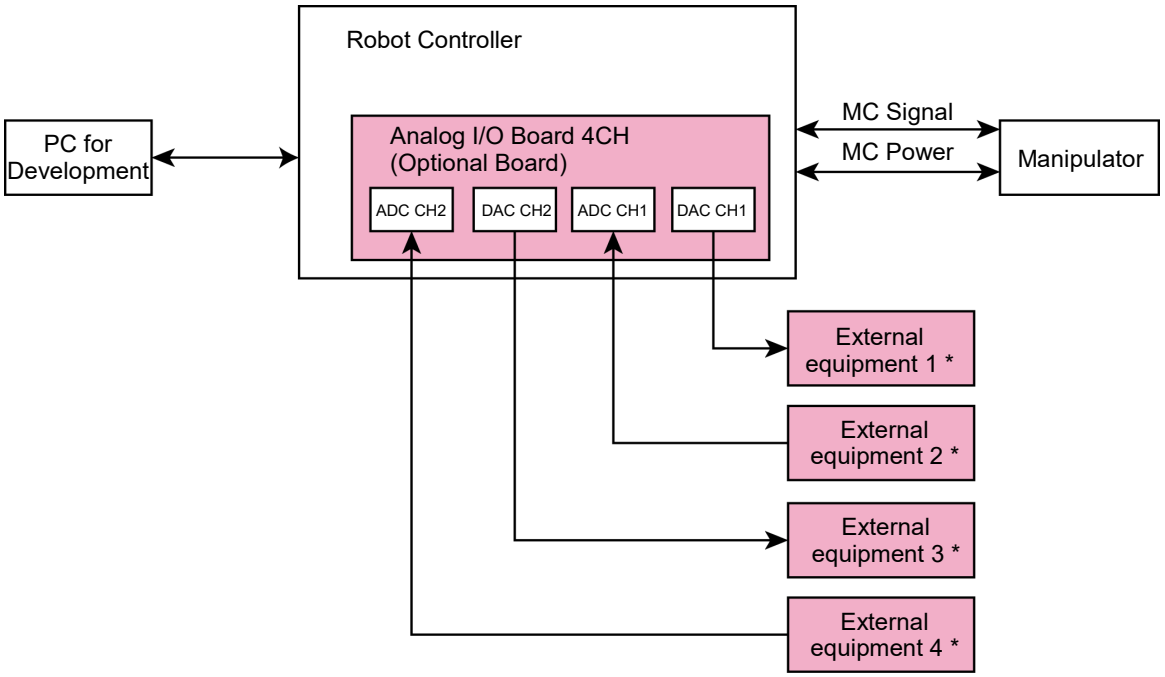
Analog input/output function is available when mounting analog I/O board to the option slot.
You can install a maximum of four analog I/O boards in the option slot.

Analog I/O Board (1CH): “DAC: 1ch” is available for one board.

Analog I/O Board (4CH): “DAC: 2ch, ADC: 2ch” is available for one board.

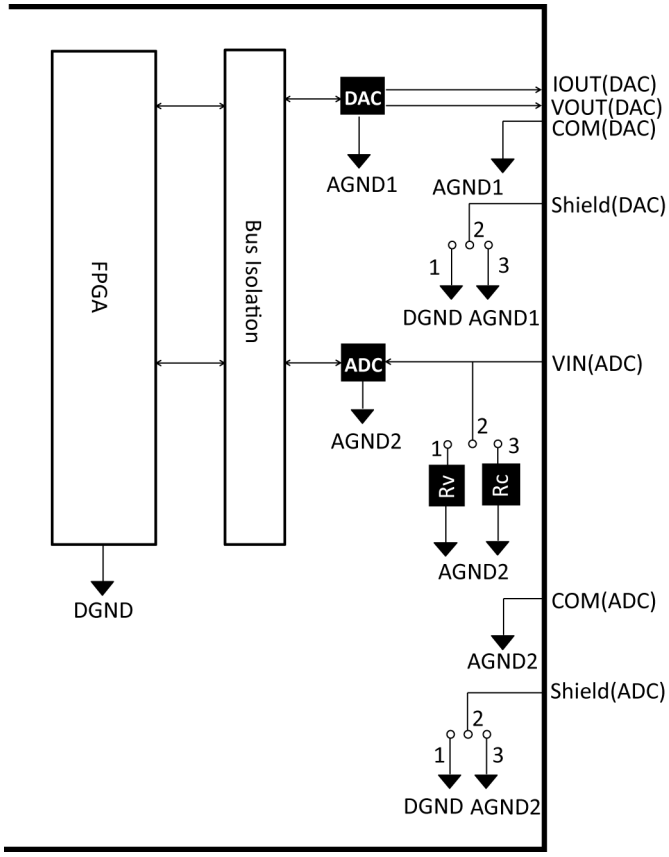
DAC: Analog signal output (voltage/current)
ADC: Analog signal input (voltage/current)

Connection Example of Analog I/O Board (4CH)



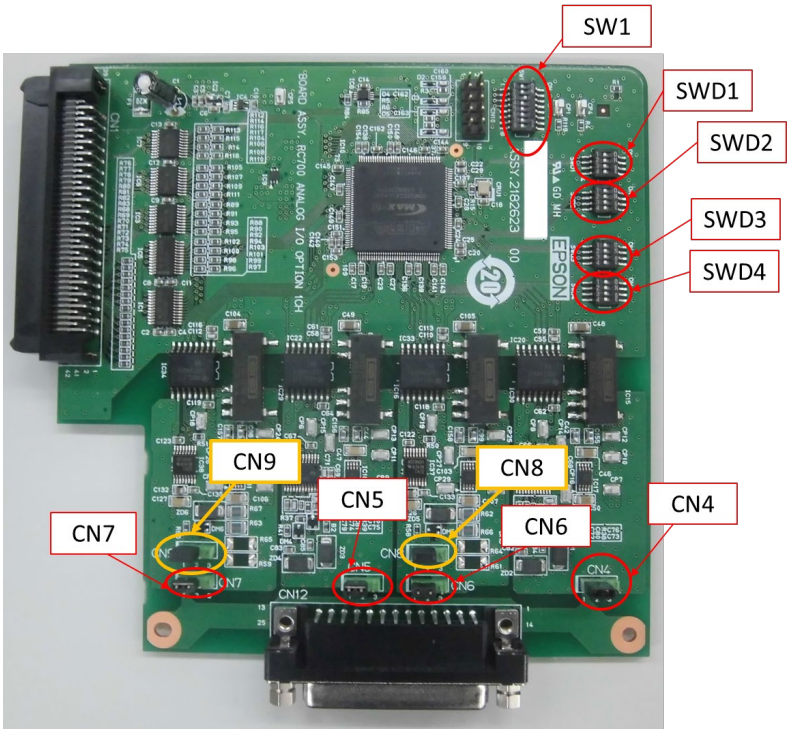
*: Voltage/Current Input

Overview of Analog I/O Board Circuit



Rv: Voltage Input Terminating Resistance (100kΩ),
Rc: Current Input Terminating Resistance

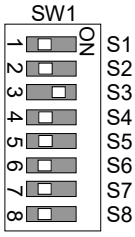
13.5.2 Board Configuration (Analog I/O Board)



Configuration of Switch and Jumper

(1) Address Configuration (SW1): Address of the option board

Configuration	S1	S2	S3	S4	S5	S6	S7	S8
The first board	Off	Off	On	Off	Off	Off	Off	Off
The second board	Off	On	Off	Off	Off	Off	Off	Off
The third board	On	Off	Off	Off	Off	Off	Off	Off
The fourth board	Off	Off	Off	Off	Off	Off	Off	Off



S5 to S8: Not used. Please turn them OFF.

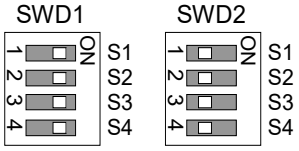
(2) Analog Output: Configuration

Voltage Current Switching Switch (SWD1): Voltage/current output

Range Switching Switch (SWD2): Output range

Channel	Output Mode	Range Configuration	SWD1 (voltage current switching)				SWD2 (range switching)			
			S1	S2	S3	S4	S1	S2	S3	S4
DAC 1ch	Voltage output mode	±5V	On	Not Use	Not Use	Not Use	On	On	Not Use	Not Use
		±10V	On				On	Off		
		0~5V	On				Off	On		
		0~10V *	On				Off	Off		
	Current output mode	0~20mA	Off				On	On		
		4~20mA	Off				Off	Off		
DAC 2ch	Voltage output mode	±5V	Not Use	On	Not Use	Not Use	Not Use	Not Use	On	On
		±10V		On					On	Off
		0~5V		On					Off	On
		0~10V *		On					Off	Off
	Current output mode	0~20mA		Off					On	On
		4~20mA		Off					Off	Off

*: Default: DAC default configuration (voltage output: 0 to 10V)



(3) Analog Input: Configuration

Voltage Current Switching Jumper (CN8/CN9): Current input/voltage input
1-2 pin short: Voltage input configuration
2-3 pin short: Current input configuration

Voltage Current Switching Switch (SWD1): Voltage/current input

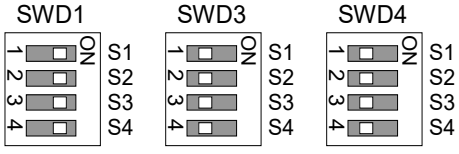
Range Switching Switch (SWD3): Input range

Channel	Input Mode	Range Configuration	SWD1(voltage current switching)				CN8 (voltage current switching)	CN9 (voltage current switching)
			S1	S2	S3	S4		
ADC 1ch	Voltage Input Mode	±5.12V	Not Use	Not Use	On	Not Use	1-2 short	Not Use
		±10.24V			On		1-2 short	
		0 to 5.12V			On		1-2 short	
		0 to 10.24V *			On		1-2 short	
	Current Input Mode	0 to 24mA			Off		2-3 short	
ADC 2ch	Voltage Input Mode	±5.12V	Not Use	Not Use	Not Use	On	Not Use	1-2 short
		±10.24V				On		1-2 short
		0 to 5.12V				On		1-2 short
		0 to 10.24V *				On		1-2 short
	Current Input Mode	0 to 24mA				Off		2-3 short

Channel	Input Mode	Range Configuration	SWD3 (range switching)				SWD4
			S1	S2	S3	S4	
ADC 1ch	Voltage Input Mode	±5.12V	On	On	Not Use	Not Use	Off
		±10.24V	On	Off			
		0 to 5.12V	Off	On			
		0 to 10.24V *	Off	Off			
	Current Input Mode	0 to 24mA	Off	On			
ADC 2ch	Voltage Input Mode	±5.12V	Not Use	Not Use	On	On	Off
		±10.24V			On	Off	
		0 to 5.12V			Off	On	
		0 to 10.24V *			Off	Off	
	Current Input Mode	0 to 24mA			Off	On	

SWD4: Not used. Please turn them OFF.

*: Default: ADC default configuration (voltage input: 0 to 10.24V)



(4) Shield Configuration

"Frame Ground" and "User Ground" of the shield: CN4, CN5, CN6, and CN7

1-2 pin short : Frame ground (FG) shield configuration.
When you want to spread the shield noise to the robot controller side.

2-3 pin short : User ground (UG) shield configuration.
When you want to insulate the shield by external connection device and robot controller.
Or when you want to spread the shield noise to the external connection device side.

User Ground (UG): Analog ground (AGND) on the external connection device side.

Frame Ground (FG): Digital ground (DGND) inside the robot controller.

Channel	Configuration	CN4	CN5	CN6	CN7
DAC1ch	FG Shield*	1-2 short	Not Use	Not Use	Not Use
	UG Shield	2-3 short			
DAC2ch	FG Shield *	Not Use	1-2 short	Not Use	Not Use
	UG Shield		2-3 short		
ADC1ch	FG Shield *	Not Use	Not Use	1-2 short	Not Use
	UG Shield			2-3 short	
ADC2ch	FG Shield *	Not Use	Not Use	Not Use	1-2 short
	UG Shield				2-3 short

*: Default

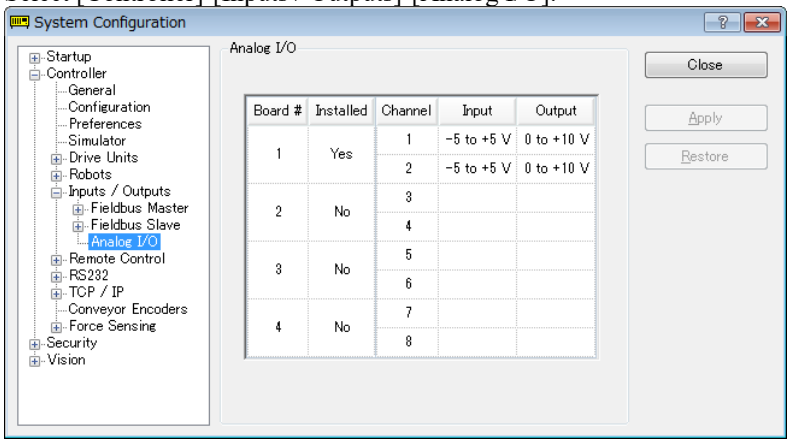
13.5.3 Confirmation with EPSON RC+ (Analog I/O Board)

The Controller software automatically identifies the analog I/O board when mounting analog I/O board to the optional unit of the controller. Therefore, no software configuration is needed.

Correct identification can be confirmed from EPSON RC+.

(1) Select the EPSON RC+ 7.0 menu-[Setup]-[System Configuration] to display the [System Configuration] dialog.

(2) Select [Controller]-[Inputs / Outputs]-[Analog I/O].



13.5.4 Input Circuit (Analog I/O Board)

Input resolution : 16 bit

Input range : voltage input: 0 to 5.12V, 0 to 10.24V, ±5.12V, ±10.24V
current input: 0 to 24mA

Input impedance : 100kΩ (±5%)

Absolute rated voltage of input pins
: ±11V

Insulation specification : insulation between channels, bus insulation

13.5.5 Output Circuit (Analog I/O Board)

Output resolution : 16 bit

Output range : voltage output: 0 to 5V, 0 to 10V, ±5V, ±10V
current output: 0 to 20mA, 4 to 20mA

Output impedance : 2Ω (TYP)

Absolute rated voltage of output pins
: ±11V

Insulation specification : insulation between channels, bus insulation

CAUTION	■ Analog input/output may not function properly if performing improper wiring or configuration. ■ Operating under external noise may affect to the analog input/output. Make sure the noise environment such as weather the cable is shielded. ■ Use the shield/twist cable. ■ Never apply a voltage or current which is out of input/output range to the analog input/output pin. Applying a voltage more than ±11V may result in malfunction of the board. ■ Improper wiring or short circuit may cause malfunction of the parts on the board and then improper function of the robot system.

13.5.6 Pin Assignments (Analog I/O Board)

1CH Specifications

Pin No.	Signal	Pin No.	Signal
1	VOUT (DAC 1ch)	20	Shield (DAC 1ch)
2	COM (DAC 1ch)	21	IOUT (DAC 1ch)
3	Shield (DAC 1ch)	22	COM (DAC 1ch)
4	Not Used	23	Not Used
5	Not Used	24	Not Used
6	Not Used	25	Not Used
7	Not Used	26	Not Used
8	Not Used	27	Not Used
9	Not Used	28	Not Used
10	Not Used	29	Not Used
11	Not Used	30	Not Used
12	Not Used	31	Not Used
13	Not Used	32	Not Used
14	Not Used	33	Not Used
15	Not Used	34	Not Used
16	Not Used	35	Not Used
17	Not Used	36	Not Used
18	Not Used	37	Not Used
19	Not Used		

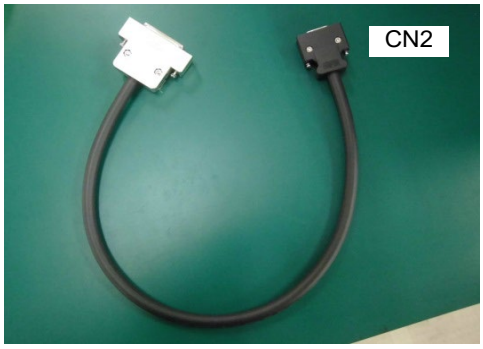
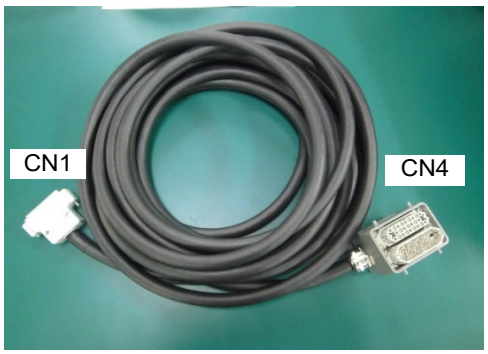
4CH Specifications

Pin No.	Signal	Pin No.	Signal
1	VOUT (DAC 1ch)	20	Shield (DAC 1ch)
2	COM (DAC 1ch)	21	IOUT (DAC 1ch)
3	Shield (DAC 1ch)	22	COM (DAC 1ch)
4	Not Used	23	Not Used
5	Not Used	24	Not Used
6	Not Used	25	Not Used
7	VIN (ADC 1ch)	26	Shield (ADC 1ch)
8	COM (ADC 1ch)	27	Not Used
9	Not Used	28	Not Used
10	Not Used	29	Not Used
11	VOUT (DAC 2ch)	30	Shield (DAC 2ch)
12	COM (DAC 2ch)	31	IOUT (DAC 2ch)
13	Shield (DAC 2ch)	32	COM (DAC 2ch)
14	Not Used	33	Not Used
15	Not Used	34	Not Used
16	Not Used	35	Not Used
17	Not Used	36	Not Used
18	VIN (ADC 2ch)	37	Shield (ADC 2ch)
19	COM (ADC 2ch)		

13.6 EUROMAP67 Board

EUROMAP67 is a standardized interface for communications between Western-manufactured injection molding machines (IMM) and robots.

List of accessories

Parts Code	Parts	Note
2194667	EUROMAP67 Cable1 	Emergency Stop cable (CN2)
2194668	EUROMAP67 Cable2 	Connection cable Robot Controller(CN1) - IMM(CN4)
2165789	EUROMAP67Emergency Connector Plug	For emergency stop switch wiring Soldering plug (CN3)
2194882	EUROMAP67Emergency Connector Shell	For emergency stop switch wiring Shell kit (CN3)

Wire the emergency stop switch wiring connector (CN3).

Reference: *9.3 Pin Assignments*

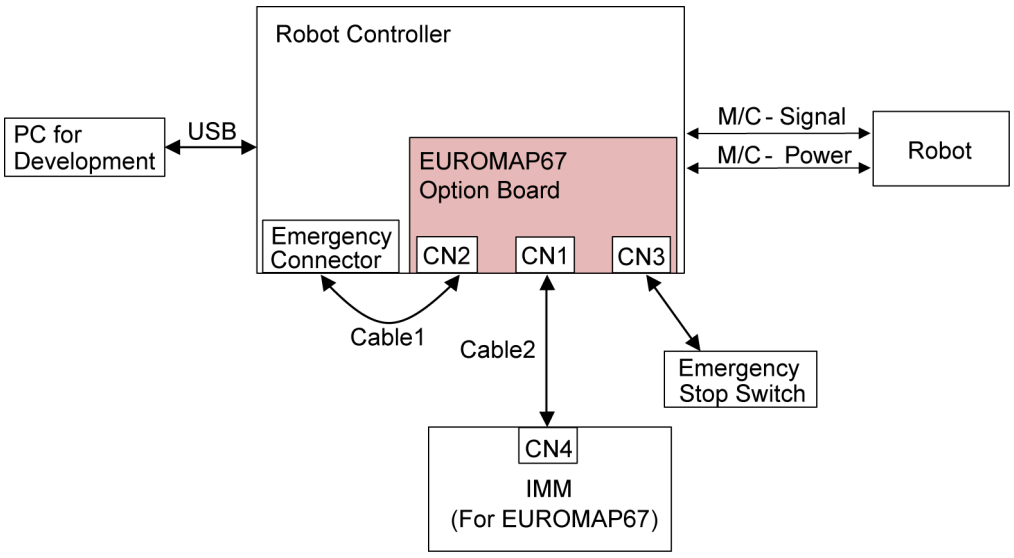
The connector signal placement is described below.

Reference: *13.6.11 Emergency stop connecter Pin Assignments*

List of connectors used

Connecter No.	Manufacturer	Model
CN1	JAE	DD-50PF-N
CN2	3M	10126-3000PE, 10326-52K0-008
CN3 (accessory)	3M	10136-3000PE, 10336-52K0-008
CN4	Tyco	T1319320125-000 / T2020252201-000 / T2020252101-000

Connection outline



(IMM: Injection Molding Machine)

13.6.1 Notes on the EUROMAP67 Board

The EUROMAP67 board contains 15 inputs and 16 outputs on a single substrate.

You can install up to one EUROMAP67 board in the controller.

The input and output bit numbers are assigned as follows.

Input Bit #	Output Bit #	Supported Hardware
192 to 206	192 to 208	The 1st EUROMAP67 board

Outputs No.205 and No.237 are not used. However, the EUROMAP standard describes that they may potentially be used in the future.

EUROMAP67 pin definitions

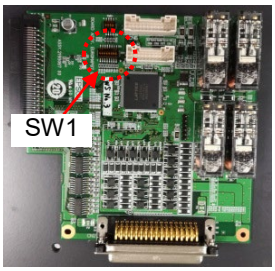
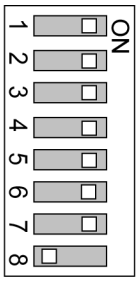
EUROMAP connecter (CN4) Pin No.	Signal Name	Note
ZA1 ZC1	Emergency stop of machine channel1	
ZA2 ZC2	Emergency stop of machine channel2	
ZA3 ZC3	Safety devices of machine channel1	
ZA4 ZC4	Safety devices of machine channel2	
ZA5	Reject	I/O Input (*1)
ZA6	Mold closed	I/O Input (*1)
ZA7	Mold open position	I/O Input (*1)
ZA8	Intermediate mold opening position	I/O Input (*1)
ZA9	Supply from handling device / robot	24V DC (Robot → IMM)
ZB2	Enable operation with handing device / robot	I/O Input (*1)
ZB3	Ejector back position	I/O Input (*1)
ZB4	Ejector forward position	I/O Input (*1)

EUROMAP connecter (CN4) Pin No.	Signal Name	Note
ZB5	Core pullers 1 in position 1	I/O Input (*1)
ZB6	Core pullers 1 in position 2	I/O Input (*1)
ZB7	Core pullers 2 in position 1	I/O Input (*1)
ZB8	Core pullers 2 in position 2	I/O Input (*1)
ZC5	Reserved for future use by EUROMAP	I/O Input (*1)
ZC6	Reserved for future use by EUROMAP	I/O Input (*1)
ZC7	Reserved for future use by EUROMAP	I/O Input (*1)
ZC8	Not fixed by EUROMAP, manufacturer dependent	I/O Input (*1)
ZC9	Supply from handling device / robot	0V (Robot → IMM)
A1 C1	Emergency stop of robot channel1	
A2 C2	Emergency stop of robot channel2	
A3 C3	Mold area free	
A4 C4	Reserved for future use by EUROMAP	
A5	Not fixed by EUROMAP, manufacturer dependent	I/O Input (*1)
A6	Enable mold closure	I/O Input (*1)
A7	Enable full mold opening	I/O Input (*1)
A8	Reserved for future use by EUROMAP	I/O Input (*1)
A9	Supply from IMM	24V DC (IMM → Robot) (*1)
B2	Robot operation mode	I/O Input (*1)
B3	Enable ejector back	I/O Input (*1)
B4	Enable ejector forward	I/O Input (*1)
B5	Enable movement of core pullers 1 to position 1	I/O Input (*1)
B6	Enable movement of core pullers 1 to position 2	I/O Input (*1)
B7	Enable movement of core pullers 2 to position 1	I/O Input (*1)
B8	Enable movement of core pullers 2 to position 2	I/O Input (*1)
C5	Not fixed by EUROMAP, manufacturer dependent	I/O Input (*1)
C6	Reserved for future use by EUROMAP	I/O Input (*1)
C7	Reserved for future use by EUROMAP	I/O Input (*1)
C8	Not fixed by EUROMAP, manufacturer dependent	I/O Input (*1)
C9	Supply from IMM	0V (IMM → Robot)

*1: DO NOT input a voltage which exceeds 24V. Board may get damage and burnout.

13.6.2 Board Settings (EUROMAP67 Board)

Configure DIP-Switch (SW1) to enable the robot controller to recognize the EUROMAP67 board.

Board Appearance	Switch setting: Setup the SW1
	

13.6.3 Installation (EUROMAP67 Board)

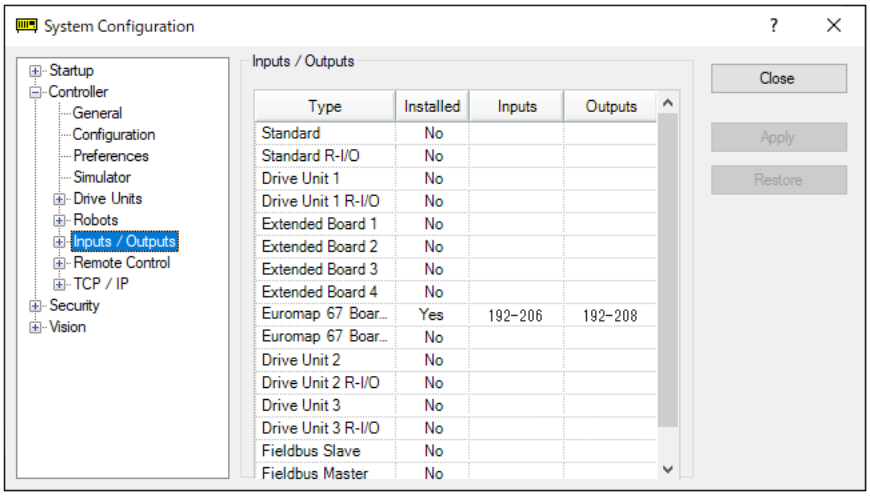
For the install procedure, refer to the *RC90 series Maintenance Manual.*

13.6.4 Confirming with EPSON RC+ 7.0 (EUROMAP67 Board)

The controller software will automatically recognize the EUROMAP67 board when attaching it to the optional unit. This eliminates the need to configure software settings.

You can check whether the software has correctly recognized the EUROMAP67 interface on the EPSON RC+ 7.0 screen.

(1) Select EPSON RC+ 7.0 menu - [Setup] - [System Configuration] to display the [System Configuration] dialog box.



(2) Select [Controller] - [Inputs/Outputs].

(3) Check that the [Installed] is “Yes”.
The EUROMAP67 board has been recognized by the controller software.
You can now select the compatible inputs and outputs to use.

13.6.5 Sample Project (EUROMAP67 Board)

You can use a sample project of EUROMAP 67 board.

The following describes procedures to use the sample project.

(1) Select EPSON RC+ 7.0 menu - [Project]-[Open..].

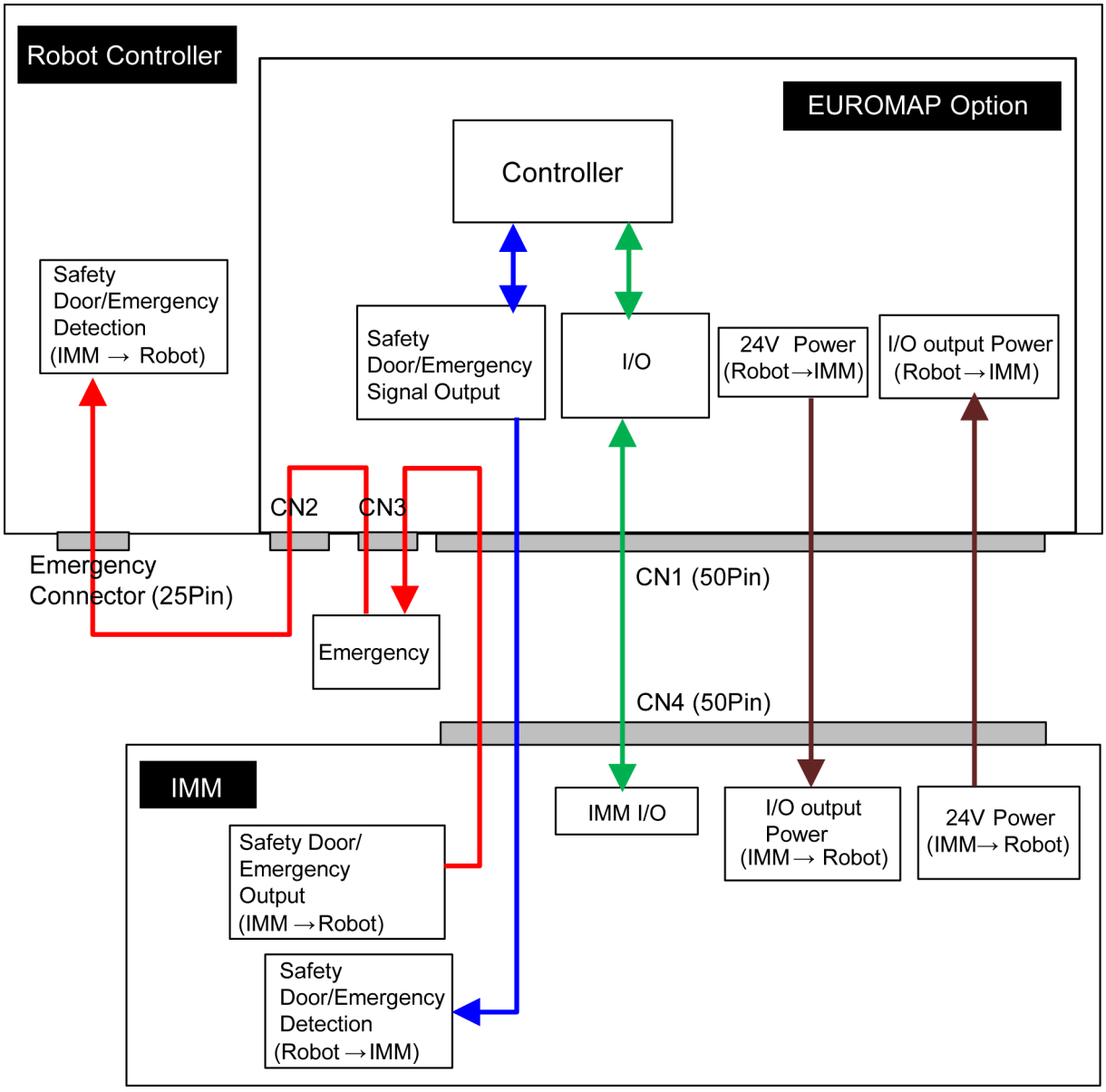
(2) Select [Projects] - [Samples] - [Euromap67Demo1].

(3) Click the <Open> button.

(4) Modify the project depending on the IMM.

13.6.6 Circuit Overview (EUROMAP67 Board)

EUROMAP67 Board: System diagram



13.6.7 Input Circuit (EUROMAP67 Board)

Input Voltage Range : + 12 to 24 V ±10 %

ON Voltage : + 10.8 V (MIN.)

OFF Voltage : + 5 V (MAX.)

Input Current : 10 mA TYP / + 24 V input

13.6.8 Output Circuit (EUROMAP67 Board)

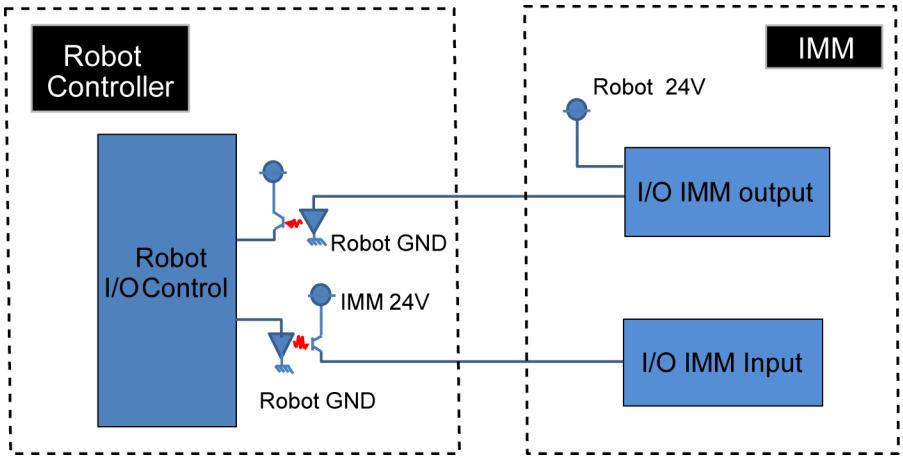
Rated Output Voltage : + 12 V to 24 V ± 10 %

Maximum Output Current : TYP 100 mA / 1 output

Output Driver : PhotoMOS relay

On-resistance (average) : 23.5Ω or less

EUROMAP67 board input/output circuit overview



CAUTION	■ Input/output circuits do not have a built-in protection circuit to prevent short circuits or reverse connections. Take care to avoid wiring mistakes. Wiring mistakes may damage board parts and prevent the robot system from functioning properly. ■ Do not use a higher than rated voltage or current. Doing so may damage board parts and prevent the robot system from functioning properly. ■ Note that the I/O logic for controlling the IMM will vary depending on the molding machine. Confirm the proper logic to use before creating programs.

13.6.9 Emergency Stop, Safeguard (EUROMAP67 Board)

When the emergency stop switch is held down on the robot controller:

A function is used to communicate the emergency stop instruction to the IMM.

The emergency stop instruction is communicated using a safety relay. A deposition detection function is also provided.

→ When deposition is detected, the robot controller will set the emergency stop instruction communicated to the IMM to OPEN.

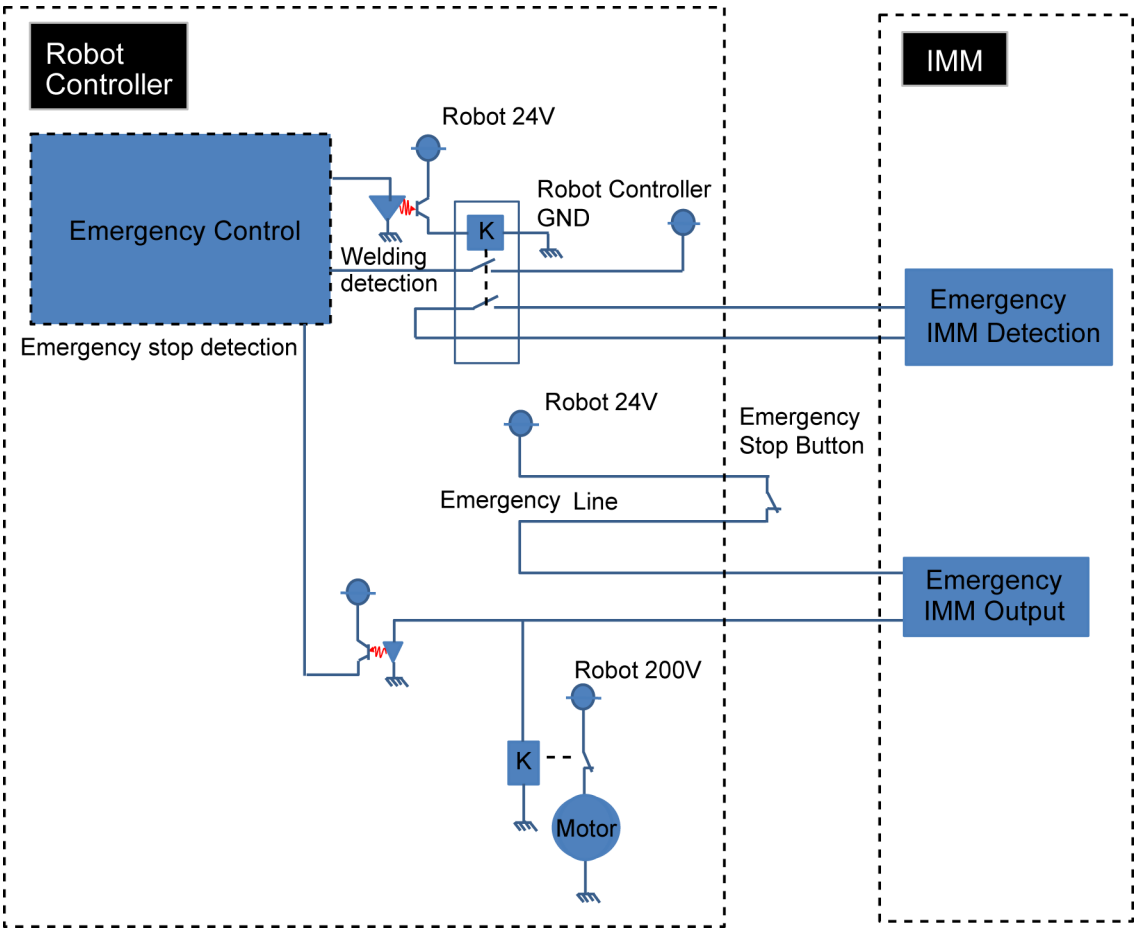
When an emergency stop occurs on the IMM side:

A function is used to communicate the emergency stop instruction to the robot controller.

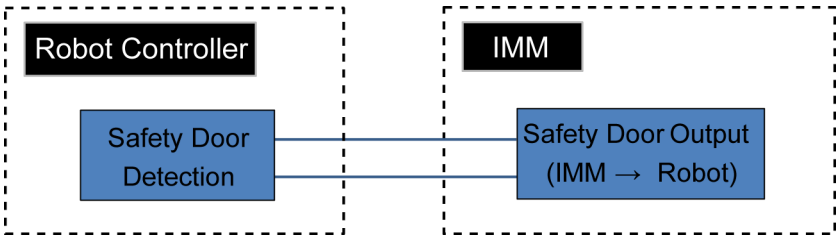
When the safety door has been opened on the IMM side:

A function is used to communicate the open safeguard instruction to the robot controller.

EUROMAP67 Board: Overview of Emergency stop circuit



EUROMAP67 Board: Overview of Safety Door circuit



13.6.10 I/O Pin Assignments (EUROMAP67 Board)

Signal Name	D-Sub connecter (CN1) Pin No.	EUROMAP67 connecter (CN4) Pin No.
Input No.192	33	ZA5 (*1)
Input No.193	32	ZA6 (*1)
Input No.194	50	ZA7 (*1)
Input No.195	49	ZA8 (*1)
Input No.196	48	ZB2 (*1)
Input No.197	47	ZB3 (*1)
Input No.198	31	ZB4 (*1)
Input No.199	30	ZB5 (*1)
Input No.200	15	ZB6 (*1)
Input No.201	14	ZB7 (*1)
Input No.202	46	ZB8 (*1)
Input No.203	45	ZC5 (*1)
Input No.204	29	ZC6 (*1)
Input No.205	28	ZC7 (*1)
Input No.206	13	ZC8 (*1)
Output No.192	41	A6 (*1)
Output No.193	7	A7 (*1)
Output No.194	8	A8 (*1)
Output No.195	24	B2 (*1)
Output No.196	25	B3 (*1)
Output No.197	42	B4 (*1)
Output No.198	43	B5 (*1)
Output No.199	9	B6 (*1)
Output No.200	10	B7 (*1)
Output No.201	26	B8 (*1)
Output No.202	27	C6 (*1)
Output No.203	11	C7 (*1)
Output No.204	4/3	A3/C3 (ModuleArea+/ModuleArea-)
Output No.205 (Not Used)	37/36	A4/C4 (Not Used)
Output No.206	12	C8 (*1)
Output No.207	40	A5 (*1)
Output No.208	44	C5 (*1)
24V (IMM)	1	A9 (*1)
GND (IMM)	2	C9 (*1)
Emergency1 (IMM)	39/38	ZA1/ZC1
Emergency2 (IMM)	21/20	ZA2/ZC2
Safety1 (IMM)	6/5	ZA3/ZC3
Safety2 (IMM)	23/22	ZA4/ZC4
24V (Robot)	17	ZA9
GND (Robot)	16	ZC9
Emergency1 (Robot)	35/34	A1/C1
Emergency2 (Robot)	19/18	A2/C2

*1: DO NOT input a voltage which exceeds 24V. Board may get damage and burnout.

13.6.11 Emergency stop connecter Pin Assignments (EUROMAP67 Board)

Emergency stop connecter (CN2) Pin Assignment table of the EUROMAP67 board.

Emergency stop connecter (CN2) Pin No.	Signal Name	Function
1	ESW11	Emergency Stop switch contact
2	ESW12	Emergency Stop switch contact
3	ESTOP1+	Emergency Stop circuit 1+
4	ESTOP1-	Emergency Stop circuit 1-
5	Not Used	-
6	Not Used	-
7	SD11	Safety Door input 1
8	SD12	Safety Door input 1
9	24V	24V output
10	24V	24V output
11	24VGND	24VGND output
12	24VGND	24VGND output
13	Not Used	-
14	ESW21	Emergency Stop switch contact
15	ESW22	Emergency Stop switch contact
16	ESTOP2+	Emergency Stop circuit 2+
17	ESTOP2-	Emergency Stop circuit 2-
18	SDLATCH1	Safety Door Latch Release
19	SDLATCH2	Safety Door Latch Release
20	SD21	Safety Door input 2
21	SD22	Safety Door input 2
22	24V	24V output
23	24V	24V output
24	24VGND	24VGND output
25	24VGND	24VGND output
26	Not Used	-

Emergency stop connecter (CN3) Pin Assignment table of the EUROMAP67 board.

Emergency stop connecter (CN3) Pin No.	Signal Name	Function
1	ESW11	Emergency Stop switch contact
2	ESW12	Emergency Stop switch contact
3	ESTOP1+	Emergency Stop circuit 1+
4	ESTOP1-	Emergency Stop circuit 1-
5	Not Used	-
6	Not Used	-
7	SD11	Safety Door input 1
8	SD12	Safety Door input 1
9	24V	24V output
10	24V	24V output
11	24VGND	24VGND output
12	24VGND	24VGND output
13	Not Used	-
14	ESW21	Emergency Stop switch contact
15	ESW22	Emergency Stop switch contact
16	ESTOP2+	Emergency Stop circuit 2+
17	ESTOP2-	Emergency Stop circuit 2-
18	SDLATCH1	Safety Door Latch Release
19	SDLATCH2	Safety Door Latch Release
20	SD21	Safety Door input 2
21	SD22	Safety Door input 2
22	24V	24V output
23	24V	24V output
24	24VGND	24VGND output
25	24VGND	24VGND output
26	Not Used	-
27	Not Used	-
28	Not Used	-
29	Not Used	-
30	Not Used	-
31	Not Used	-
32	Not Used	-
33	Not Used	-
34	Not Used	-
35	Not Used	-
36	Not Used	-

Regular Inspection

Performing inspection properly is essential to prevent trouble and ensure safety.
This volume describes the inspection schedule and contents.
Inspect according to the schedule.

1. Regular Inspection for RC90-B

This chapter describes maintenance inspection procedures. Performing maintenance inspection properly is essential to prevent trouble and ensure safety.
Be sure to perform the maintenance inspections in accordance with the schedule.

1.1 Inspections and Schedules

Inspection Point	Frequency	Method	Check Item
Controller	12 months	Turn ON/OFF motor and Restart	Activates without any error
Emergency Stop Button	12 months	When the motor turned on, activate the emergency stop button	The LED lamp on the Manipulator turns off and E-STOP LED on the Controller turns on When the Controller is connected to EPSON RC+, "EStop" in red letters appears on the status bar
Safeguard	12 months	When the motor turned on, turn on the safeguard	The LED lamp on the Manipulator turns off When the Controller is connected to EPSON RC+, "Safety" in blue letters appears on the status bar
Fan Filter	1 months	Visual inspection and cleaning	No stain
Fan (Front)	1 months	Checking the operating sound	No noise
Fan (Rear)	1 months	Checking the operating sound	No noise
Battery	12 months	Visual inspection of seven-segment and LED	No alarm occurred

Appendix

Appendix A: Open Source Software License

(for RC90 series)

(1) This product includes open source software programs listed below according to the license terms of each open source software program.

(2) We provide the source code of the Open Source Programs (each is defined below) until five (5) years after the discontinuation of same model of this option product. If you desire to receive the source code above, please contact the "SUPPLIER" in the first pages of *Safety Manual*. You shall comply with the license terms of each open source software program.

(3) The open source software programs are WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. See the license agreements of each open source software program for more details, which are described below.

(4) OpenSSL toolkit

The T series product includes software developed by the OpenSSL project for use in the OpenSSL Toolkit (http://www.openssl.org/).

This product includes cryptographic software written by Eric Young (eay@cryptsoft.com).

(5) cJSON

This library is available under the MIT License.

Copyright (c) 2009-2017 Dave Gamble and cJSON contributors

https://opensource.org/licenses/mit-license.php

(6) Libxml2

This library is available under the MIT License.

Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.

https://opensource.org/licenses/mit-license.php

(7) OPC UA Stack

This module is released under the RCL Licence.

Portions copyright (C) by OPC Foundation, Inc. and licensed under the Reciprocal Community License (RCL)

https://opcfoundation.org/license/rcl.html